

7. Which of the following assumptions has been used to obtain the relation $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ for a transformer ?

- (A) The resistance of the primary coil is large.
- (B) The same flux links both the primary and the secondary coils.
- (C) The resistance of secondary coil is large.
- (D) The transformer is 100% efficient.

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7. Which of the following assumptions has been used to obtain the relation $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ for a transformer ?
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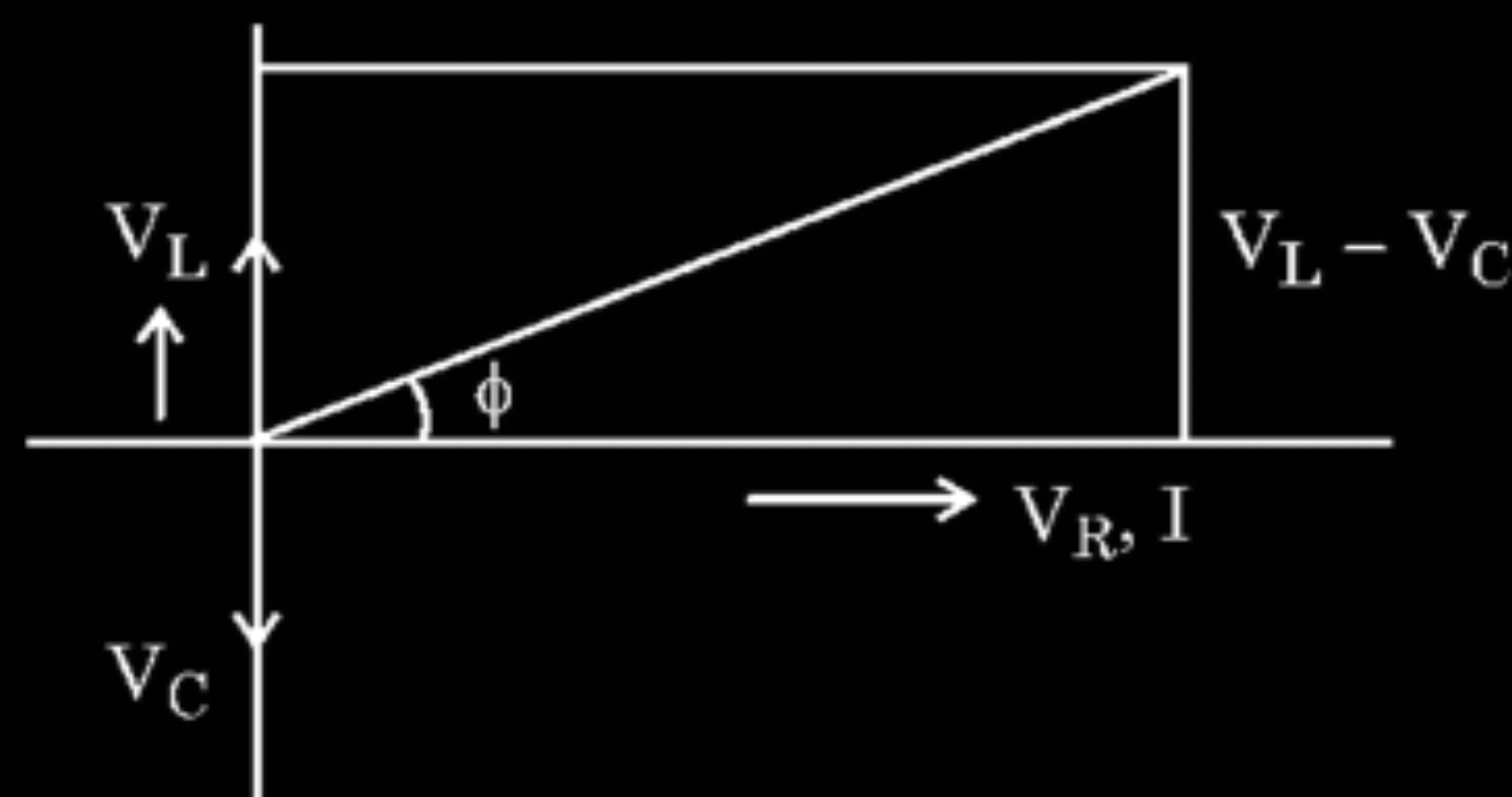
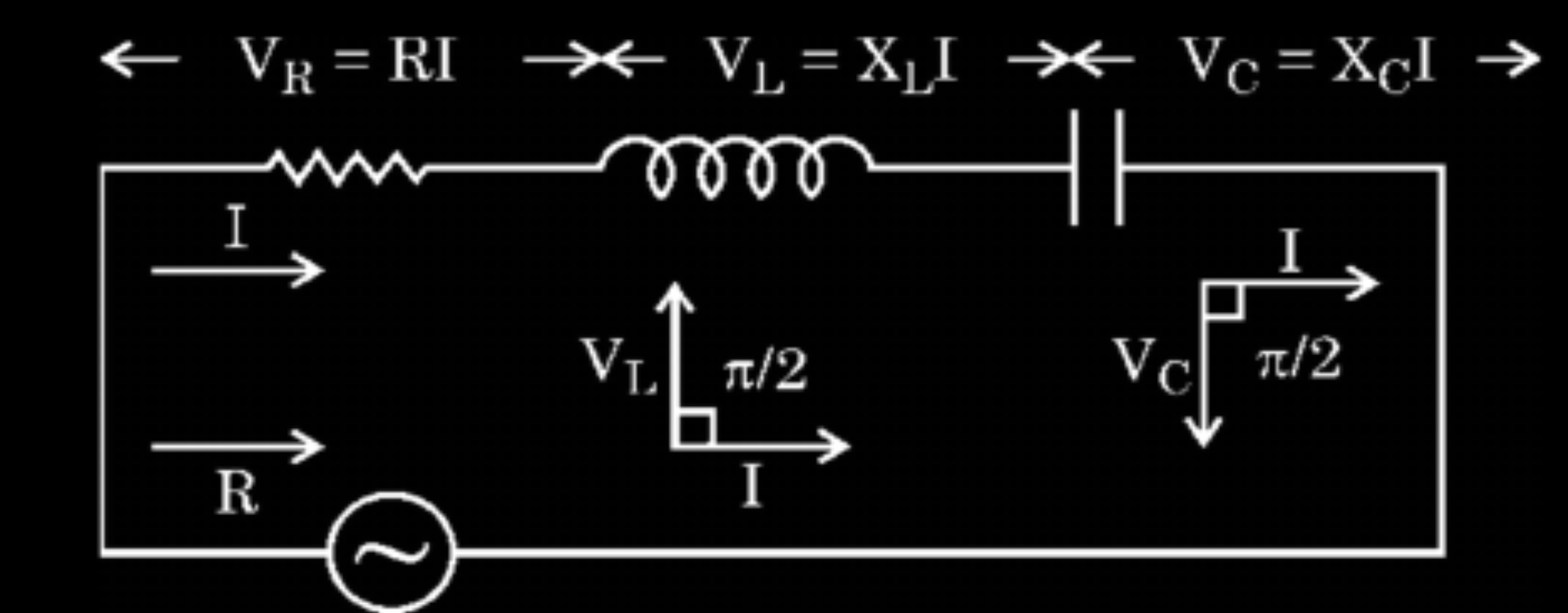
(B) The same flux links both the primary and secondary coils

- 23.** (a) A $20\ \mu\text{F}$ capacitor is connected to a $220\ \text{V}$, $\frac{125}{\pi}\ \text{Hz}$ ac source.
- (i) Find the capacitive reactance and the peak value of current in the circuit.
 - (ii) If the frequency of the ac source is doubled, by what factor will the capacitive reactance and the current be changed ? 3

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- 33.** (a) (i) A series combination of L, C and R is connected to an ac source $V = V_m \sin \omega t$. Obtain :
- (I) the impedance of the circuit using phasor diagram,
 - (II) the expression for the instantaneous current I, and
 - (III) the phase relationship of current to the applied voltage.
- (ii) Define power factor of an ac circuit. State the conditions under which it is :
- (I) maximum,
 - (II) minimum.

(a) (i) (I) Let resistance R, inductor L and capacitor C be connected in series with an alternate e.m.f.



$$\mathcal{E} = \sqrt{I^2[R^2 + (X_L - X_c)^2]}$$

$$Z = \frac{\mathcal{E}}{I}$$

$$Z = \sqrt{R^2 + (X_L - X_c)^2}$$

(II) $I_m = \frac{V_m}{Z} \sin(\omega t - \phi)$ 1/2

(III) Phase angle $\tan \phi = \frac{V_L - V_C}{V_R}$ 1/2

(ii) $P = E_V I_V \cos \phi$ where P = true power and $E_V I_V$ is the apparent power. ϕ is phase difference between current and voltage.
Power factor is defined as the ratio of true power to the apparent power.

Power factor = $\frac{\text{True Power (P)}}{\text{Apparent power (E_V I_V)}} = \cos \phi$

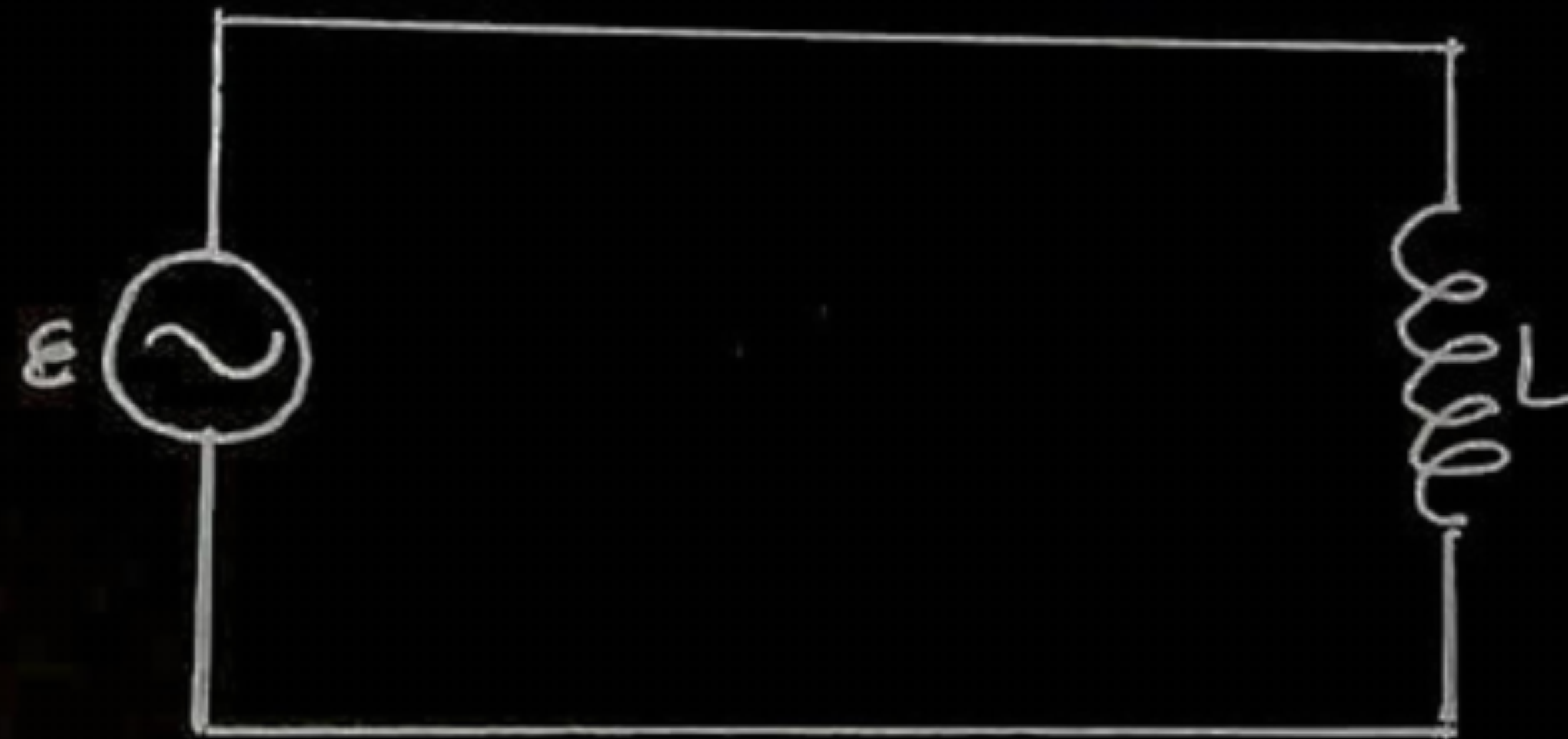
(i) Power factor is maximum when $\phi = 0$ purely resistive circuit(It occurs when $\omega = \frac{1}{\sqrt{LC}}$ at resonance in an LCR series circuit). 1/2

(ii) Power factor is minimum when $\phi = \frac{\pi}{2}$, in pure inductive or capacitive circuit. 1/2

- (b) (i) Prove that the voltage is ahead of the current in phase by $\frac{\pi}{2}$ rad in an ac circuit containing an ideal inductor. CBSE 2025 C

(i) Let the voltage across the source be

$$v = v_m \sin \omega t$$



Using Kirchhoff's loop rule

$$v - L \frac{di}{dt} = 0$$

$$\frac{di}{dt} = \frac{v}{L} = \frac{v_m}{L} \sin \omega t$$

$$di = \frac{v_m}{L} \sin \omega t dt$$

Integrating

$$i = \frac{v_m}{\omega L} \cos \omega t$$

$$i = i_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

This shows that current lags behind the voltage by $\frac{\pi}{2}$ rad.

$\therefore$ Voltage is ahead of current in phase by $\frac{\pi}{2}$ rad

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. The average value of voltage of 220 V ac mains during its positive half cycle will be :

(A) 156 V

(B) 198 V

(C) 220 V

(D) zero

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(C) 220 V

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5. Which of the following statements about transformers is ***not*** correct ?
- (A) For ideal transformers, the output power is equal to the input power.
 - (B) A transformer can work for both alternating and direct voltages.
 - (C) For a step-up transformer, the voltage across the secondary coil is higher but the current in it is lower.
 - (D) For an ideal transformer, the magnetic flux linked with the secondary is equal to that linked with the primary.

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(B) A transformer can work for both alternating and direct voltages.

16. *Assertion (A) :* The reactance of an inductor decreases with the increase in the frequency of alternating voltage applied.

Reason (R) : The reactance of an inductor $X_L = \frac{1}{\omega L}$.

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Reason (R) : The reactance of an inductor $X_L = \frac{1}{\omega L}$.

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(D) Both Assertion (A) and Reason (R) are false.

8. An alternating current is given by $I = I_0 \cos (100\pi)t$. The least time the current takes to decrease from its maximum value to zero will be :

(A) $\left(\frac{1}{200}\right)\text{s}$

(B) $\left(\frac{1}{150}\right)\text{s}$

(C) $\left(\frac{1}{100}\right)\text{s}$

(D) $\left(\frac{1}{50}\right)\text{s}$

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(D) $\left(\frac{1}{50}\right)\text{s}$

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(A) 1/200 s

- 11.** A capacitor and an inductor are connected in series across an ac source of voltage of variable frequency. The frequency is increased continuously. The nature of the circuit before and after the resonance will be :
- (A) inductive only
 - (B) capacitive only
 - (C) capacitive and inductive respectively
 - (D) inductive and capacitive respectively

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- (B) capacitive only
- (C) capacitive and inductive respectively
- (D) inductive and capacitive respectively

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(C) capacitive and inductive respectively

- 27.** An ac source of voltage $V = V_0 \sin \omega t$ is connected to a circuit element X. It is observed that the current flowing through X varies as $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$.
- (a) Identify the element X and write the expression for its reactance.
 - (b) Plot a graph to show the variation of reactance of the element with the frequency of the applied voltage.
 - (c) Draw plots showing the variation of voltage and current with time over one cycle of applied ac.

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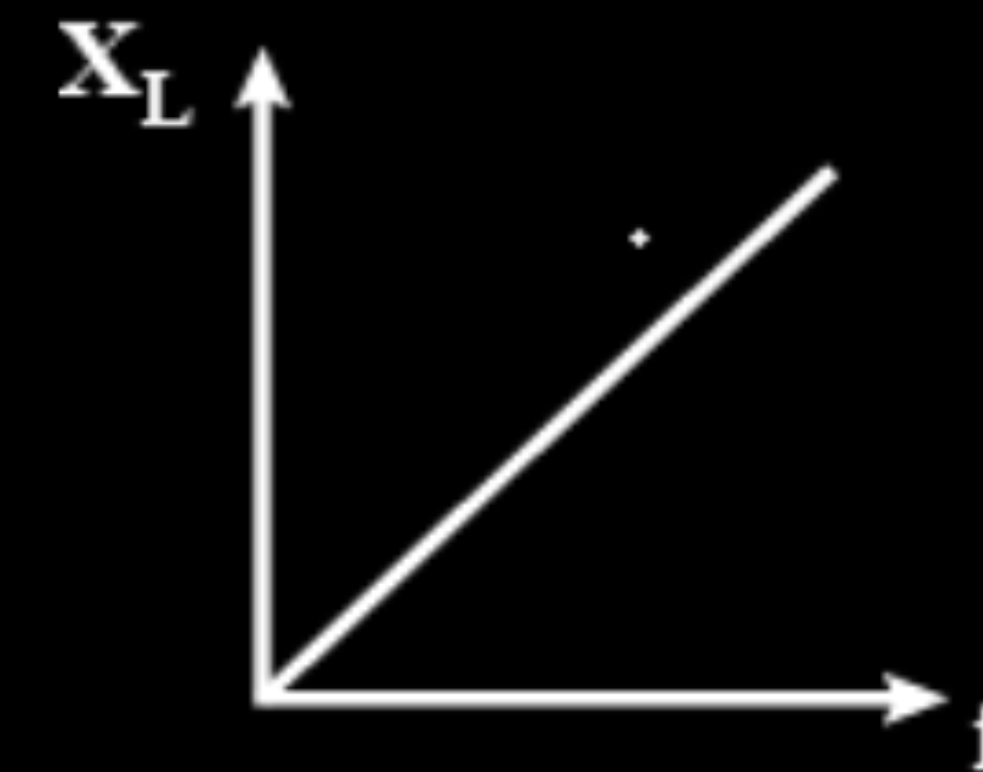
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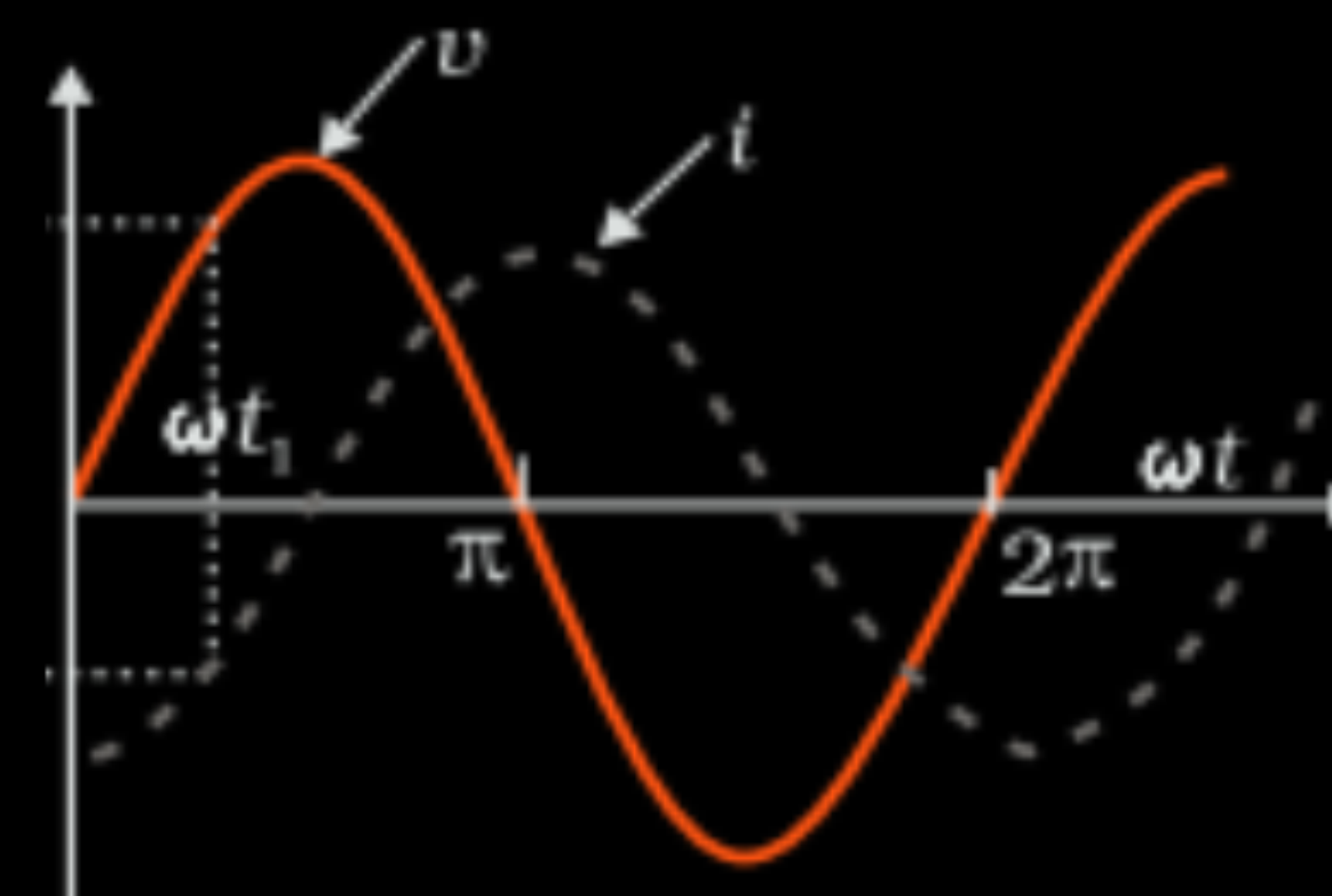
(a) Inductor

$$X_L = 2\pi fL$$

(b)



(c)



$\frac{1}{2}$

$\frac{1}{2}$

1

1

6. An ac source is connected to a resistor and an inductor in series. The voltage across the resistor and inductor are 8 V and 6 V respectively. The voltage of the source is :

(A) 10 V

(B) 12 V

(C) 14 V

(D) 16 V

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(A) 10 V

(B) 12 V

(C) 14 V

(D) 16 V

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(A) 10 V

- (ii) A resistor of $400\ \Omega$, an inductor of $\left(\frac{5}{\pi}\right)\text{H}$ and a capacitor of $\left(\frac{50}{\pi}\right)\mu\text{F}$ are joined in series across an ac source $v = 140 \sin(100\pi)t\ \text{V}$. Find the rms voltages across these three circuit elements. The algebraic sum of these voltages is more than the rms voltage of source. Explain.

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$$(i) \quad Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$= \sqrt{(400)^2 + \left(100\pi \times \frac{5}{\pi} - \frac{1}{100\pi \times \frac{50}{\pi} \times 10^{-6}} \right)^2}$$

$$= 500 \Omega$$

$$I_{rms} = \frac{V_{rms}}{Z}$$

$$I_{rms} = \frac{140}{\sqrt{2} \times 500} = \frac{0.28}{\sqrt{2}} \text{ A}$$

$$(V_{rms})_R = I_{rms} R$$

$$= \frac{0.28}{\sqrt{2}} \times 400$$

$$= \frac{112}{\sqrt{2}} = 56\sqrt{2} \text{ V}$$

$$(V_{rms})_L = I_{rms} \omega L$$

$$= \frac{0.28}{\sqrt{2}} \times 500$$

$$= \frac{140}{\sqrt{2}} = 70\sqrt{2} \text{ V}$$

$$(V_{rms})_C = I_{rms} \frac{1}{\omega C}$$

$$= \frac{0.28}{\sqrt{2}} \times 200$$

$$= \frac{56}{\sqrt{2}} = 28\sqrt{2} \text{ V}$$

The algebraic sum of voltages is more than the rms voltage of source because voltages across R, L and C are not in phase.

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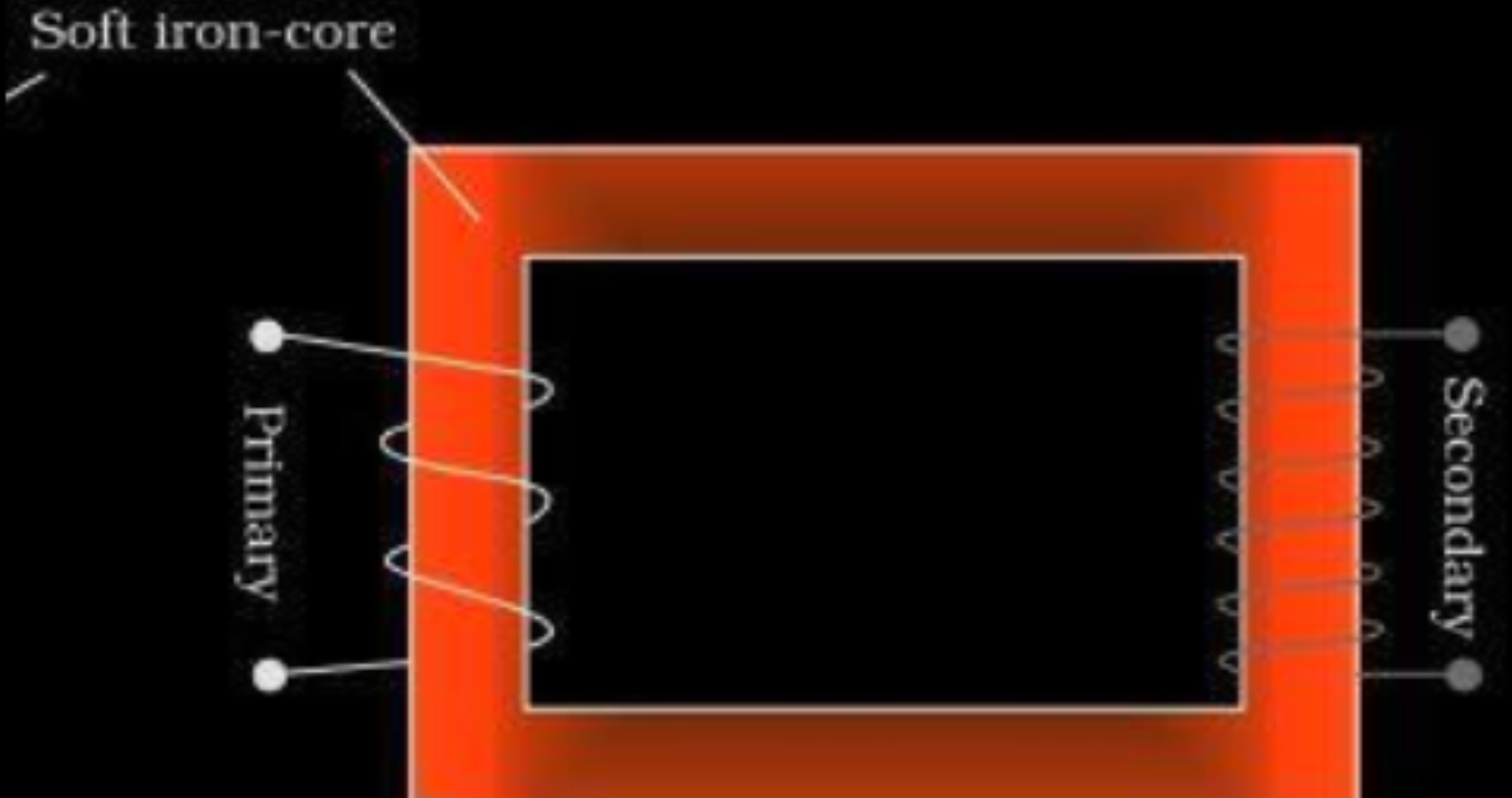
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- (b) (i) Write the principle of working of a transformer. With the help of a labelled diagram, explain the working of a step-up transformer.
- (ii) An ideal transformer is designed to convert 50 V into 250 V. It draws 200 W power from an ac source whose instantaneous voltage is given by $v_i = 20 \sin (100\pi)t$ V.
- Find :
- (I) rms value of input current.
 - (II) expression for instantaneous output voltage.
 - (III) expression for instantaneous output current.

5

(i) Principle: It works on the principle of mutual induction.	1
	1
Working: When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it. Since the no. of turns are more in secondary windings an emf induced is proportional to the no. of turns. Therefore more emf is developed across the secondary windings.	1

(ii) $P_i = V_p I_p$	$\frac{1}{2}$
$200 = \frac{20}{\sqrt{2}} I_p$	$\frac{1}{2}$
$I_p = 10\sqrt{2} \text{ A}$	
$\frac{V_o}{V_i} = \frac{250}{50}$	
$5 = \frac{V_o}{V_i}$	
$V_o = 100 \sin(100\pi) \text{ t V}$	$\frac{1}{2}$
$P_o = (V_o)_{rms} (I_o)_{rms}$	
$200 = \frac{100}{\sqrt{2}} (I_o)_{rms}$	
$(I_o)_{rms} = 2\sqrt{2} \text{ A}$	
$\therefore I_o = (2\sqrt{2})\sqrt{2} \sin(100\pi) \text{ t}$	
$I_o = 4 \sin(100\pi) \text{ t A}$	$\frac{1}{2}$

5. A step up transformer – 1
- (A) decreases the current to transmit power over short distances with minimum loss.
 - (B) increases the current to transmit power over short distances with minimum loss.
 - (C) increases the voltage to transmit power over short distances with minimum loss.
 - (D) increases the voltage to transmit power over long distances with minimum loss.

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 - (B) increases the current to transmit power over short distances with minimum loss.
 - (C) increases the voltage to transmit power over short distances with minimum loss.
 - (D) increases the voltage to transmit power over long distances with minimum loss.

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(D) increases the voltage to transmit power over long distance with minimum loss.

14. **Assertion (A)** : At resonance the impedance of a series LCR circuit is $Z = R$.

1

Reason (R) : $Z = \sqrt{R^2 + (X_C - X_L)^2}$ and at resonance the inductive reactance (X_L) is equal to the capacitive reactance (X_C).

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(A) Both assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the assertion (A).

25. (a) Explain how the reactance of a capacitor and that of an inductor change with the frequency of ac source. 3
- (b) An ideal inductor of self-inductance 0.5 H is connected to an ac source of peak voltage 314 V and frequency 50 Hz . Calculate the peak value of current in the circuit. **CBSE 2025**

25. (a) Explain how the reactance of a capacitor and that of an inductor change with the frequency of ac source. 3
- (b) An ideal inductor of self-inductance 0.5 H is connected to an ac source of peak voltage 314 V and frequency 50 Hz. Calculate the peak value of current in the circuit. CBSE 2025

a)	$X_c = \frac{1}{\omega C} = \frac{1}{2\pi\nu C}$ $X_c \propto \frac{1}{\nu}$ If frequency increases, X_c decreases and vice-versa	$\frac{1}{2}$
	$X_L = \omega L = 2\pi\nu L$ $X_L \propto \nu$ If frequency increases, X_L increases and vice-versa	$\frac{1}{2}$
b)	$I_o = \frac{V_o}{2\pi\nu L}$ $I_o = \frac{314}{2 \times 3.14 \times 50 \times 0.5} = 2A$	$\frac{1}{2}$
		$\frac{1}{2}$

2. A capacitor of capacitance C has reactance X in an ac circuit. If the capacitance and the frequency of the applied voltage are doubled, the new reactance will become :

(A) $4X$

(B) $2X$

(C) $\frac{X}{2}$

(D) $\frac{X}{4}$

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(D) $\frac{X}{4}$

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(D) $\frac{X}{4}$

6. A transformer is a device used for converting :
- (A) high ac voltage and large ac current to low ac voltage and small ac current
 - (B) high ac voltage and small ac current to low ac voltage and small ac current
 - (C) low ac voltage and large ac current to high ac voltage and small ac current
 - (D) low ac voltage and small ac current to high ac voltage and large ac current

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16. *Assertion (A) :* A series LCR circuit behaves as a pure resistive circuit at resonance.

Reason (R) : At resonance, $X_L = X_C$ gives $\omega = \frac{1}{\sqrt{LC}}$.

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(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of the Assertion (A)

- (b) (i) Differentiate between peak and rms values of alternating current. How are they related ?
- (ii) A current element X is connected across an ac source of emf $V = V_0 \sin 2\pi\nu t$. It is found that the voltage leads the current in phase by $\frac{\pi}{2}$ radian. If element X was replaced by element Y, the voltage lags behind the current in phase by $\frac{\pi}{2}$ radian.
- (I) Identify elements X and Y by drawing phasor diagrams.
- (II) Obtain the condition of resonance when both elements X and Y are connected in series to the source and obtain expression for resonant frequency. What is the impedance value in this case ?

(i)

Peak value - It is the maximum value of Alternating current.
rms value - It is the equivalent dc current that would produce the same average power loss as alternating current.

Alternatively-

Peak value - It is the maximum value of Alternating current.
rms value- It is the effective value of an ac representing the equivalent dc, that would produce the same heating effect in same resistor in same time period.

Relation $I_{rms} = \frac{I_o}{\sqrt{2}}$

(II) Impedance of the circuit

$$Z = (X_L - X_C)$$

At resonance $Z = 0$

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}, \quad \omega = \frac{1}{\sqrt{LC}}$$

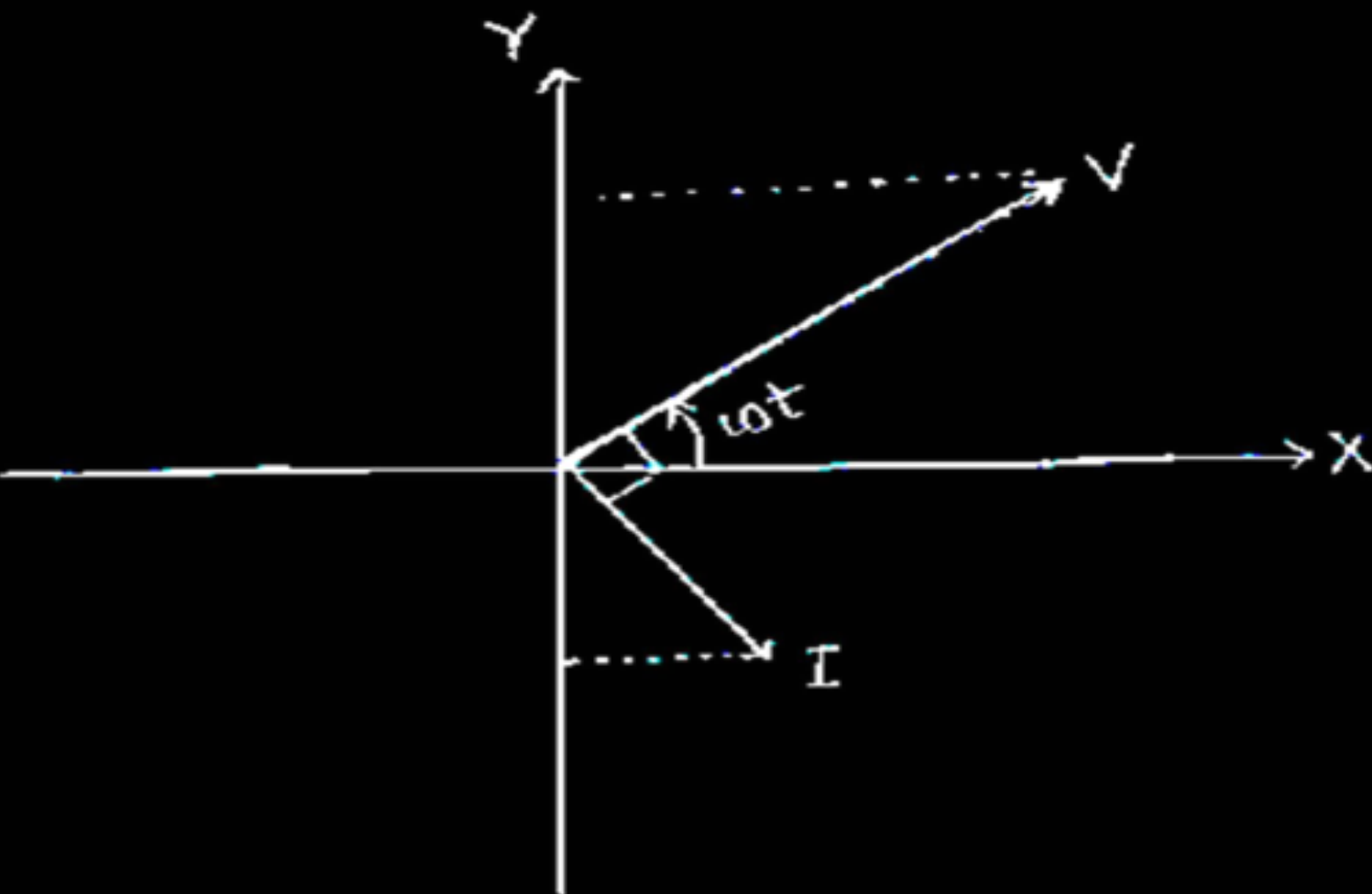
$$\nu = \frac{1}{2\pi\sqrt{LC}}$$

Impedance at resonance

$$Z=0$$

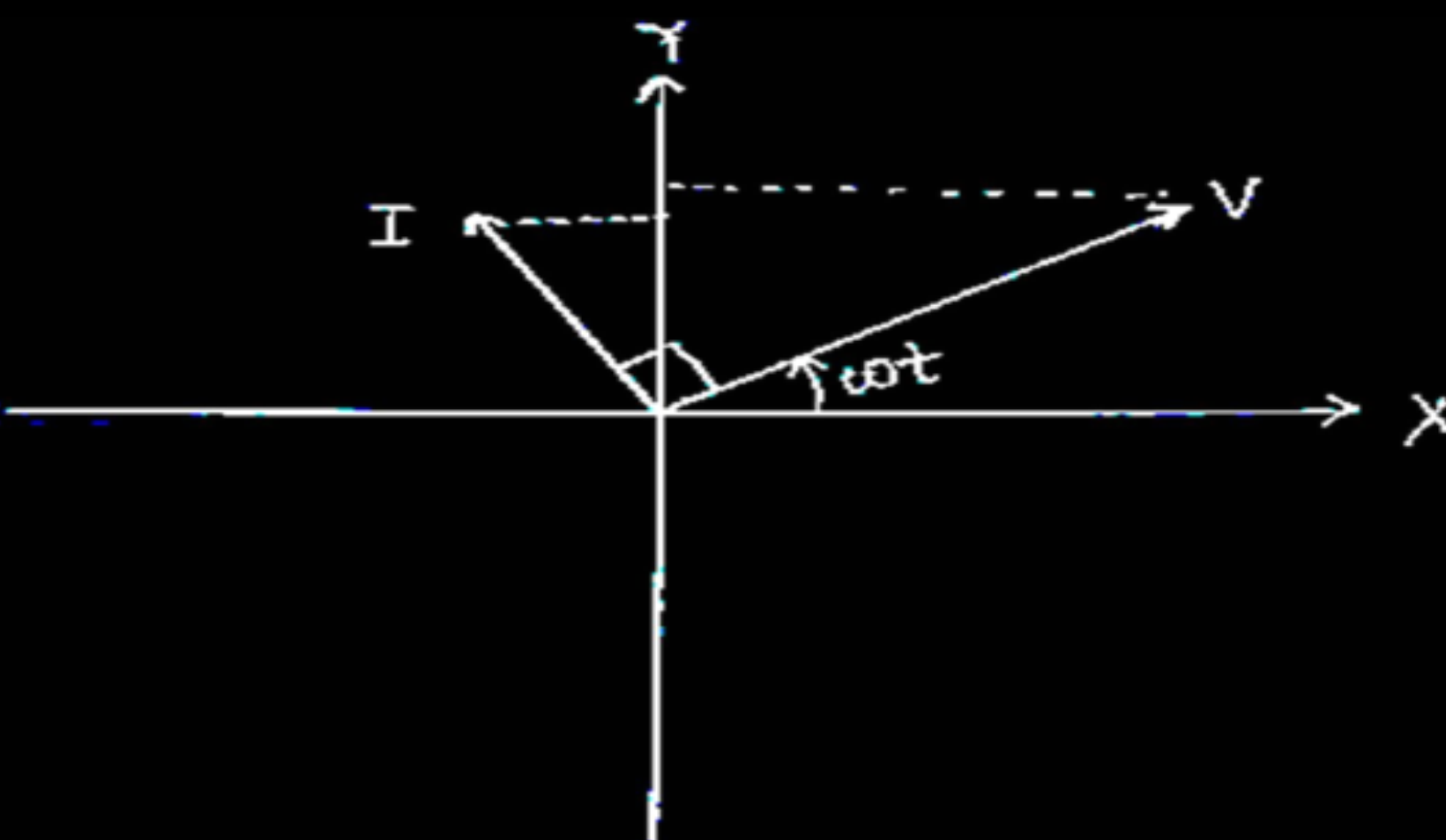
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(ii) (I) X- Inductor (L)



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Y- Capacitor (C)



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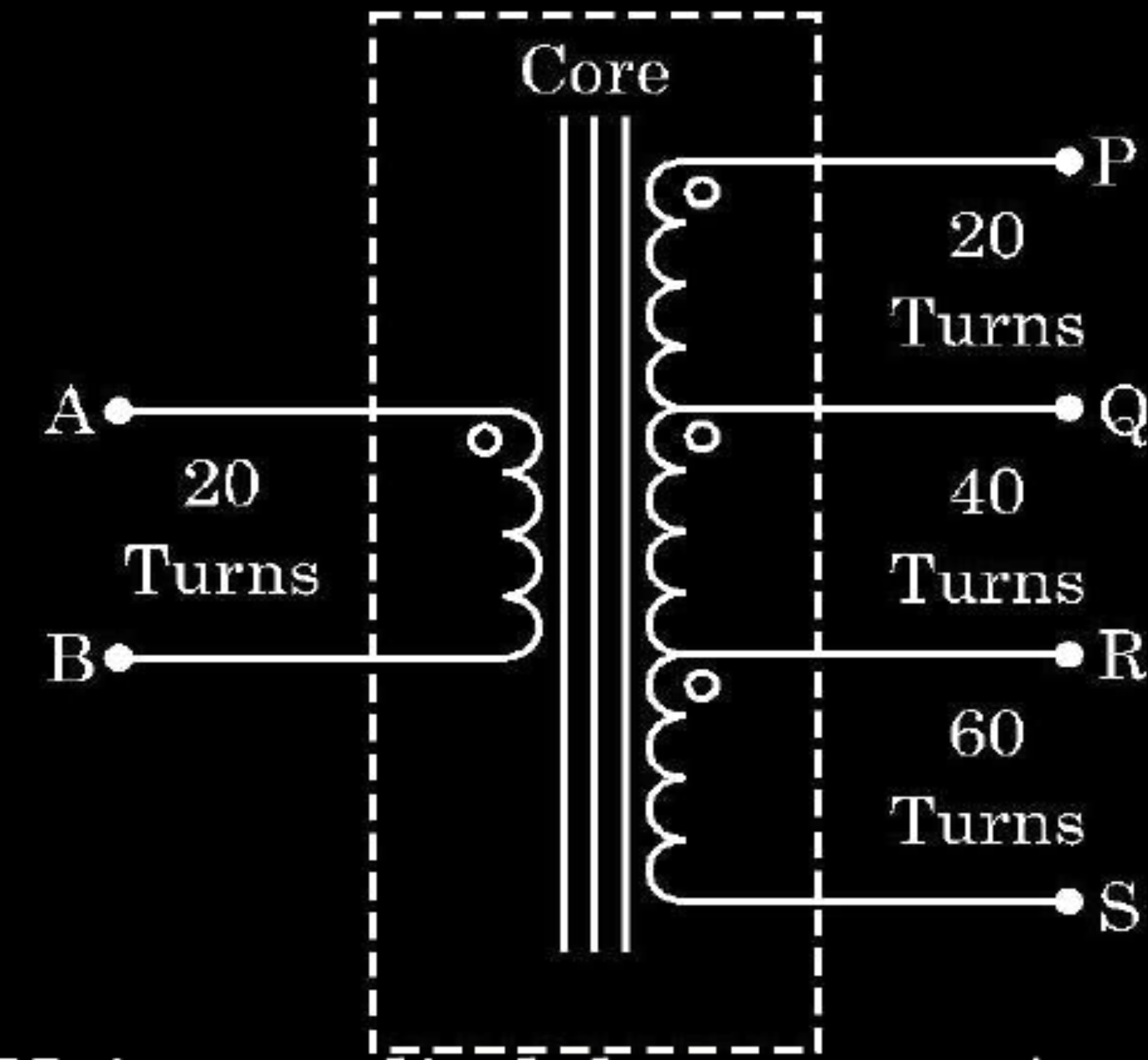
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7. The number of turns between different pairs of output terminals are shown for a step-up transformer. 1

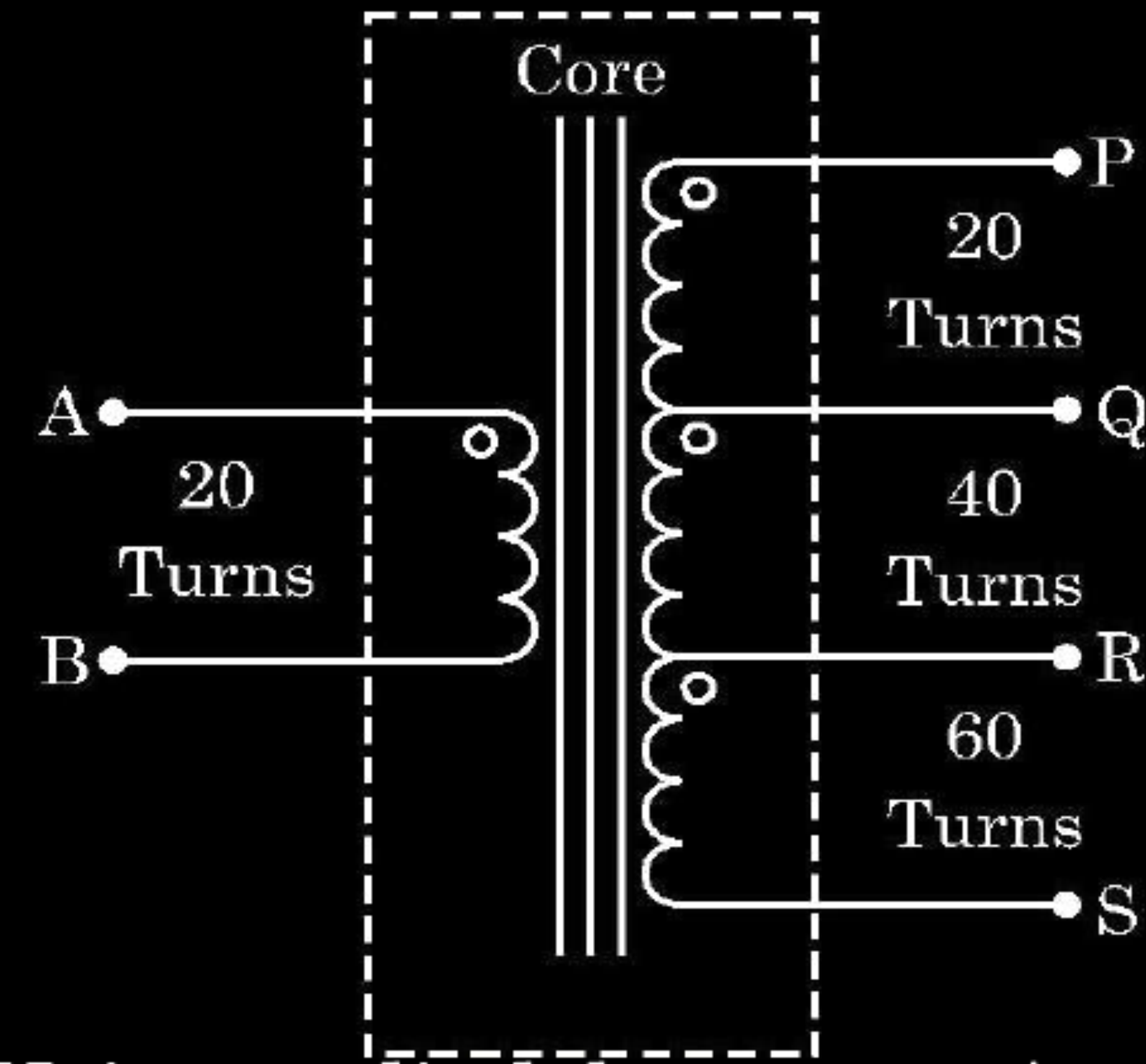


Input voltage of 20 V is applied between A and B. Between which two terminals will the output be 120 V ?

- (A) P and Q
(B) Q and S
(C) P and R
(D) P and S

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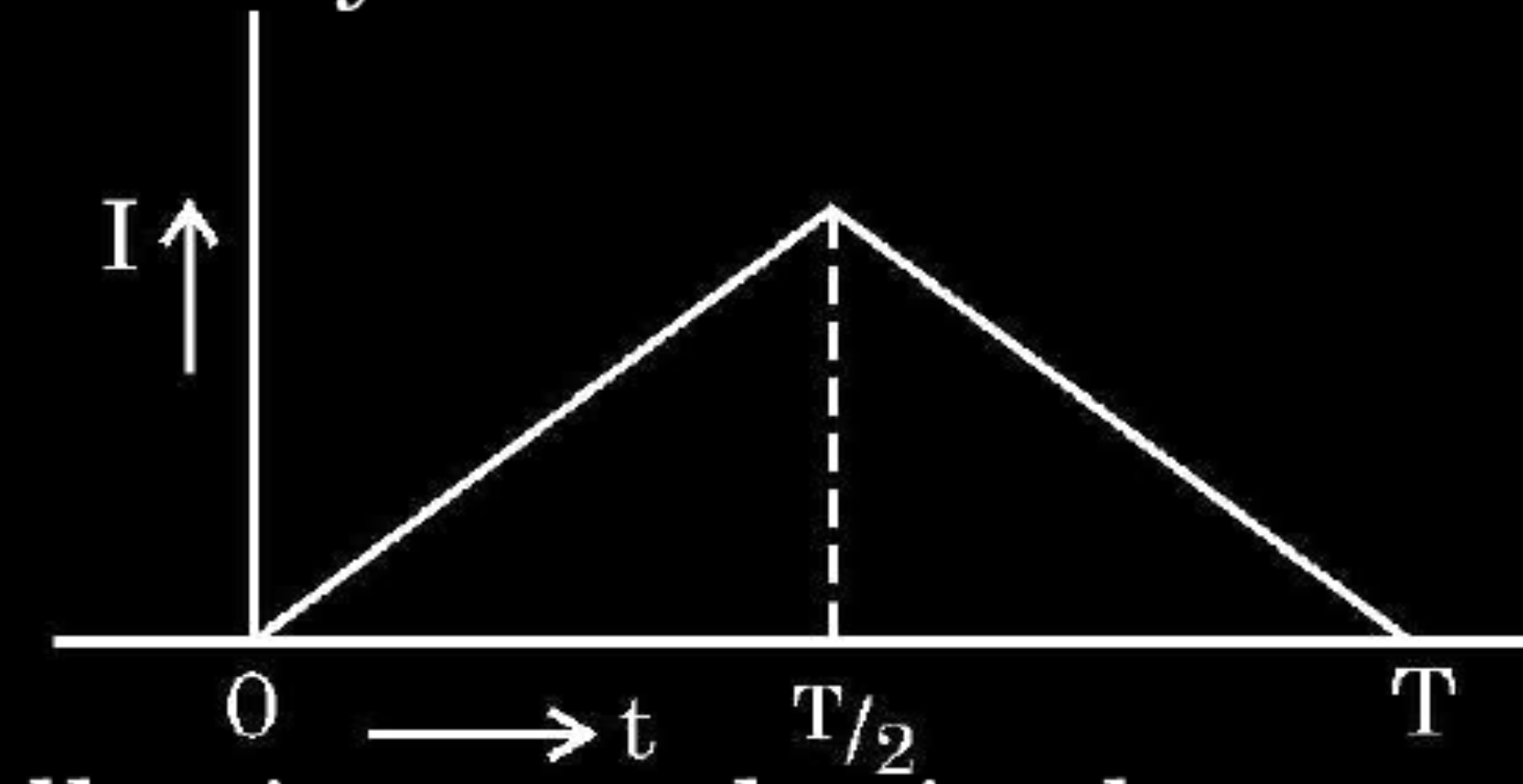
- (A) P and Q (B) Q and S
(C) P and R (D) P and S

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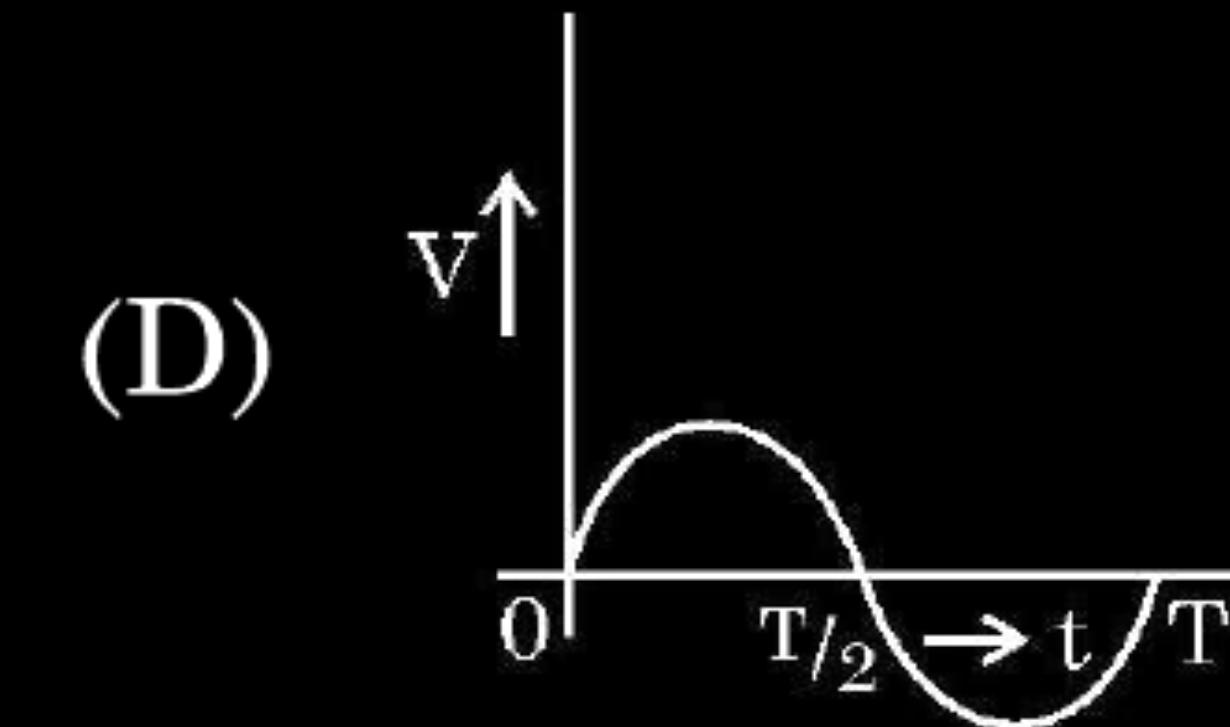
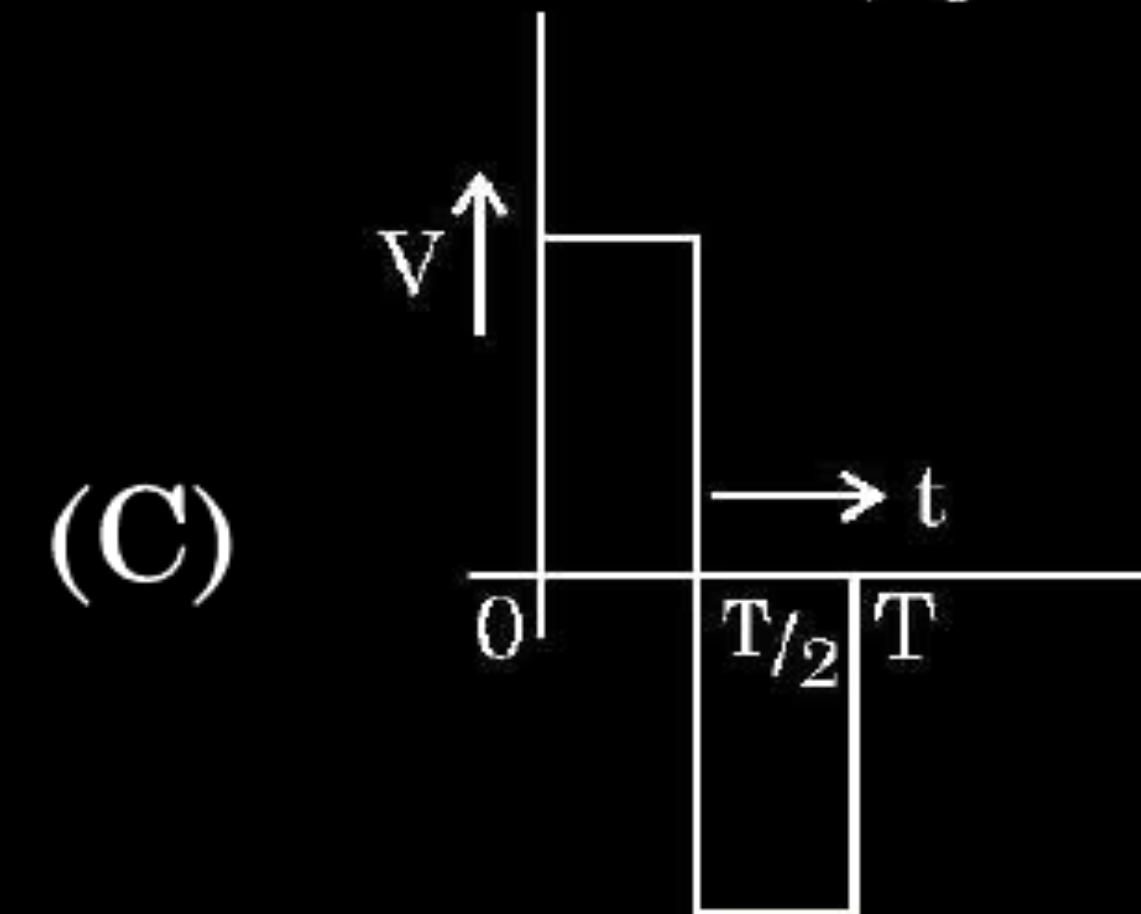
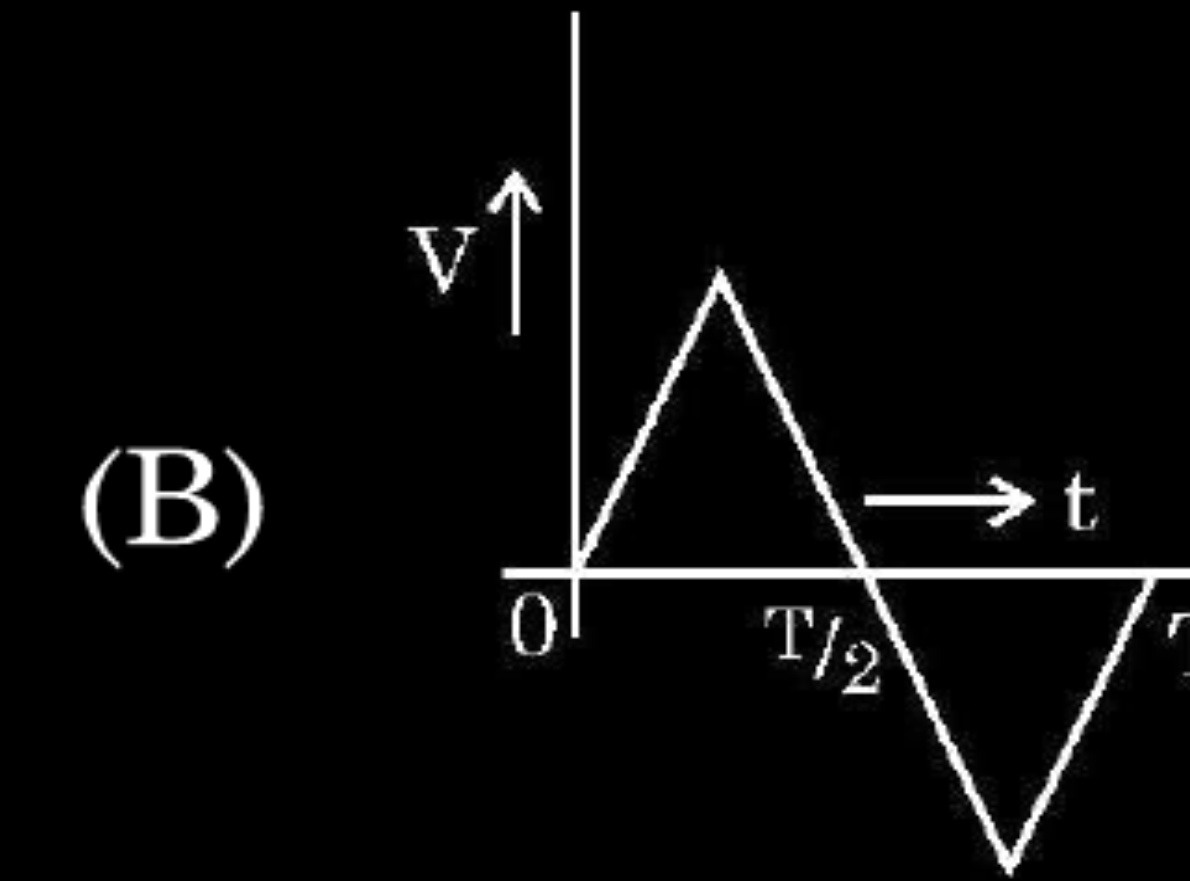
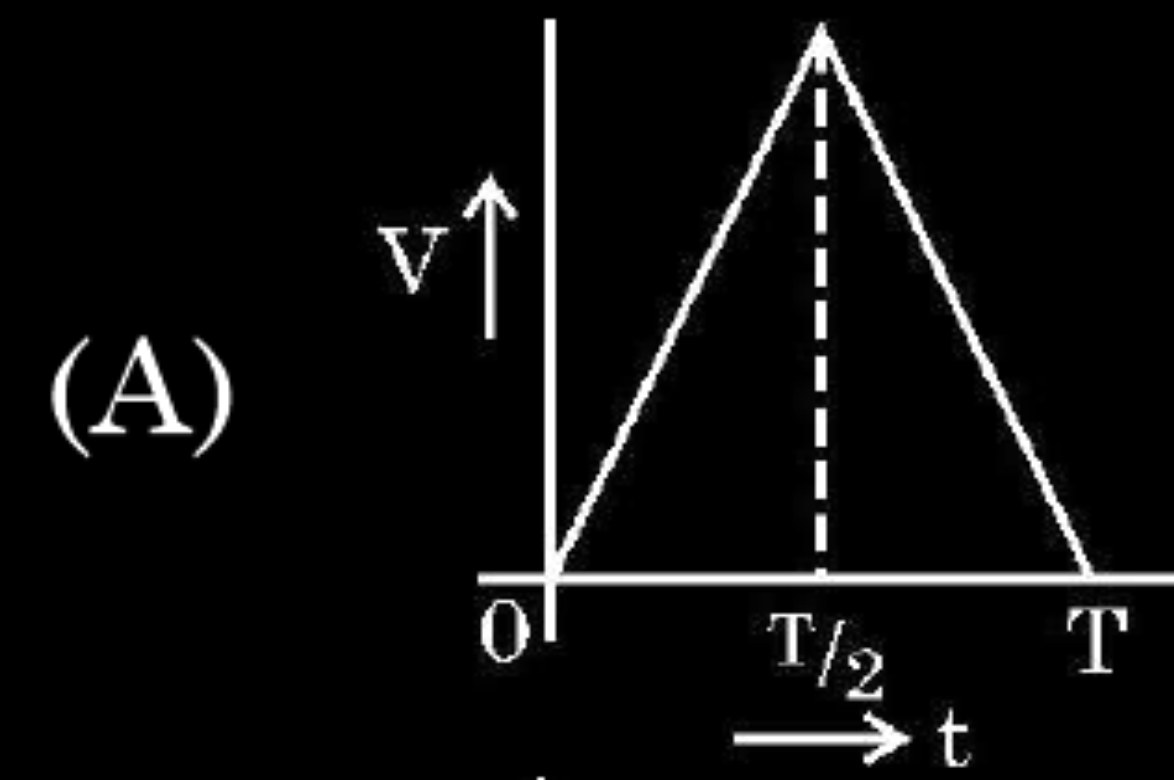
(D) P and S

8. The alternating current I in an inductor is observed to vary with time t as shown in the graph for a cycle.

1



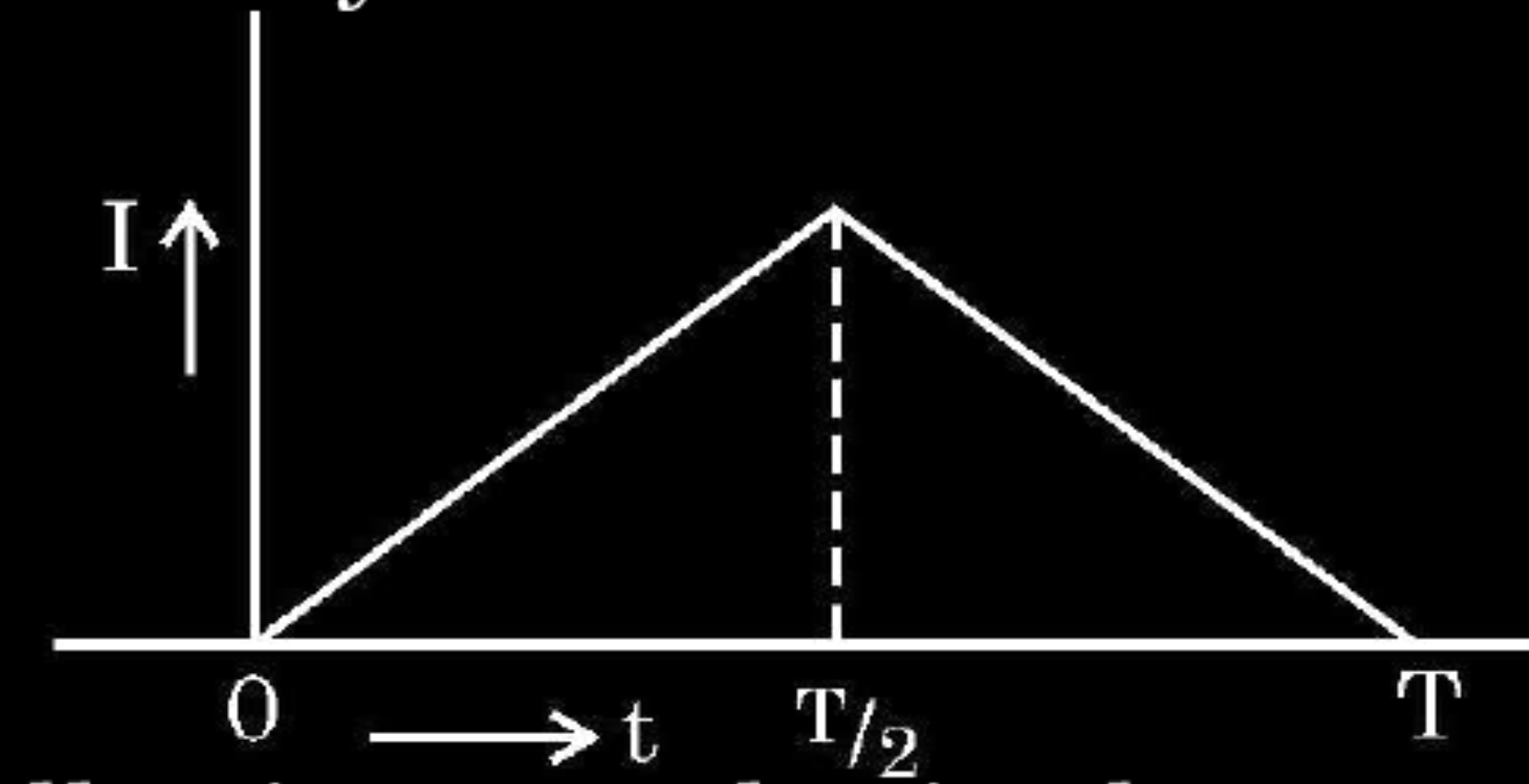
Which one of the following graphs is the correct representation of wave form of voltage V with time t ?



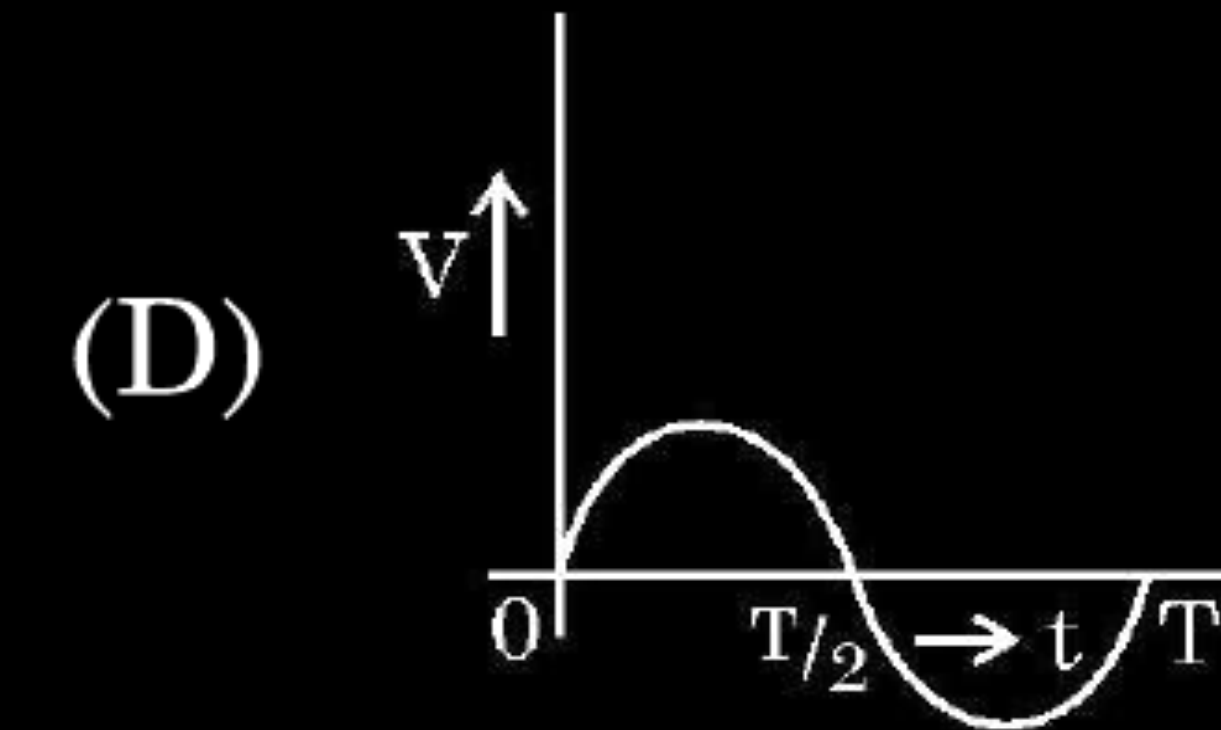
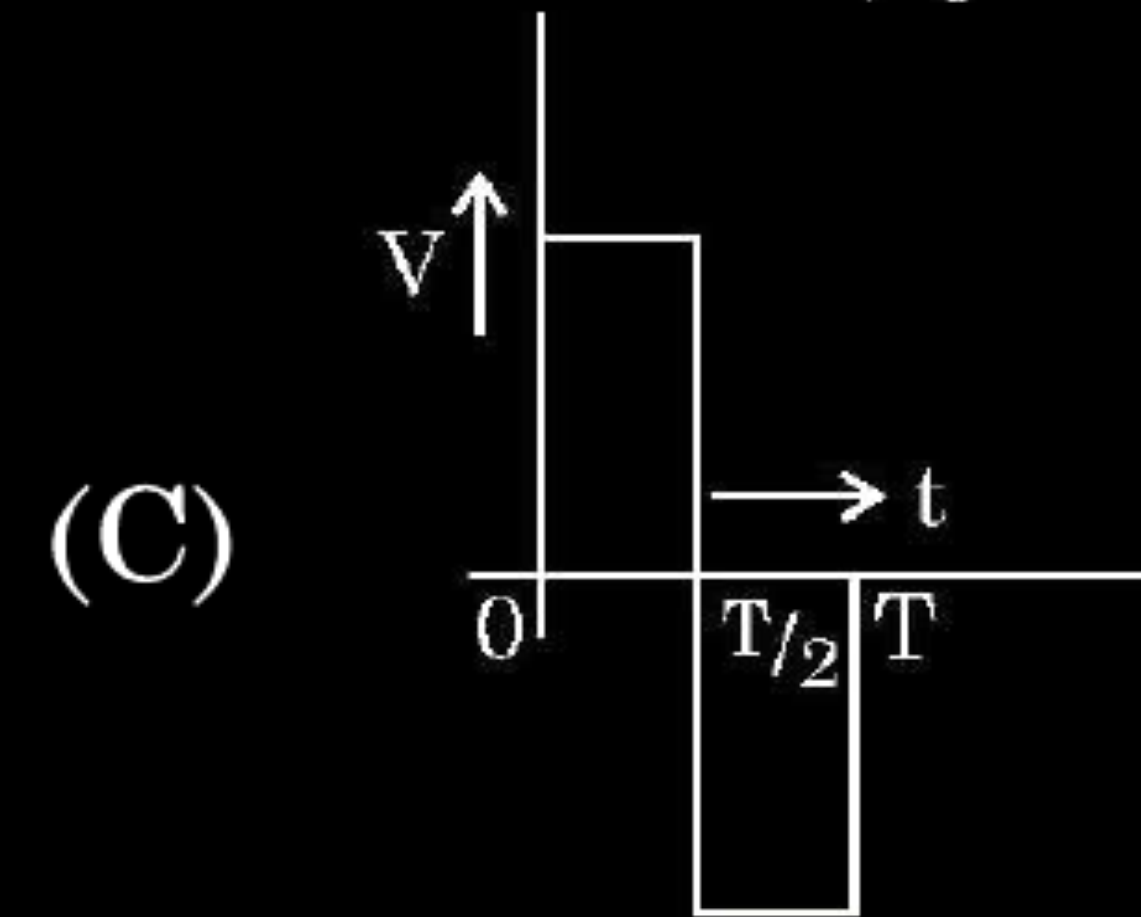
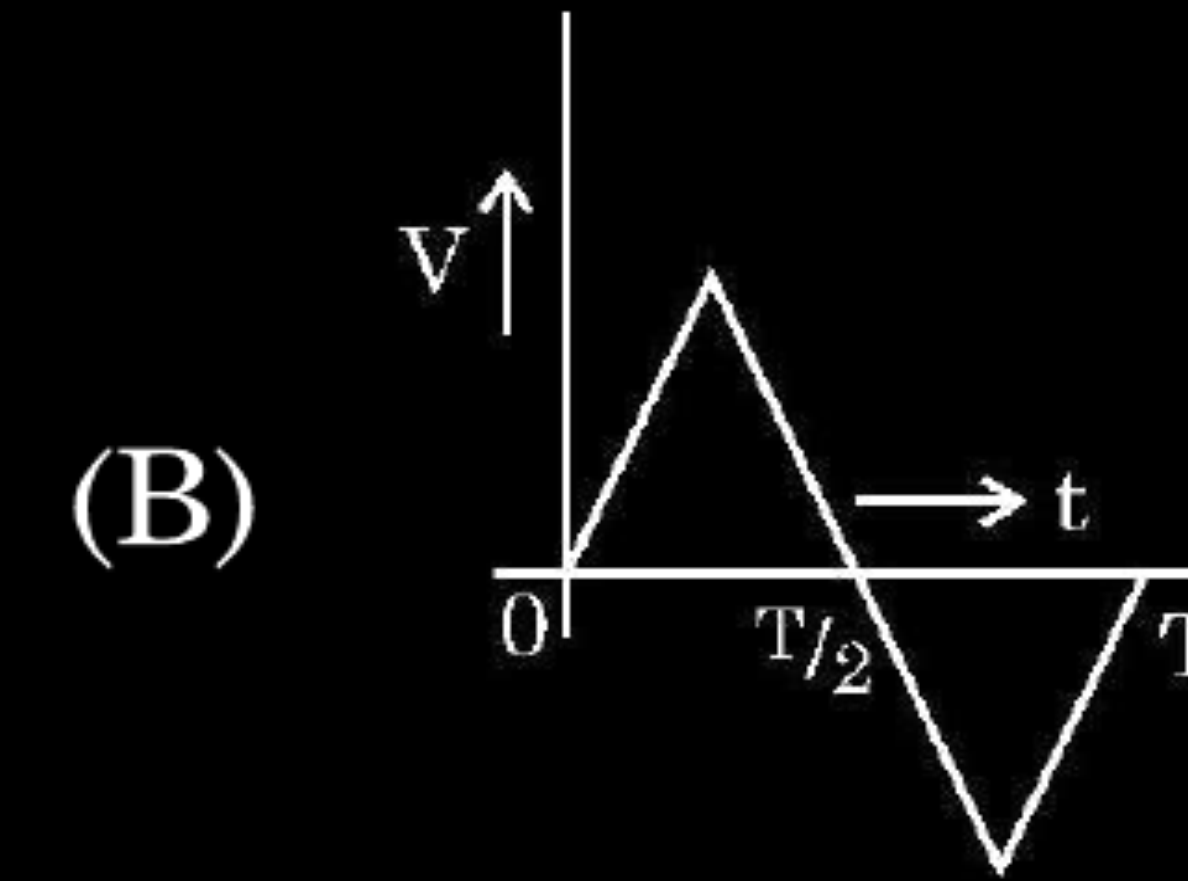
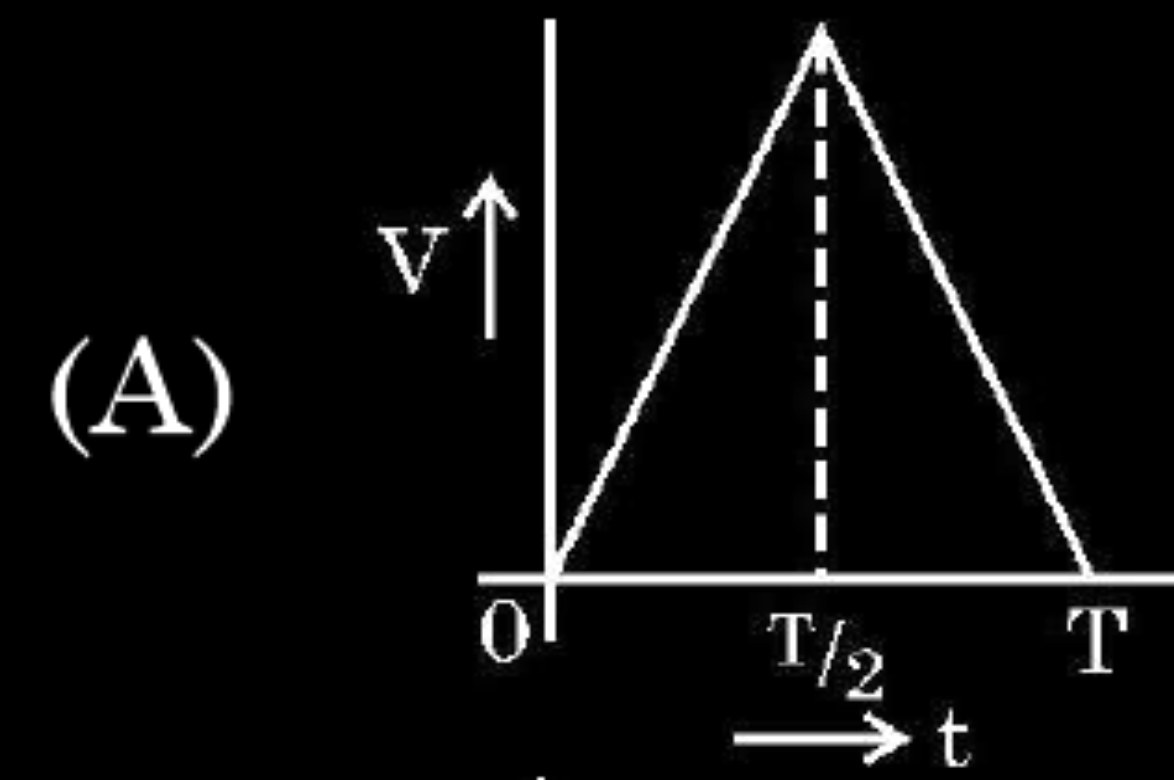
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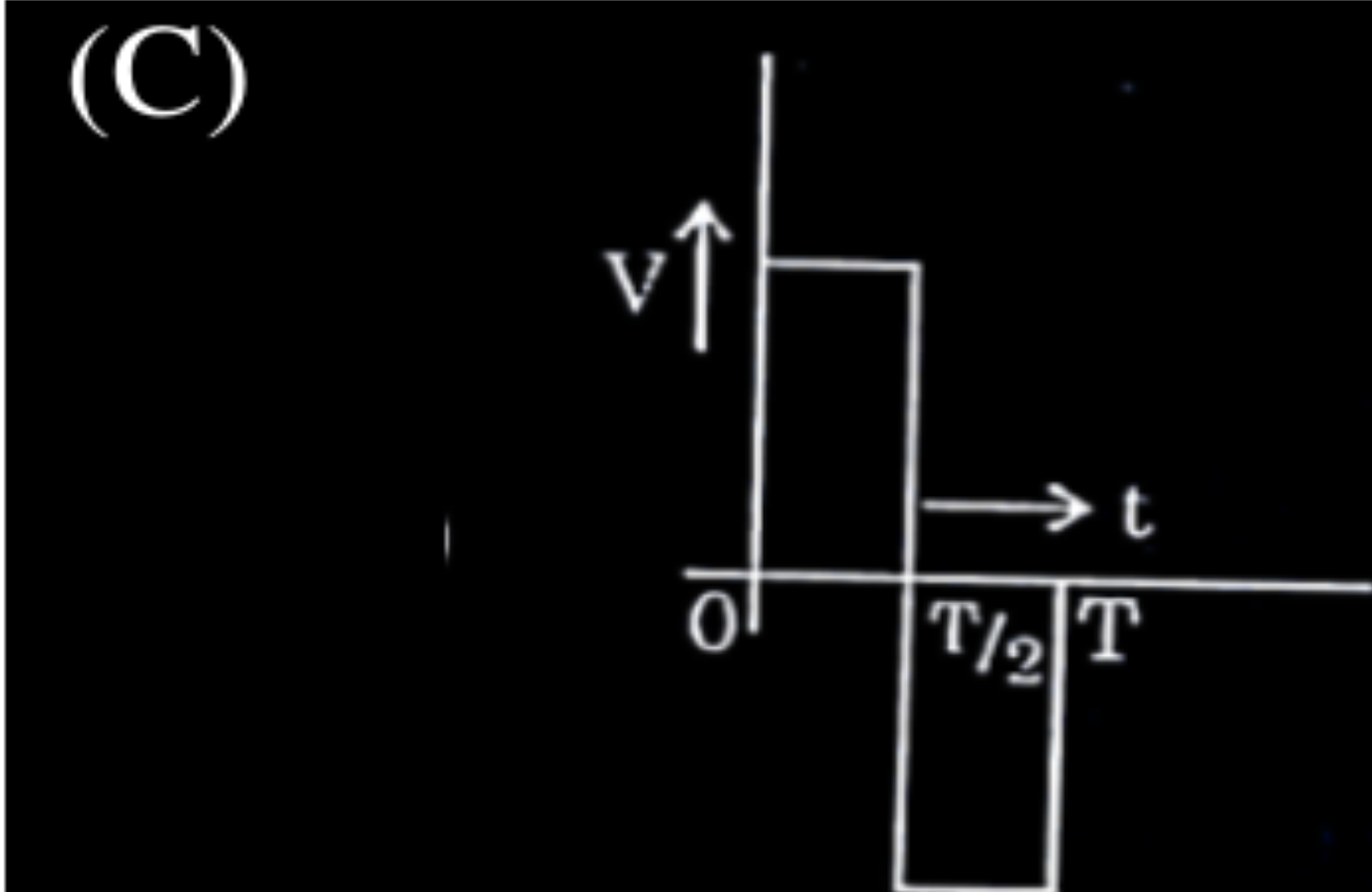
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Which one of the following graphs is the correct representation of wave form of voltage V with time t ?



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13. **Assertion (A)** : In an ideal step-down transformer, the electrical energy is not lost.

1

Reason (R) : In a step-down transformer, voltage decreases but the current increases.

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Reason (R) : In a step-down transformer, voltage decreases but the current increases.

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(B) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

24. (a) ac voltage of frequency ω is applied across a series LCR circuit. Draw the phasor diagram and obtain the impedance of the circuit. 3
- (b) Discuss 'resonance' in a series LCR circuit and write the expression for resonant frequency.

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- (b) Discuss 'resonance' in a series LCR circuit and write the expression for resonant frequency.

3

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$$\vec{V}_L + \vec{V}_R + \vec{V}_C = \vec{V}$$

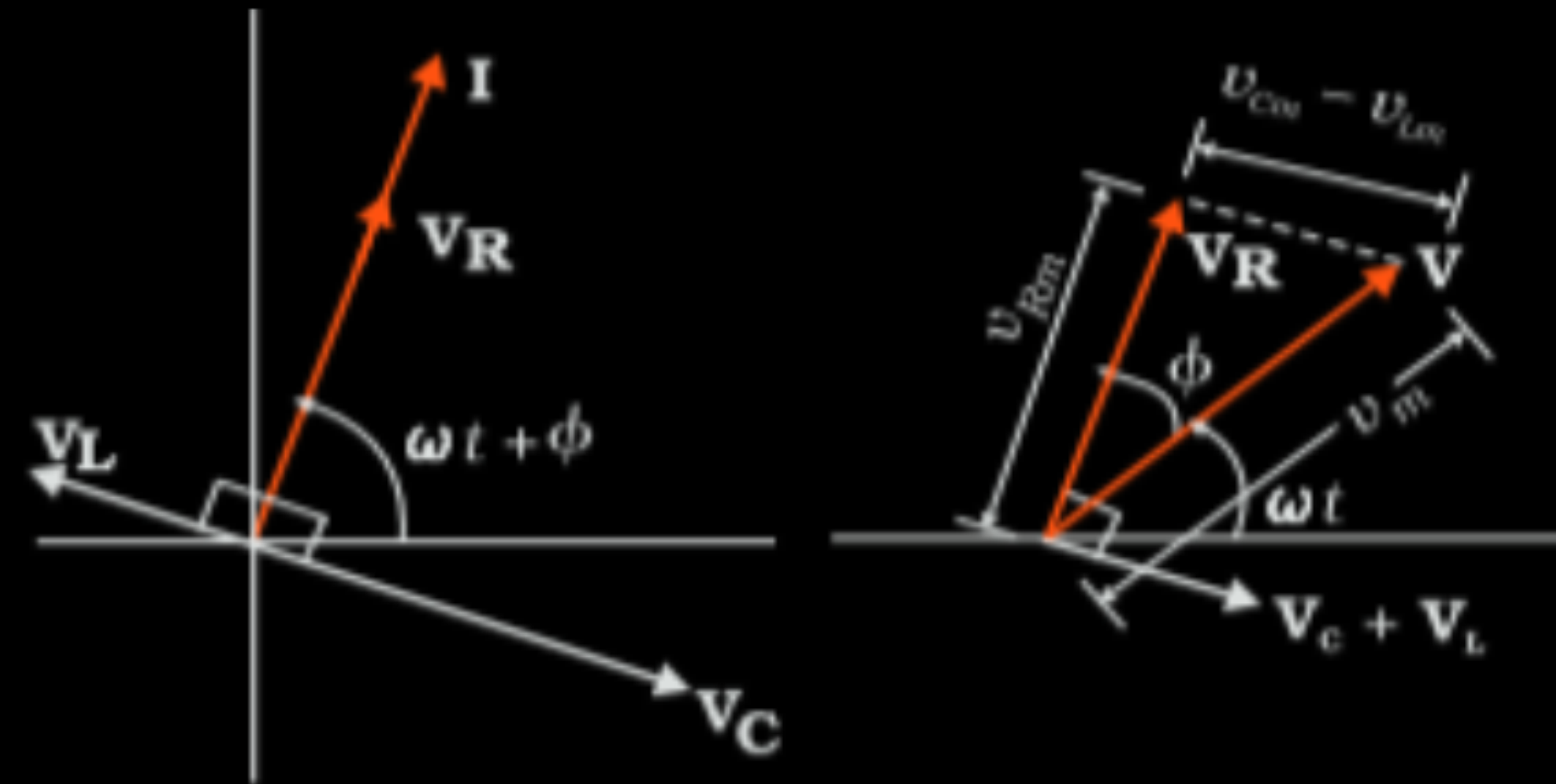
$$V_m^2 = V_{Rm}^2 + (V_{Cm} - V_{Lm})^2$$

$$= (i_m R)^2 + (i_m X_C - i_m X_L)^2$$

$$V_m^2 = i_m^2 [R^2 + (X_C - X_L)^2]$$

$$i_m = \frac{V_m}{\sqrt{R^2 + (X_C - X_L)^2}}$$

$$\Rightarrow Z = \sqrt{R^2 + (X_C - X_L)^2}$$



(b) When $X_L = X_C$, the impedance of the series LCR circuit is minimum and the current flowing through it is maximum.

$$\omega_o = \frac{1}{\sqrt{LC}}$$

1

1/2

1/2

1/2

1/2

24. Differentiate between the peak value and root mean square value of an alternating current. Derive the expression for the root mean square value of alternating current, in terms of its peak value.

3

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24. Differentiate between the peak value and root mean square value of an alternating current. Derive the expression for the root mean square value of alternating current, in terms of its peak value.

3

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Peak value is the maximum value of the alternating current.

$\frac{1}{2}$

rms current is the equivalent dc current that would produce the same average power loss as the alternating current.

$\frac{1}{2}$

Alternatively: -

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

Alternatively: -

$$I_{\text{rms}} = 0.707 I_0$$

The instantaneous power dissipated in the resistor is $P = i^2 R = i_m^2 R \sin^2 \omega t$

$1/2$

The average power over a cycle is: -

$$\bar{P} = \langle i^2 R \rangle = \langle i_m^2 R \sin^2 \omega t \rangle$$

$1/2$

$$\langle \sin^2 \omega t \rangle = \frac{1}{2}$$

$$\bar{P} = \frac{1}{2} i_m^2 R = I_{rms}^2 R$$

$1/2$

$$I_{rms} = \sqrt{\frac{i_m^2}{2}} = \frac{i_m}{\sqrt{2}}$$

$1/2$

Alternatively: -

$$i = i_o \sin \omega t$$

$$I_{rms}^2 = \frac{1}{T} \int_0^T i_o^2 \sin^2 \omega t dt$$

$1/2$

$$= \frac{i_o^2}{T} \int_0^T \left(\frac{1 - \cos 2\omega t}{2} \right) dt$$

$1/2$

$$I_{rms}^2 = \frac{i_o^2}{2T} (T - 0) \quad \left[\text{As } \int_0^T \frac{\cos 2\omega t}{2} dt = 0 \right]$$

$1/2$

$$I_{rms} = \frac{i_o}{\sqrt{2}}$$

$1/2$

7. A transformer is connected to a 200 V ac source. The transformer supplies 3000 V to a device. If the number of turns in the primary coil is 450, then the number of turns in its secondary coil is –

(A) 30
(C) 4500

(B) 450
(D) 6750

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7. A transformer is connected to a 200 V ac source. The transformer supplies 3000 V to a device. If the number of turns in the primary coil is 450, then the number of turns in its secondary coil is –

(A) 30
(C) 4500

(B) 450
(D) 6750

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1

(D) 6750

4. The ratio of the number of turns of the primary to the secondary coils in an ideal transformer is 20 : 1. If 240 V ac is applied from a source to the primary coil of transformer and a $6.0\ \Omega$ resistor is connected across the output terminals, then current drawn by the transformer from the source will be -

(A) 4.0 A

(B) 3.8 A

(C) 0.97 A

(D) 0.10 A

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4. The ratio of the number of turns of the primary to the secondary coils in an ideal transformer is 20 : 1. If 240 V ac is applied from a source to the primary coil of transformer and a $6.0\ \Omega$ resistor is connected across the output terminals, then current drawn by the transformer from the source will be -

(A) 4.0 A

(B) 3.8 A

(C) 0.97 A

(D) 0.10 A

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1

(D) 0.1 A

6. A voltage $v = v_0 \sin \omega t$ applied to a circuit drives a current $i = i_0 \sin (\omega t + \phi)$ in the circuit. The average power consumed in the circuit over a cycle is 1

(A) Zero

(B) $i_0 v_0 \cos \phi$

(C) $\frac{i_0 v_0}{2}$

(D) $\frac{i_0 v_0}{2} \cos \phi$

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6. A voltage $v = v_0 \sin \omega t$ applied to a circuit drives a current $i = i_0 \sin (\omega t + \phi)$ in the circuit. The average power consumed in the circuit over a cycle is 1

(A) Zero

(B) $i_0 v_0 \cos \phi$

(C) $\frac{i_0 v_0}{2}$

(D) $\frac{i_0 v_0}{2} \cos \phi$

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(D) $\frac{i_0 v_0}{2} \cos \phi$

24. An ac source of voltage $v = v_m \sin \omega t$ is connected to a series combination of LCR circuit. Draw the phasor diagram. Using it obtain an expression for the impedance of the circuit and the phase difference between applied voltage and the current.

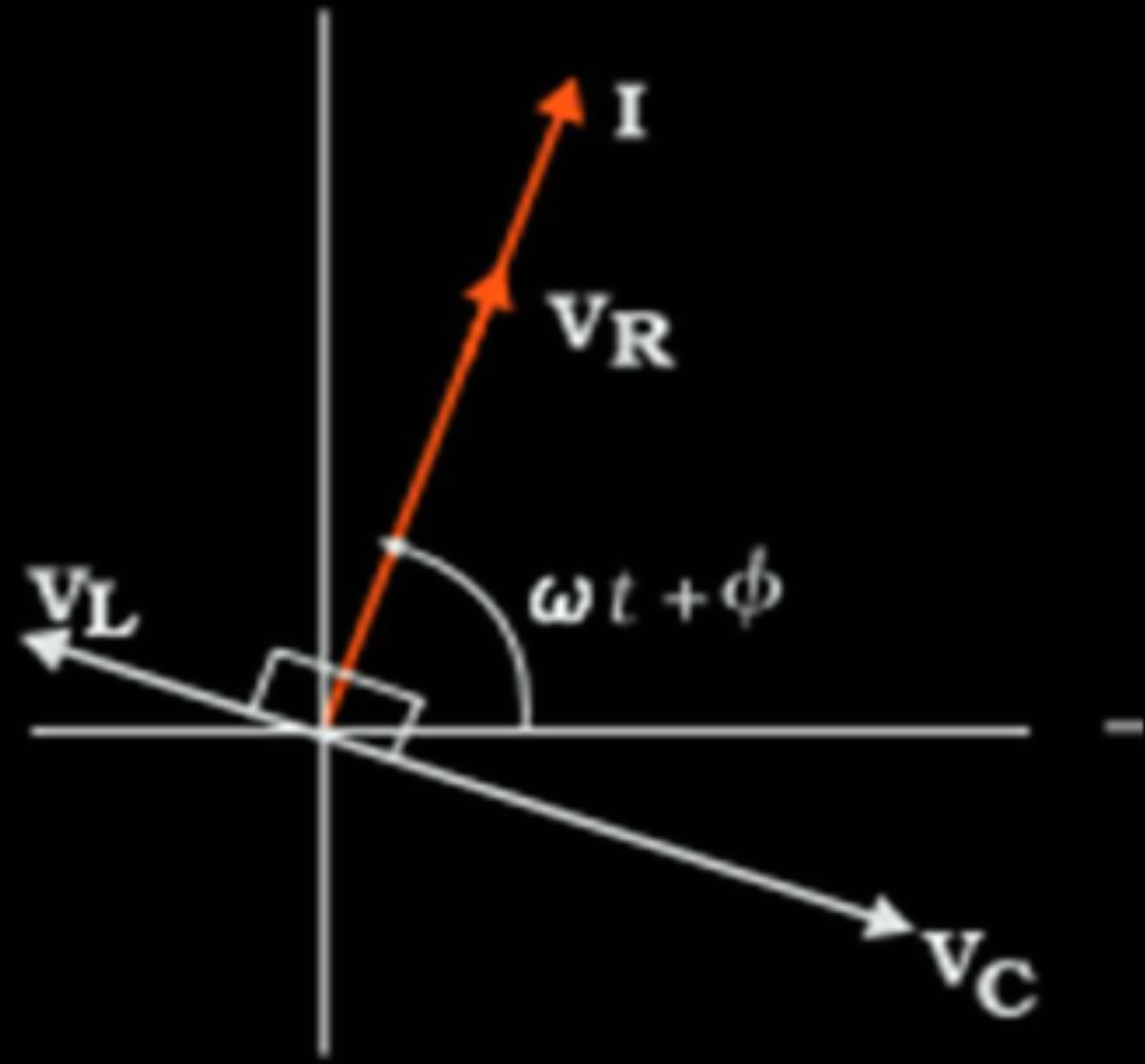
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24. An ac source of voltage $v = v_m \sin \omega t$ is connected to a series combination of LCR circuit. Draw the phasor diagram. Using it obtain an expression for the impedance of the circuit and the phase difference between applied voltage and the current.

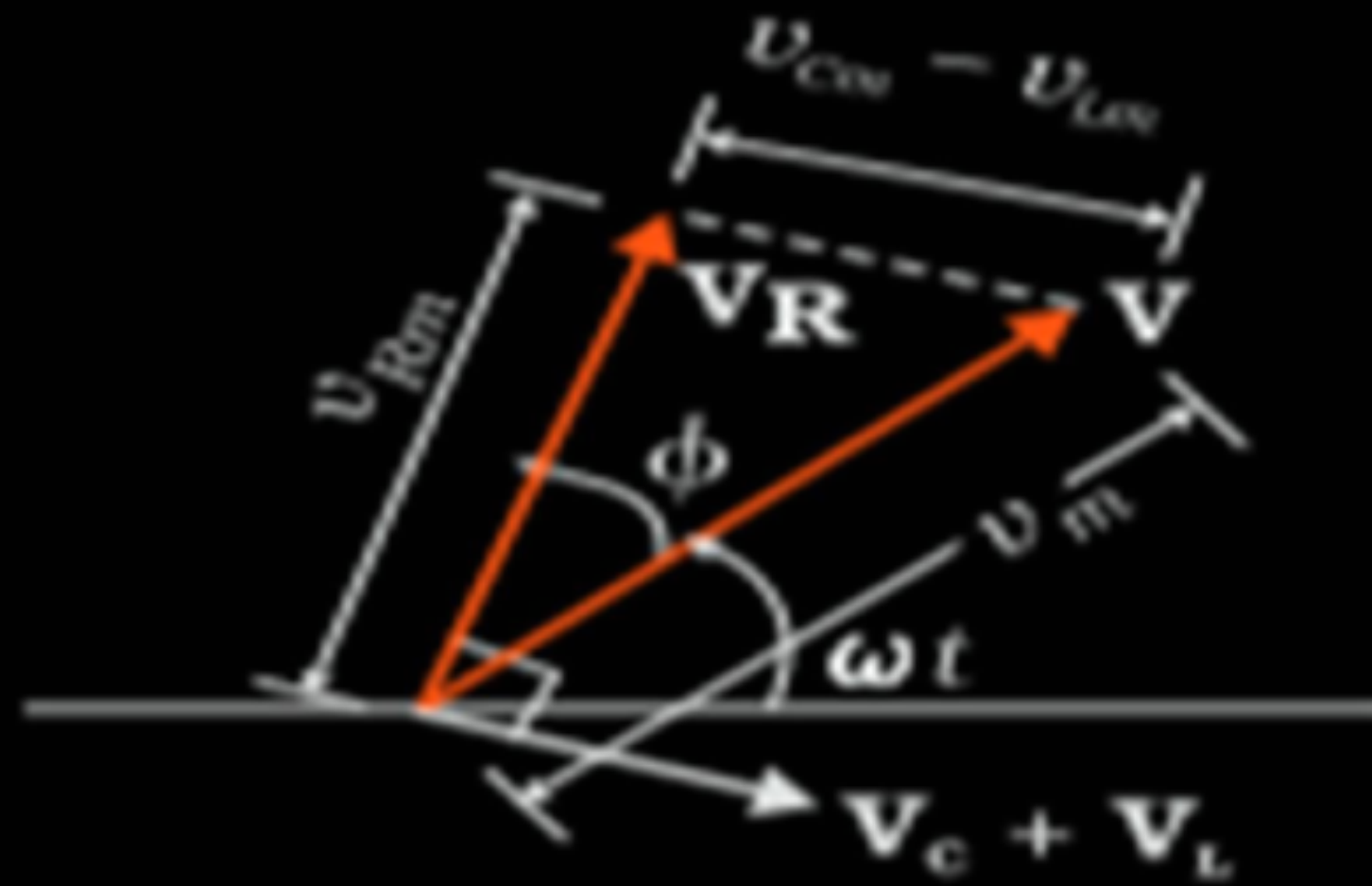
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3

a)



b)



$\frac{1}{2} + \frac{1}{2}$

$$V_{Rm} = i_m R, V_{Cm} = i_m X_c, V_{Lm} = i_m X_L$$

From Phasor diagram

$$V_m^2 = V_{Rm}^2 + (V_{Cm} - V_{Lm})^2$$

$$V^2 = (i R)^2 + (i X_c - i X_L)^2$$

$$= (i_m)^2 [R^2 + (X_c - X_L)^2]$$

$$\text{Or } i_m = \frac{V_m}{\sqrt{R^2 + (X_c - X_L)^2}}$$

$$\therefore i_m = \frac{V_m}{Z}$$

$$\therefore Z = \sqrt{R^2 + (X_c - X_L)^2}$$

From phasor diagram

$$\tan \theta = \frac{V_{Cm} - V_{Lm}}{V_{Rm}}$$

$$= \frac{X_c - X_L}{R}$$

$$\therefore \theta = \tan^{-1} \left(\frac{X_c - X_L}{R} \right)$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2. A capacitor of $5\ \mu\text{F}$ is connected to an ac source of $200\ \text{V}$, $\frac{1000}{\pi}$ Hz through a resistor of $100\ \Omega$. The phase difference between the voltage (V) applied and current (I) is :
- (A) 120° (B) 90° (C) 60° (D) 45°

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2. A capacitor of $5\ \mu\text{F}$ is connected to an ac source of $200\ \text{V}$, $\frac{1000}{\pi}$ Hz through a resistor of $100\ \Omega$. The phase difference between the voltage (V) applied and current (I) is :
- (A) 120° (B) 90° (C) 60° (D) 45°

CBSE 2024 C

(D) 45°

27. An ideal resistor R , an ideal inductor L and an ideal capacitor C are connected, one by one, across the same source of ac voltage $v = v_0 \sin \omega t$. It is observed that the same current I flows in each case.
- (a) What will be the instantaneous value of current that will flow through their series combination when connected across the same source ?
 - (b) How will the current in each case be affected if the frequency of the source is increased ?

Justify your answers.

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(a) Current,
$$I = \frac{V}{Z} = \frac{V_0 \sin \omega t}{\sqrt{R^2 + (X_L - X_C)^2}}$$

For the same source of ac voltage and current is same in all the three cases.

$$R = X_L = X_C$$

When L, R and C are joined in series

$$Z = R$$

Hence , Instantaneous current,
$$I = \frac{V_0}{R} \sin \omega t$$

(b) If frequency of the source is increased, current in pure resistive circuit will remain same as ohmic resistance is independent of frequency.

Current in pure inductive circuit will decrease as

$$X_L \propto \nu$$

$$\text{so, } I \propto \frac{1}{\nu}$$

Current in pure capacitive circuit will increase as

$$I \propto \nu$$

7. An inductor of resistance $5\ \Omega$ and self-inductance $25\ \text{mH}$ is connected across an ac source of angular frequency $200\ \text{rad s}^{-1}$. The phase angle between the voltage and the current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

CBSE 2024 C

7. An inductor of resistance $5\ \Omega$ and self-inductance $25\ \text{mH}$ is connected across an ac source of angular frequency $200\ \text{rad s}^{-1}$. The phase angle between the voltage and the current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

CBSE 2024 C

(B) 45°

- 26.** A current of 1 A flows through an inductor connected to a 200 V dc source. When it is connected to 200 V, 50 Hz source, only 0.5 A current flows. Calculate the self-inductance of the inductor.

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26. A current of 1 A flows through an inductor connected to a 200 V dc source. When it is connected to 200 V, 50 Hz source, only 0.5 A current flows. Calculate the self-inductance of the inductor.

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3

$$\text{Resistance (R)} = \frac{V}{I} = \frac{200}{1} = 200\Omega$$

$\frac{1}{2}$

$$\text{Impedance(Z)} = \frac{V}{I} = \frac{200}{0.5} = 400\Omega$$

$\frac{1}{2}$

$$Z = \sqrt{R^2 + X_L^2}$$

$\frac{1}{2}$

$$(400)^2 = (200)^2 + \omega^2 L^2$$

$\frac{1}{2}$

On solving

$$L = \frac{2\sqrt{3}}{\pi} \text{ H}$$

Or
$$L = \frac{2}{\sqrt{3}} \text{ H}$$

1

2. A coil of resistance $20\ \Omega$ and self-inductance 10 mH is connected to an ac source of frequency $1000/\pi\text{ Hz}$. The phase difference between current in the circuit and the source voltage is :

- (A) 30° (B) 60° (C) 75° (D) 45°

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2. A coil of resistance $20\ \Omega$ and self-inductance 10 mH is connected to an ac source of frequency $1000/\pi\text{ Hz}$. The phase difference between current in the circuit and the source voltage is :

- (A) 30° (B) 60° (C) 75° (D) 45°

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(D) 45°

14. *Assertion (A) :* At a very high frequency, a capacitor can behave as a conductor.

Reason (R) : The capacitance of a capacitor depends only on its geometrical configuration and nature of dielectric used.

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14. *Assertion (A) :* At a very high frequency, a capacitor can behave as a conductor.

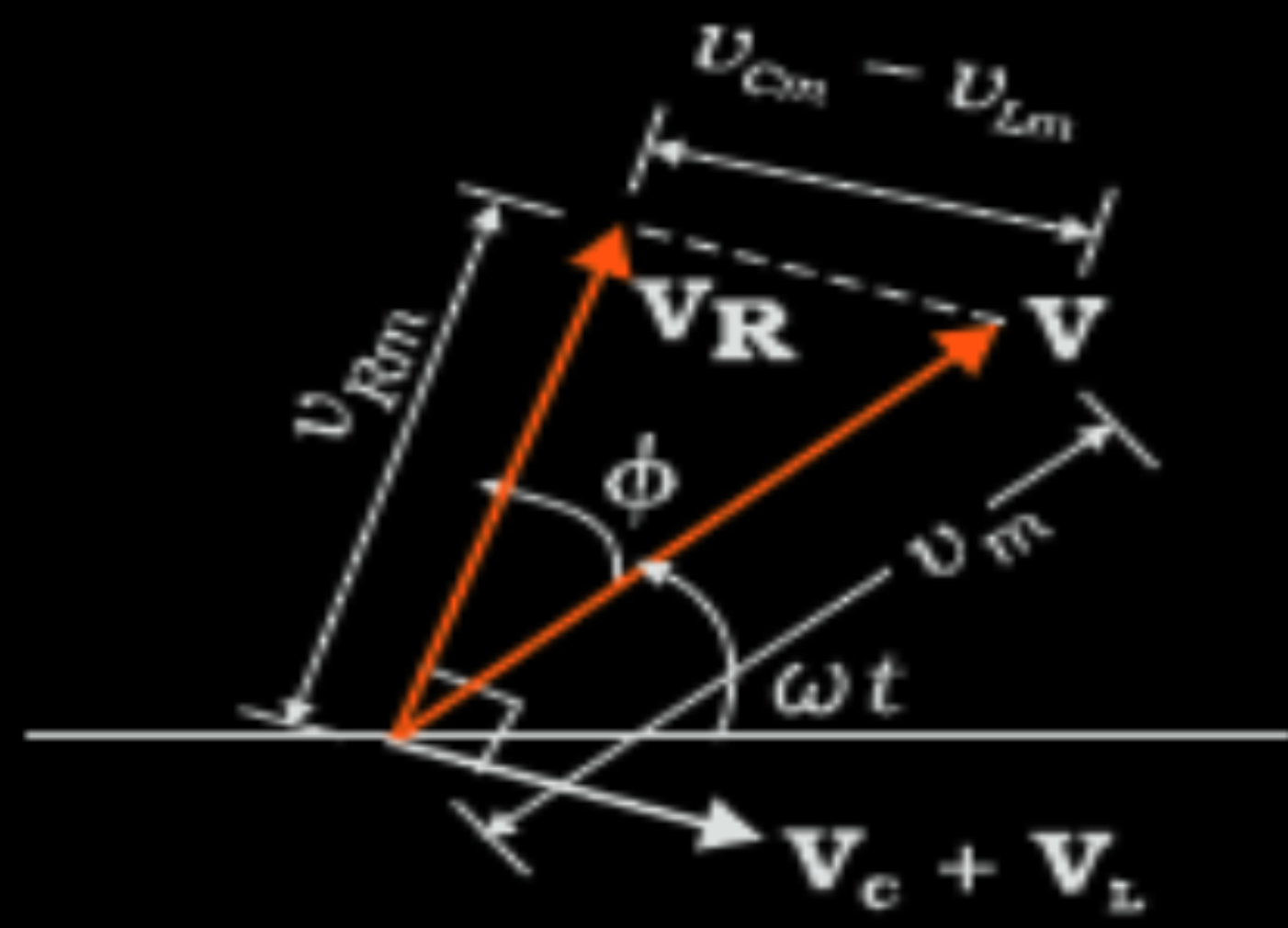
Reason (R) : The capacitance of a capacitor depends only on its geometrical configuration and nature of dielectric used.

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(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

31. (a) (i) You are given three circuit elements X, Y and Z. They are connected one by one across a given ac source. It is found that V and I are in phase for element X. V leads I by $\left(\frac{\pi}{4}\right)$ for element Y while I leads V by $\left(\frac{\pi}{4}\right)$ for element Z. Identify elements X, Y and Z.
- (ii) Establish the expression for impedance of circuit when elements X, Y and Z are connected in series to an ac source. Show the variation of current in the circuit with the frequency of the applied ac source.
- (iii) In a series LCR circuit, obtain the conditions under which (i) impedance is minimum and (ii) wattless current flows in the circuit.

- (i) X : Resistor
 Y : real inductor (such that its reactance is equal to its resistance) / Inductor
 Z : real capacitor (such that its reactance is equal to its resistance)/ Capacitor
 (ii)



From the fig.

$$V_m^2 = V_{Rm}^2 + (V_{Cm} - V_{Lm})^2$$

$$V_m^2 = (i_m R)^2 + (i_m X_C - i_m X_L)^2$$

$$\text{Impedance (Z)} = \frac{V_m}{I_m} = \sqrt{R^2 + (X_C - X_L)^2}$$

1/2

1/2

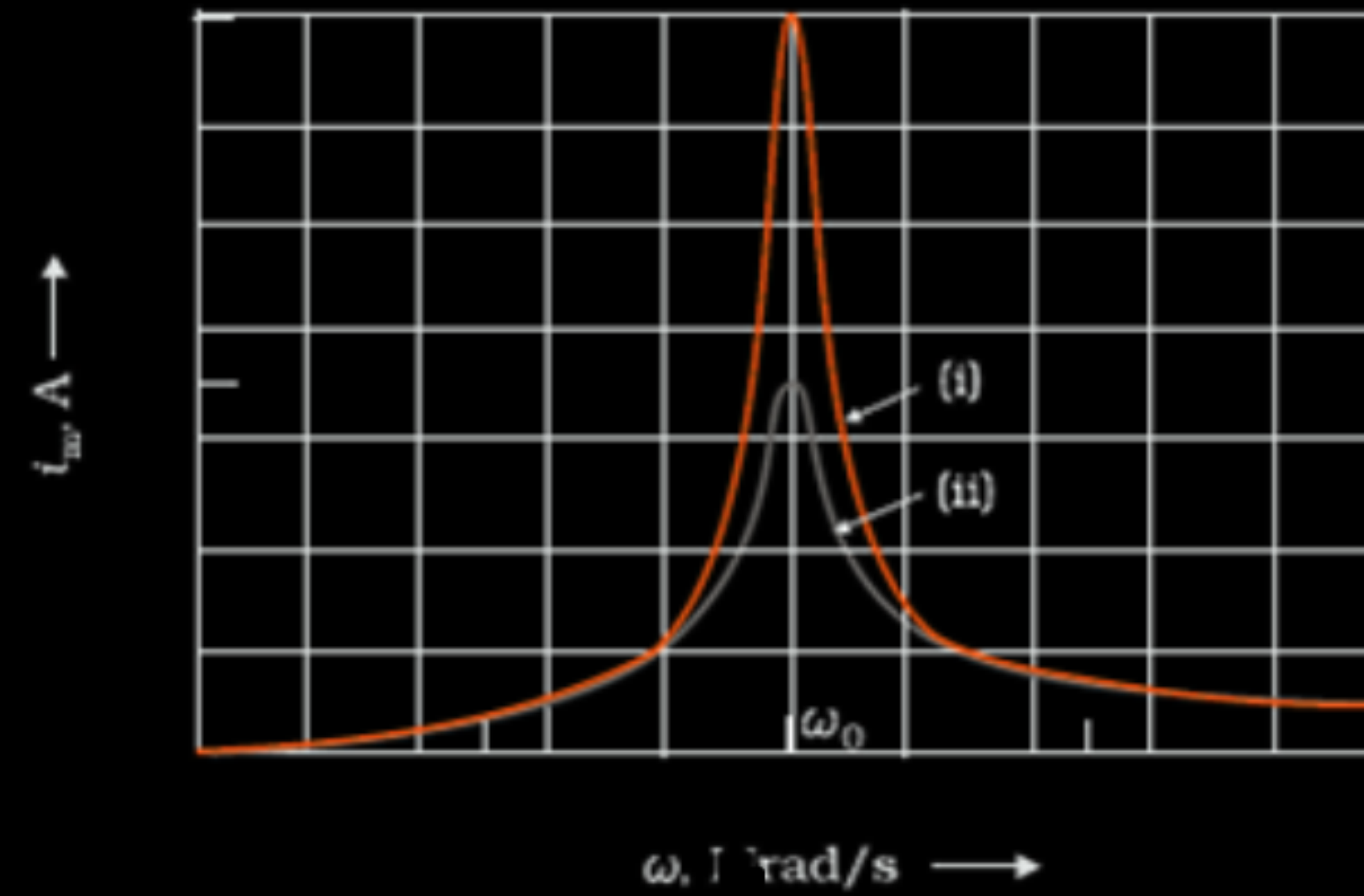
1/2

1/2

1/2

1/2

1/2



$$(iii) Z = \sqrt{R^2 + (X_C - X_L)^2}$$

For the minimum value of impedance

$$(i) X_C = X_L$$

(ii) Average power consumed in A.C. circuit over a cycle

$$P = VI \cos \phi$$

For wattless current $P = 0$

Since $V \neq 0, I \neq 0$

$$\cos \phi = 0$$

$$\text{i.e. } \phi = \frac{\pi}{2}$$

1/2

1/2

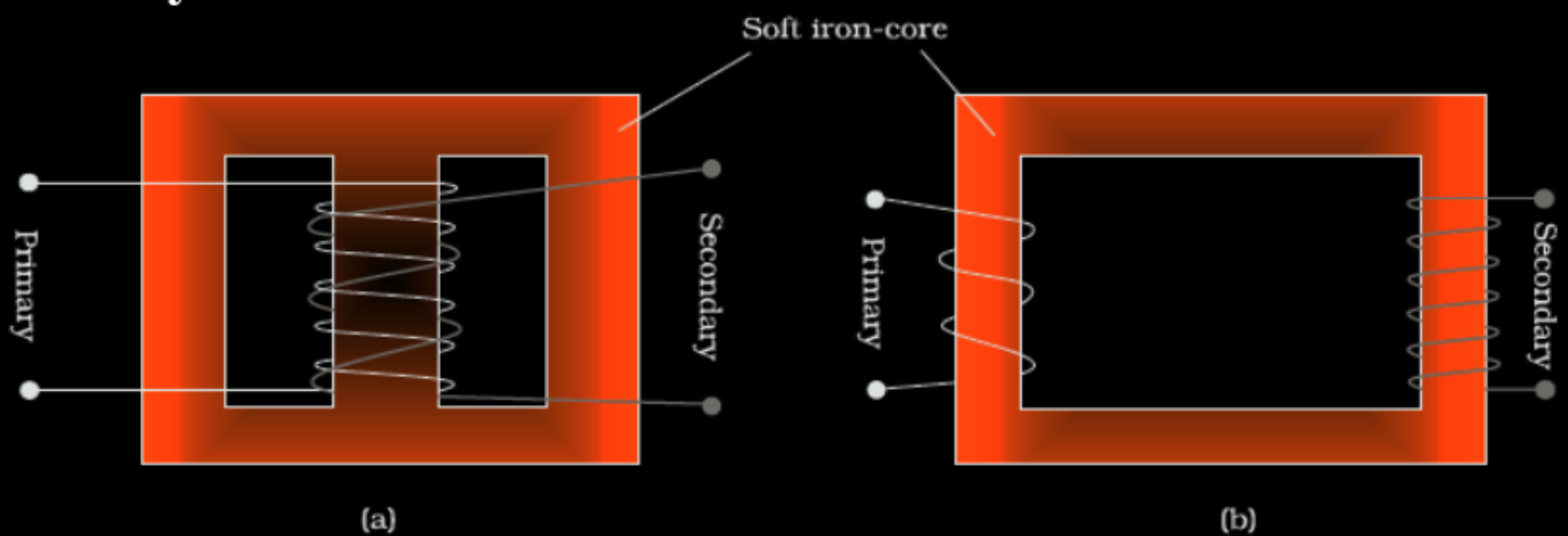
1/2

31. (b) (i) Describe the construction and working of a transformer and hence obtain the relation for $\left(\frac{V_s}{V_p}\right)$ in terms of number of turns of primary and secondary.
- (ii) Discuss four main causes of energy loss in a real transformer.

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(i) **Construction:** A transformer consists of two sets of coils, insulated from each other. They are wound on a soft- iron core, either one on top of other or on separate limbs of the core.

Alternatively



Working: When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links with the secondary and induces an e.m.f. in it.

For an ideal transformer the induced e.m.f. (ε_p) in primary coil for applied alternating voltage (V_P)

$$\varepsilon_p = V_P = -N_p \frac{d\phi}{dt} \text{ -----(1)}$$

e.m.f. induced ε_s in the secondary coil

$$\varepsilon_s = V_s = -N_s \frac{d\phi}{dt} \text{ -----(2)}$$

From eq. (1) and (2)

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

- (ii) Any four energy losses

1. Flux leakage.

2. Resistance of windings/ copper loss.

3. Eddy currents/iron loss.

4. Hysteresis.

5. Magnetostriction.

$\frac{1}{2} \times 4$

1

1

1/2

1/2

2. An ac voltage is applied across an ideal inductor. The current in it
- (A) leads the voltage by $\left(\frac{1}{4}\right)$ cycle.
 - (B) lags the voltage by $\left(\frac{1}{4}\right)$ cycle.
 - (C) leads the voltage by $\left(\frac{1}{2}\right)$ cycle.
 - (D) lags the voltage by $\left(\frac{1}{2}\right)$ cycle.

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2. An ac voltage is applied across an ideal inductor. The current in it

(A) leads the voltage by $\left(\frac{1}{4}\right)$ cycle.

(B) lags the voltage by $\left(\frac{1}{4}\right)$ cycle.

(C) leads the voltage by $\left(\frac{1}{2}\right)$ cycle.

(D) lags the voltage by $\left(\frac{1}{2}\right)$ cycle.

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(B) lags the voltage by $\left(\frac{1}{4}\right)$ cycle

9. The r.m.s. value of a current given by $i = (i_1 \cos \omega t + i_2 \sin \omega t)$ is –

(A) $\frac{1}{\sqrt{2}} (i_1 + i_2)$

(B) $\frac{1}{\sqrt{2}} (i_1 - i_2)$

(C) $\frac{1}{\sqrt{2}} \sqrt{i_1^2 + i_2^2}$

(D) $\frac{1}{\sqrt{2}} (i_1^2 + i_2^2)$

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9. The r.m.s. value of a current given by $i = (i_1 \cos \omega t + i_2 \sin \omega t)$ is –

(A) $\frac{1}{\sqrt{2}} (i_1 + i_2)$

(B) $\frac{1}{\sqrt{2}} (i_1 - i_2)$

(C) $\frac{1}{\sqrt{2}} \sqrt{(i_1^2 + i_2^2)}$

(D) $\frac{1}{\sqrt{2}} (i_1^2 + i_2^2)$

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(C) $\frac{1}{\sqrt{2}} \sqrt{(i_1^2 + i_2^2)}$

33. (a) (i) Mention the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing variation of impedance of a series LCR circuit with the frequency of the applied a.c. source.
- (ii) With the help of a suitable diagram, explain the working of a step-up transformer.
- (iii) Write two causes of energy loss in a real transformer.

5

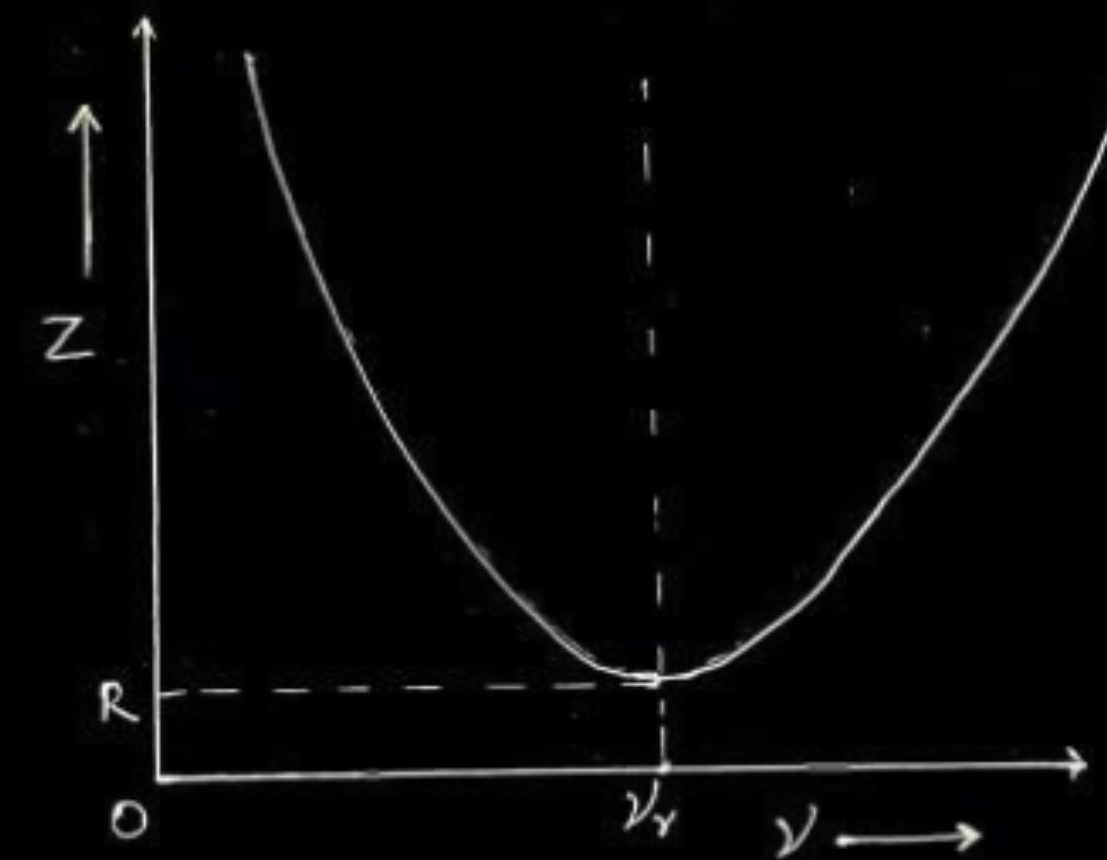
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(i) Inductance
Capacitance

$\frac{1}{2}$
 $\frac{1}{2}$

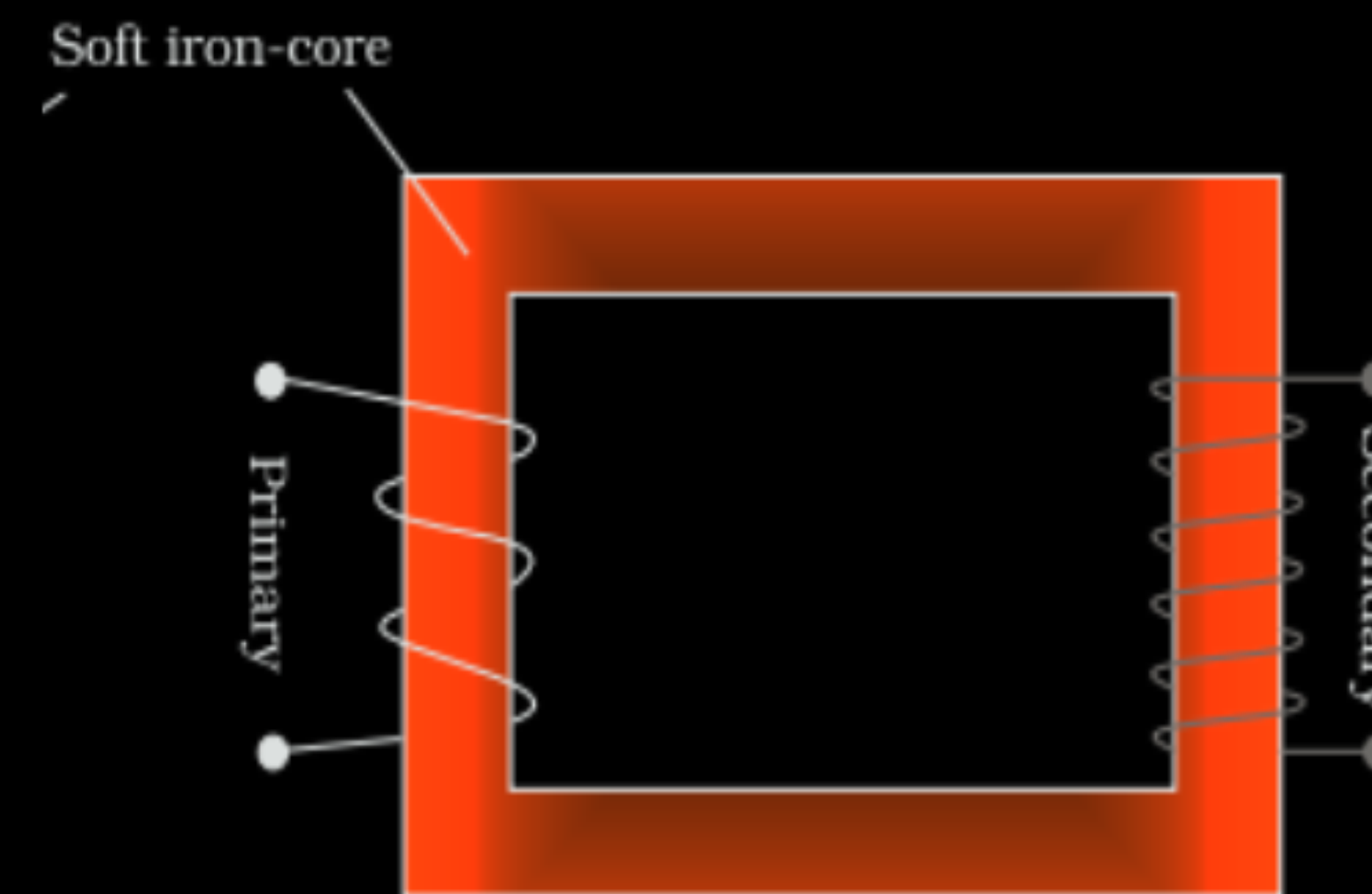
Alternatively

$$v_0 = \frac{1}{2\pi\sqrt{LC}}$$



1

(ii)



1

Working - when an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it.

1

(iii) Causes of energy loss (any two)

- (1) Flux leakage
- (2) Resistance of the windings
- (3) Hysteresis
- (4) Eddy currents

$\frac{1}{2} + \frac{1}{2}$

9. An ac source $V = 282 \sin (100 t)$ volt is connected across a $1 \mu\text{F}$ capacitor.

The rms value of current in the circuit will be (take $\sqrt{2} = 1.41$)

1

(A) 10 mA

(B) 20 mA

(C) 40 mA

(D) 80 mA

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9. An ac source $V = 282 \sin (100 t)$ volt is connected across a $1 \mu\text{F}$ capacitor.

The rms value of current in the circuit will be (take $\sqrt{2} = 1.41$)

1

(A) 10 mA

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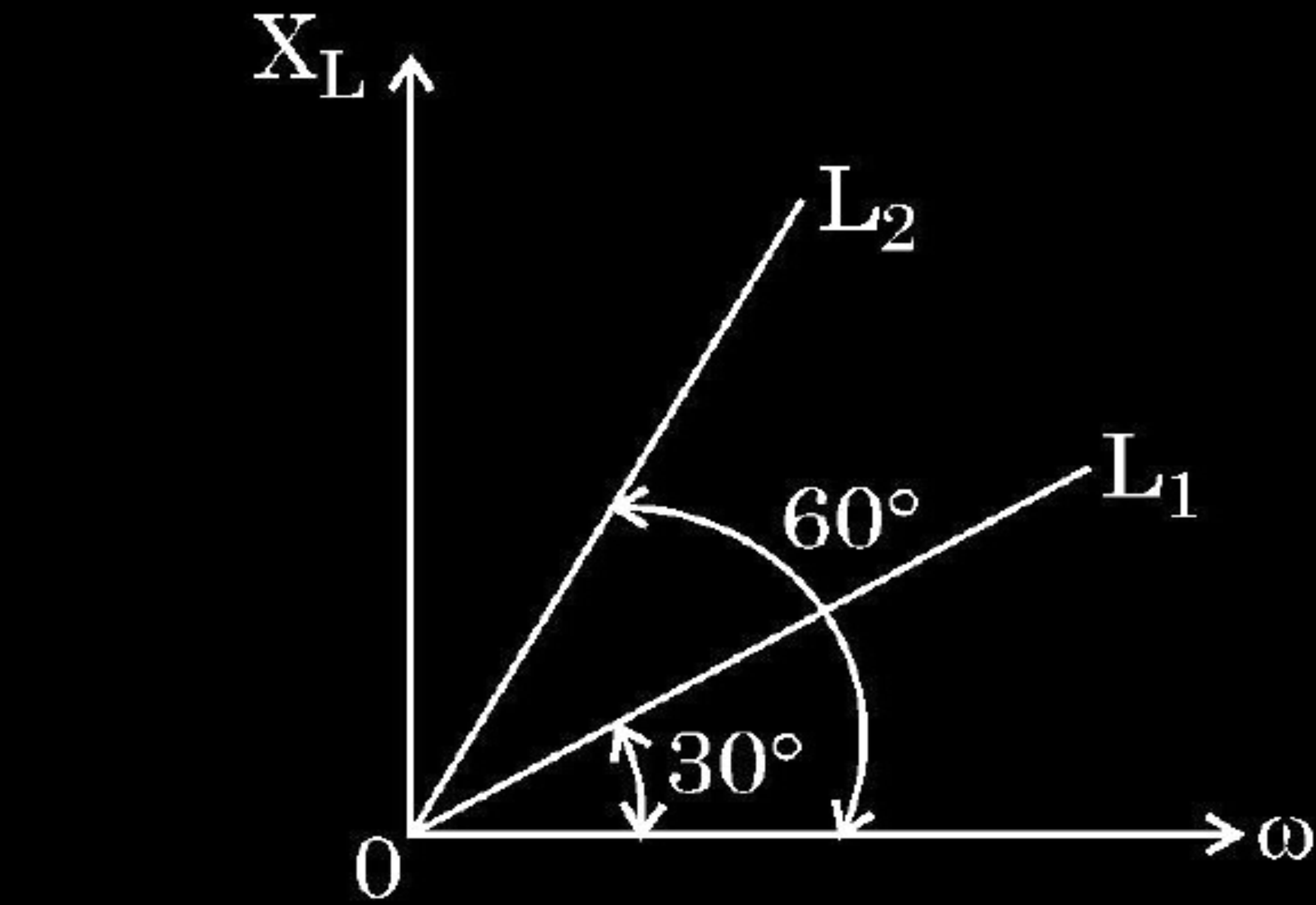
(D) 80 mA

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(B) 20 mA

9. Figure shows the variation of inductive reactance X_L of two ideal inductors of inductance L_1 and L_2 , with angular frequency ω . The value of $\frac{L_1}{L_2}$ is

1



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

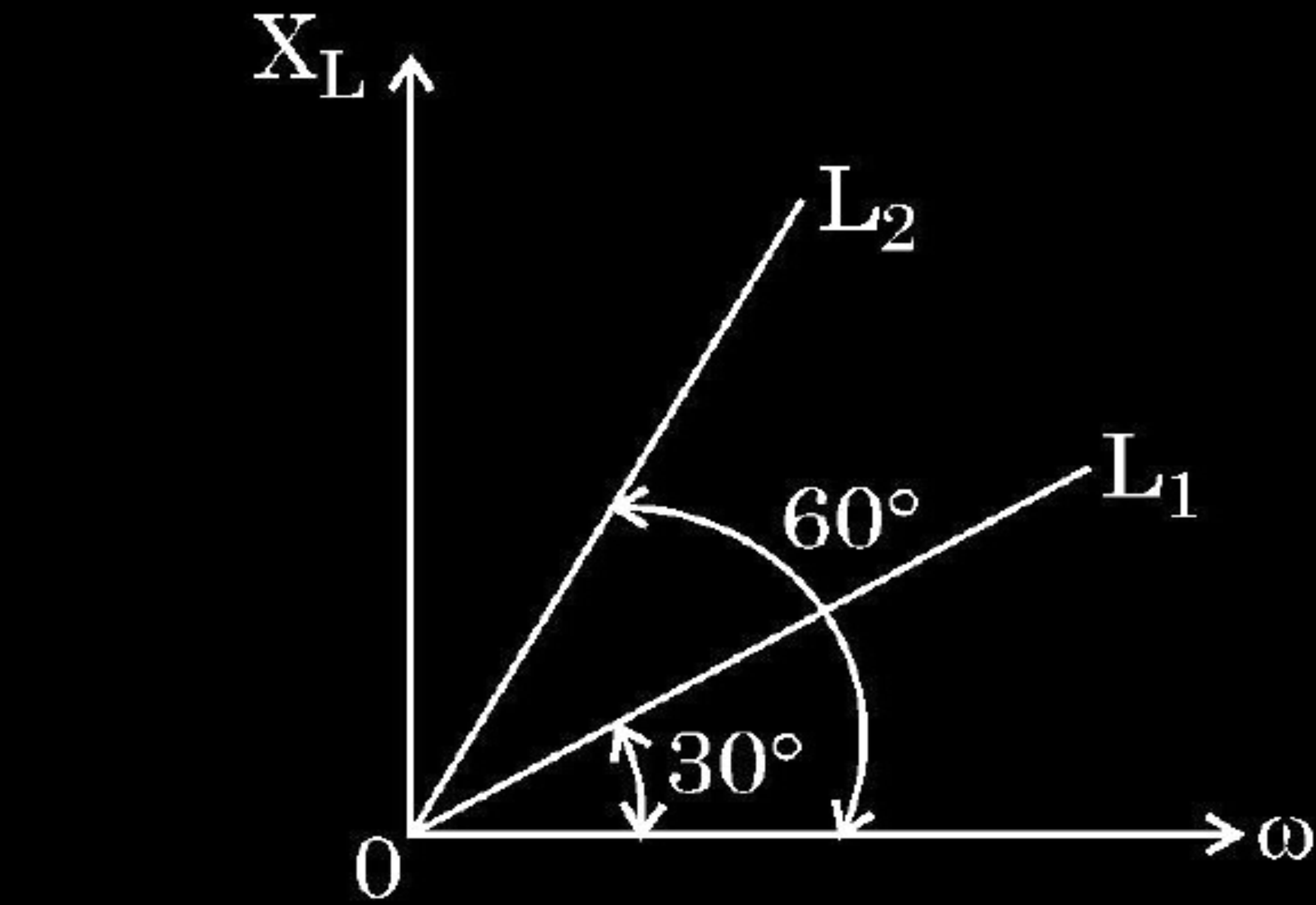
(C) 3

(D) $\frac{1}{3}$

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9. Figure shows the variation of inductive reactance X_L of two ideal inductors of inductance L_1 and L_2 , with angular frequency ω . The value of $\frac{L_1}{L_2}$ is

1



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

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(D) $\frac{1}{3}$

6. The average value of the alternating voltage $v = (157 \text{ V}) \sin \omega t$ over its first half-cycle is :

(A) 157 V

(B) $\frac{157}{\sqrt{2}} \text{ V}$

(C) 78.5 V

(D) 100 V

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6. The average value of the alternating voltage $v = (157 \text{ V}) \sin \omega t$ over its first half-cycle is :

(A) 157 V

(B) $\frac{157}{\sqrt{2}} \text{ V}$

(C) 78.5 V

(D) 100 V

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6. The average value of the alternating voltage $v = (157 \text{ V}) \sin \omega t$ over its first half-cycle is :

(A) 157 V

(B) $\frac{157}{\sqrt{2}} \text{ V}$

(C) 78.5 V

(D) 100 V

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(D) 100 V

- 24.** Distinguish between reactance and impedance of an ac circuit. Show that an ideal inductor in an ac circuit does not dissipate any power.

3

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24.

Distinguish between reactance and impedance of an ac circuit. Show that an ideal inductor in an ac circuit does not dissipate any power.

3

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Reactance- It is the measure of opposition to flow of current in ac circuit comprising Inductor or Capacitor. Impedance- It is the measure of opposition to flow of current in ac circuit comprising Resistor, Capacitor and Inductor. $\varepsilon = \varepsilon_0 \sin \omega t$ $I = I_0 \sin(\omega t - \frac{\pi}{2}) = -I_0 \cos \omega t$ $P = \varepsilon I$ $= -\varepsilon_0 I_0 \sin \omega t \cos \omega t$ $= -\frac{\varepsilon_0 I_0}{2} 2 \sin \omega t \cos \omega t$ $P = \frac{\varepsilon_0 I_0}{2} \sin 2\omega t$	1/2 1/2 1/2 1/2	$\langle P \rangle = \frac{\int_0^T P dt}{\int_0^T dt}$ $\langle P \rangle = \frac{\int_0^T \frac{\varepsilon_0 I_0}{2} \sin 2\omega t dt}{T}$ $= \frac{\varepsilon_0 I_0}{2T} \int_0^T \sin 2\omega t dt$ $= -\frac{\varepsilon_0 I_0}{2T} (\cos \omega t)_0^T = \frac{\varepsilon_0 I_0}{2T} (1 - 1)$ $\langle P \rangle = 0$ Hence average power associated with inductor is zero. Alternatively $P = \varepsilon_{rms} I_{rms} \cos \phi$ For inductive circuit $\phi = \pi / 2$ $P = \varepsilon_{rms} I_{rms} \cos \frac{\pi}{2}$ $P = 0$ 1/2 1/2 1/2 1 1/2 1/2
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24. A sinusoidal voltage is applied to an electric circuit containing a circuit element 'X' in which the current leads the voltage by $\frac{\pi}{2}$.

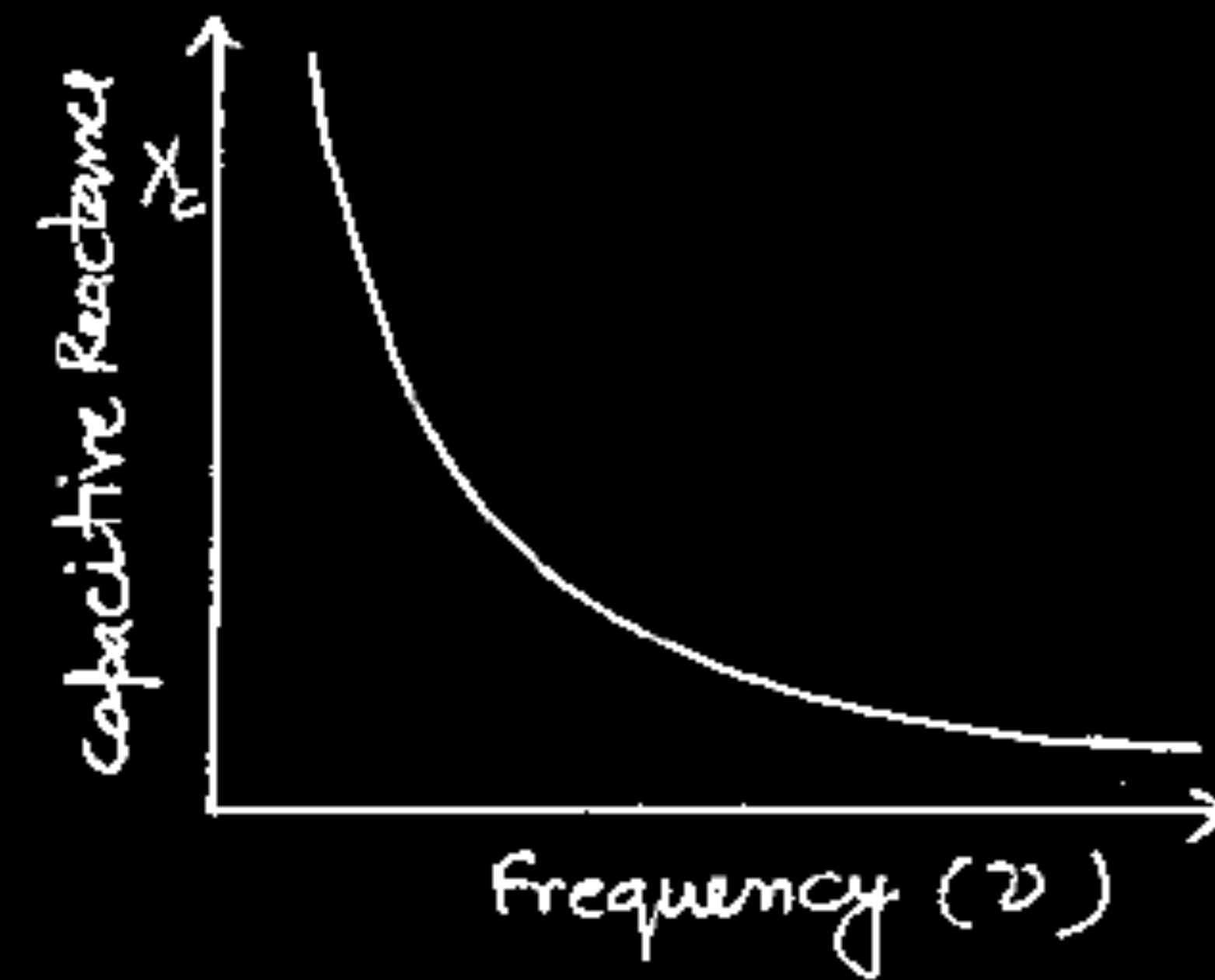
- (a) Identify the circuit element 'X' in the circuit.
- (b) Write the formula for its reactance.
- (c) Show graphically the variation of this reactance with frequency of ac voltage.
- (d) Explain the behaviour of this element when it is used in (i) an ac circuit, and (ii) a dc circuit.

3

(a) Capacitor

(b) $X_c = \frac{1}{\omega C}$

(c)



(d) (i) For ac X_c is finite and therefore allows the ac to pass.

(ii) For dc X_c is infinite and therefore does not allow the dc to pass.

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

5. Which of the following quantity/quantities remains same in primary and secondary coils of an ideal transformer ?

Current, Voltage, Power, Magnetic flux

- (A) Current only
- (B) Voltage only
- (C) Power only
- (D) Magnetic flux and Power both

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5. Which of the following quantity/quantities remains same in primary and secondary coils of an ideal transformer ?

Current, Voltage, Power, Magnetic flux

- (A) Current only
- (B) Voltage only
- (C) Power only
- (D) Magnetic flux and Power both

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(D) Magnetic Flux and Power both

6. In an ac circuit, the instantaneous values of current (in A) and voltage (in V) are $I = 5 \sin \omega t$ and $E = 200 \cos (\omega t + \frac{\pi}{3})$ respectively. The phase difference between voltage and current at any instant is :

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{4}$

(C) $\frac{5\pi}{2}$

(D) $\frac{3\pi}{2}$

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6. In an ac circuit, the instantaneous values of current (in A) and voltage (in V) are $I = 5 \sin \omega t$ and $E = 200 \cos (\omega t + \frac{\pi}{3})$ respectively. The phase difference between voltage and current at any instant is :

(A) $\frac{5\pi}{6}$

(B) $\frac{5\pi}{4}$

(C) $\frac{5\pi}{2}$

(D) $\frac{3\pi}{2}$

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(A) $\frac{5\pi}{6}$

24. An ac voltage $v_i = 140 \sin (100 \pi t)$ V is applied to the primary coil having 200 turns, of an ideal transformer and it supplies a power of 5 kW. If the secondary coil has 1000 turns, find :

- (a) the output voltage,
- (b) the instantaneous voltage across the secondary coil, and
- (c) the current in the secondary coil. (Take $\sqrt{2} = 1.4$)

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3

24. An ac voltage $v_i = 140 \sin(100 \pi t)$ V is applied to the primary coil having 200 turns, of an ideal transformer and it supplies a power of 5 kW. If the secondary coil has 1000 turns, find :

- (a) the output voltage,
- (b) the instantaneous voltage across the secondary coil, and
- (c) the current in the secondary coil. (Take $\sqrt{2} = 1.4$)

3

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(a) $V_p(\text{rms}) = \frac{140}{\sqrt{2}} = \frac{140}{1.4} = 100V$
 $\therefore V_s = \frac{N_s}{N_p} V_p = \frac{1000}{200} 100 = 500V$
 (b) $v_s = 500\sqrt{2} \sin 100 \pi t = 700 \sin 100 \pi t$
 (c) Power Output = Power Input
 $I_s = \frac{5000}{500} = 10A$

1/2

1/2

1

1

6. A series LCR circuit ($L = 2 \text{ mH}$, $C = 0.2 \text{ } \mu\text{F}$ and $R = 30 \text{ } \Omega$) is connected to an ac source of variable frequency. The impedance of this circuit will be minimum at a frequency of :

(A) $\frac{10^5}{4\pi} \text{ Hz}$

(B) $\frac{10^5}{2\pi} \text{ Hz}$

(C) $\frac{10^4}{4\pi} \text{ Hz}$

(D) $\frac{10^4}{2\pi} \text{ Hz}$

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6. A series LCR circuit ($L = 2 \text{ mH}$, $C = 0.2 \text{ } \mu\text{F}$ and $R = 30 \text{ } \Omega$) is connected to an ac source of variable frequency. The impedance of this circuit will be minimum at a frequency of :

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(C) $\frac{10^4}{4\pi} \text{ Hz}$

(D) $\frac{10^4}{2\pi} \text{ Hz}$

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(A) $\frac{10^5}{4\pi} \text{ Hz}$

6. A resistor and an ideal inductor are connected in series to a $100\sqrt{2}$ V, 50 Hz ac source. When a voltmeter is connected across the resistor or the inductor, it shows the same reading. The reading of the voltmeter is :

(A) $100\sqrt{2}$ V

(B) 100 V

(C) $50\sqrt{2}$ V

(D) 50 V

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6. A resistor and an ideal inductor are connected in series to a $100\sqrt{2}$ V, 50 Hz ac source. When a voltmeter is connected across the resistor or the inductor, it shows the same reading. The reading of the voltmeter is :

(A) $100\sqrt{2}$ V

(B) 100 V

(C) $50\sqrt{2}$ V

(D) 50 V

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(B) 100V

7. The primary and secondary coils of a transformer have 500 turns and 5000 turns respectively. The primary coil is connected to an ac source of 220 V – 50 Hz. The output across the secondary coil is :

(A) 220 V – 50 Hz

(B) 1100 V – 50 Hz

(C) 2200 V – 5 Hz

(D) 2200 V – 50 Hz

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7. The primary and secondary coils of a transformer have 500 turns and 5000 turns respectively. The primary coil is connected to an ac source of 220 V – 50 Hz. The output across the secondary coil is :

(A) 220 V – 50 Hz

(B) 1100 V – 50 Hz

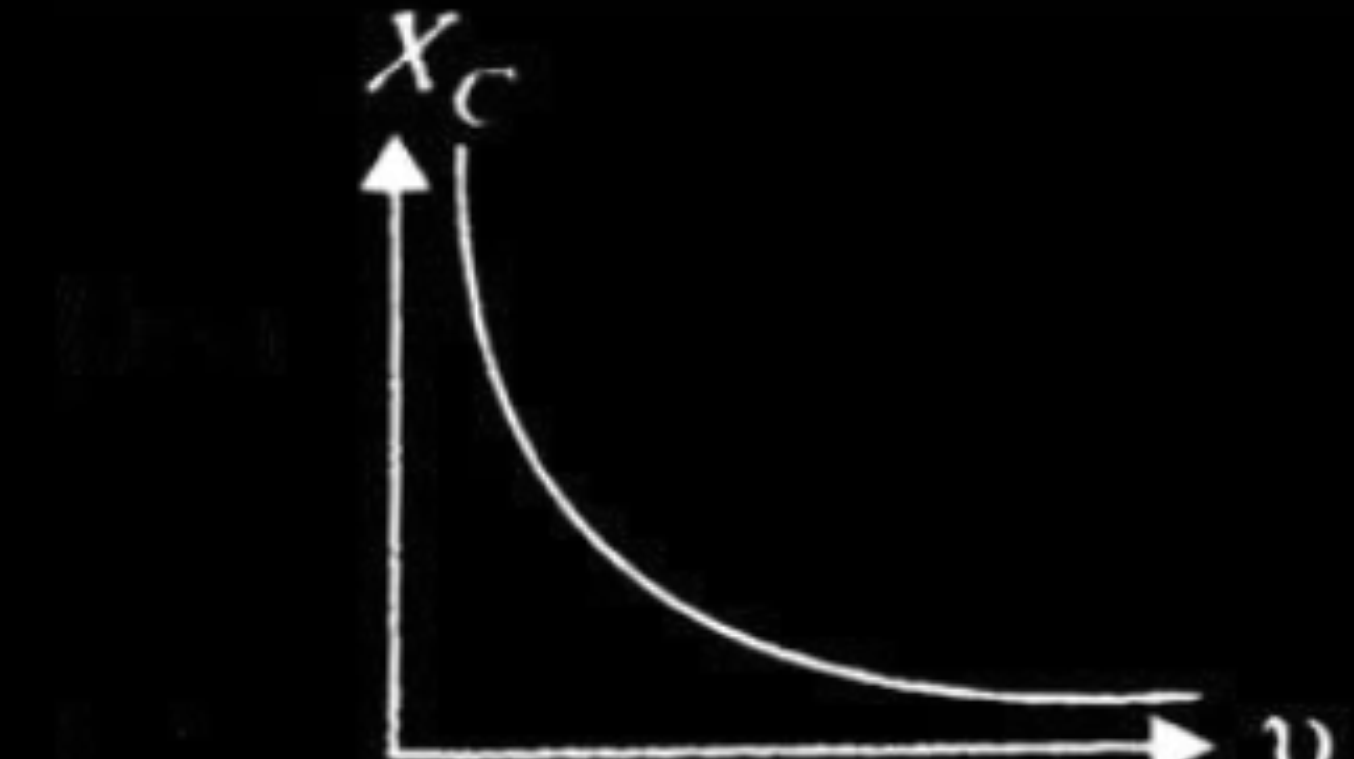
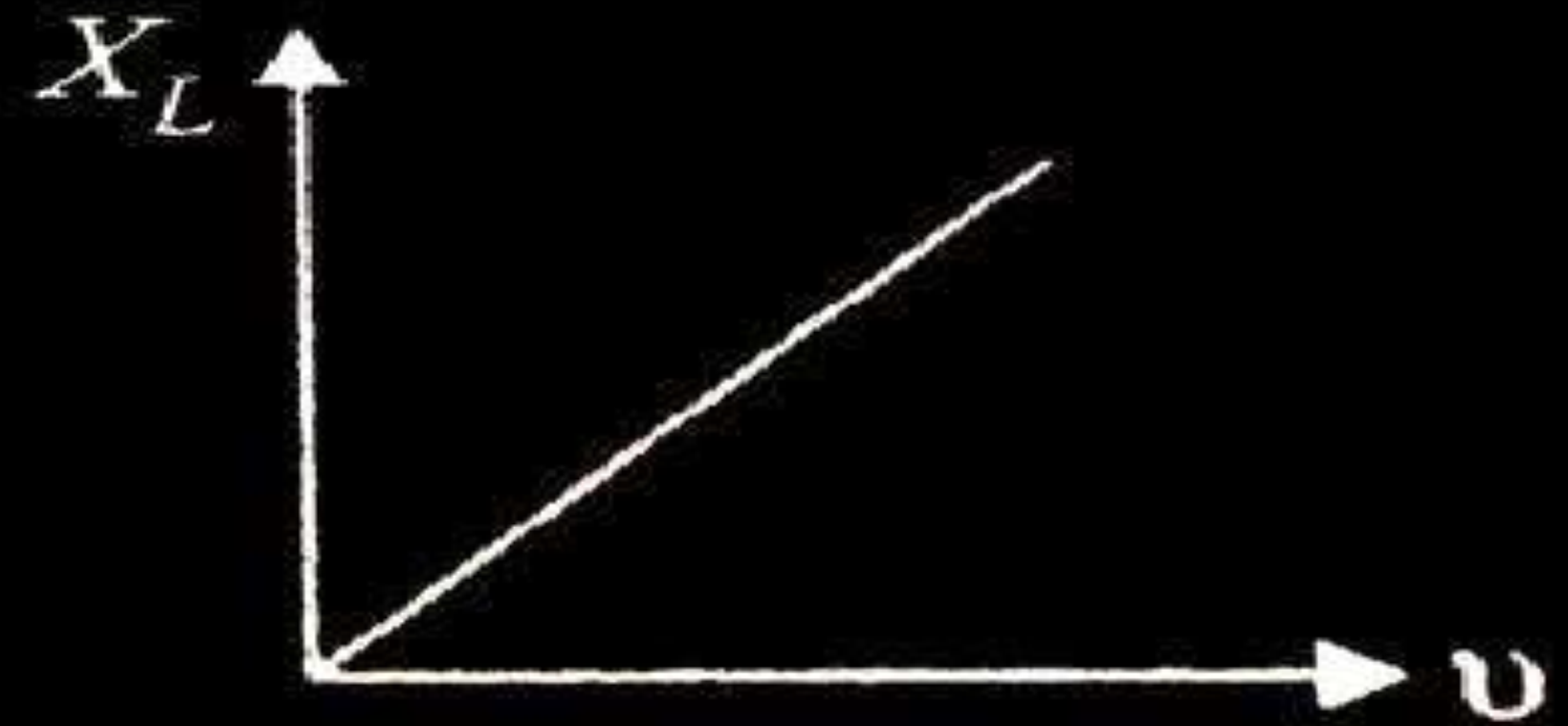
(C) 2200 V – 5 Hz

(D) 2200 V – 50 Hz

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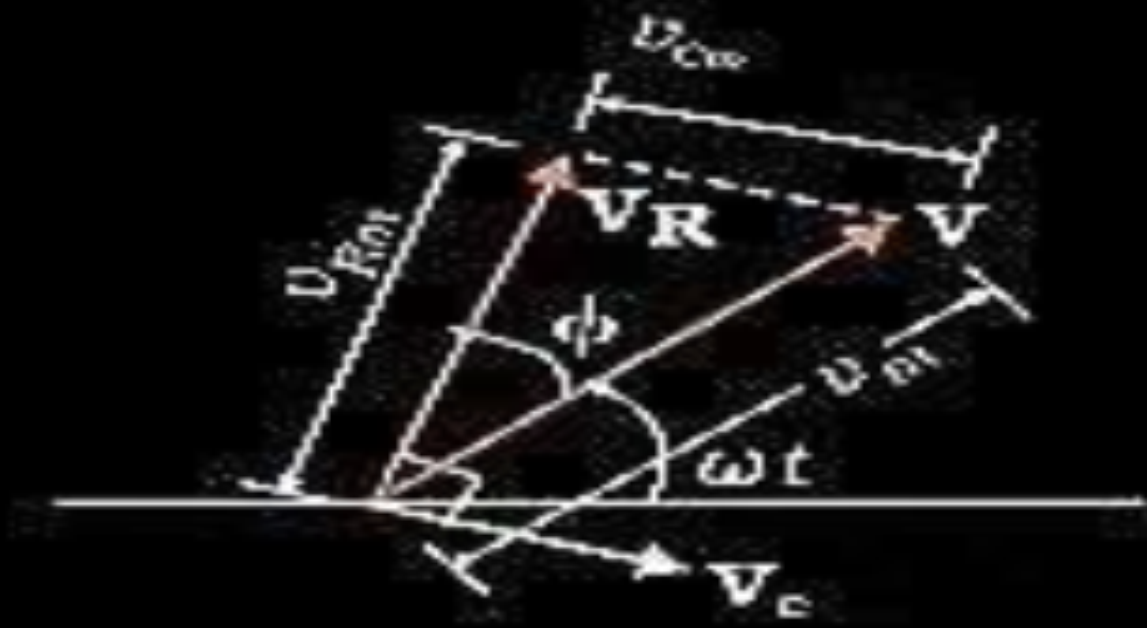
(D) 2200V - 50 HZ

- 25.** (a) Draw the graphs showing the variation of the following with the frequency of ac source in a circuit :
- (i) Resistance
 - (ii) Capacitive reactance
 - (iii) Inductive reactance
- (b) Can the voltage drop across the inductor or the capacitor in a series LCR circuit be greater than the applied voltage of the ac source ? Justify your answer.

(a) (i) 	$\frac{1}{2}$
(ii) 	$\frac{1}{2}$
(iii) 	$\frac{1}{2}$
(b) Yes, V_L and V_C can be greater than applied voltage. V_L and V_C are in opposite phase.	$\frac{1}{2}$ 1

- 31.** (a) (i) A resistor and a capacitor are connected in series to an ac source $v = v_m \sin \omega t$. Derive an expression for the impedance of the circuit.
- (ii) When does an inductor act as a conductor in a circuit ? Give reason for it.
- (iii) An electric lamp is designed to operate at 110 V dc and 11 A current. If the lamp is operated on 220 V, 50 Hz ac source with a coil in series, then find the inductance of the coil.

(i)



$$V_C + V_R = V$$

$$v_m^2 = v_{rm}^2 + v_{cm}^2$$

$$v_{rm} = i_m R$$

$$v_{cm} = i_m X_c$$

$$v_m^2 = (i_m R)^2 + (i_m X_c)^2$$

$$= i_m^2 [R^2 + X_c^2]$$

$$\Rightarrow i_m = \frac{v_m}{\sqrt{R^2 + X_c^2}}$$

$$\Rightarrow \text{Impedance } Z = \sqrt{R^2 + X_c^2}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(ii) For direct current (dc), an inductor behaves as a conductor.

$$\text{As } X_L = \omega L = 2\pi \nu L$$

$$\text{For dc } \nu = 0 \Rightarrow X_L = 0$$

Alternatively: -

$$\text{Induced emf } (\mathcal{E}) = - \frac{L dI}{dt}$$

$$\text{For dc; } dI = 0 \Rightarrow \mathcal{E} = 0$$

$$\text{(iii) } R = \frac{110}{11} = 10 \Omega$$

$$i_{rms} = \frac{v_{rms}}{\sqrt{R^2 + X_L^2}} = \frac{220}{\sqrt{100 + X_L^2}}$$

$$11 = \frac{220}{\sqrt{100 + X_L^2}}$$

$$\sqrt{100 + X_L^2} = \frac{220}{11} = 20 \Omega$$

Squaring both sides:

$$\Rightarrow 100 + X_L^2 = 400$$

$$\Rightarrow X_L^2 = 300 \Rightarrow X_L = 10\sqrt{3} \Omega$$

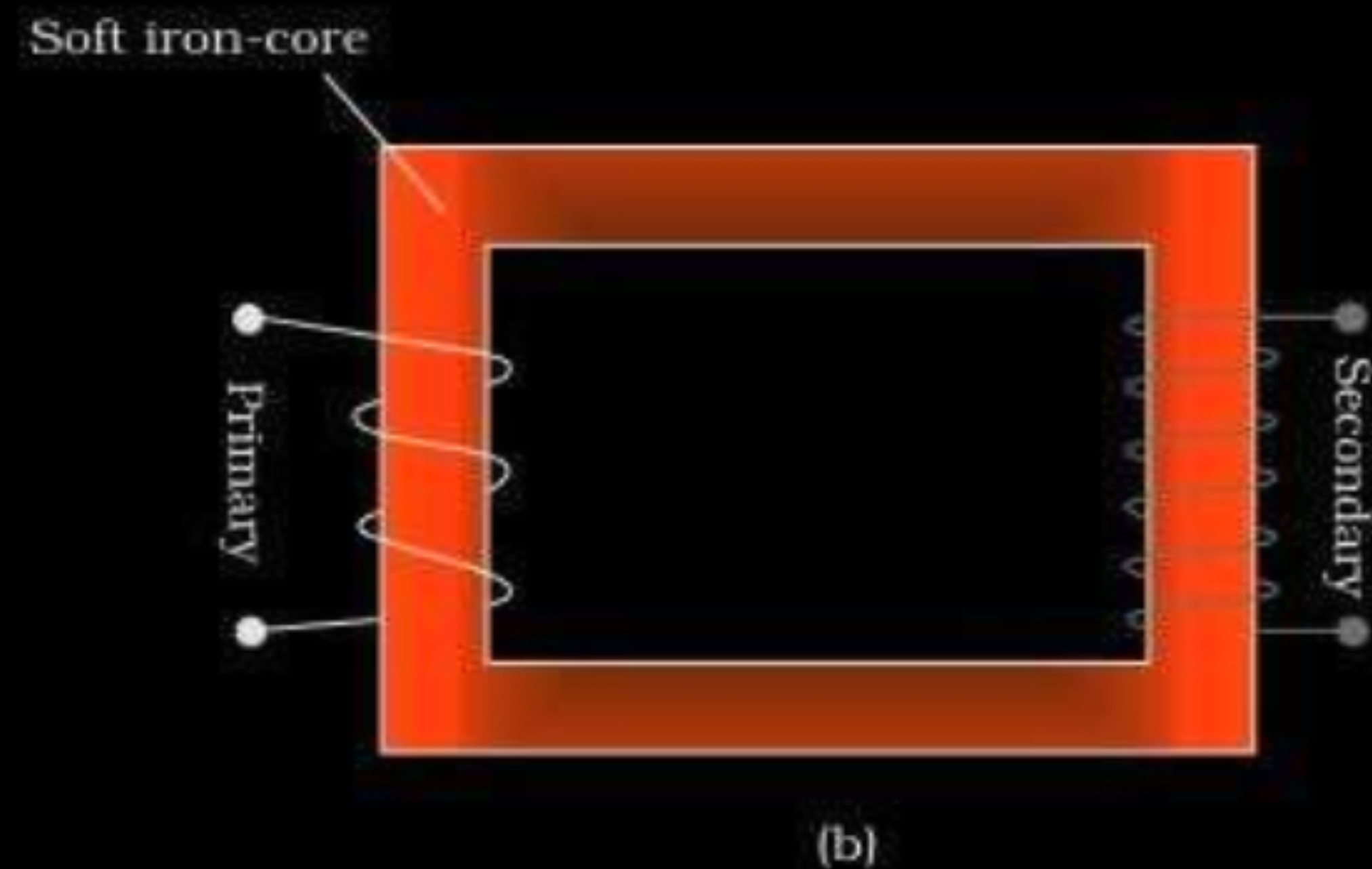
$$X_L = 2\pi fL \Rightarrow 10\sqrt{3} = 2\pi \times 50 \times L$$

$$L = \frac{\sqrt{3}}{10\pi} H$$

1

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (b) (i) Draw a labelled diagram of a step-up transformer and describe its working principle. Explain any three causes for energy losses in a real transformer.
- (ii) A step-up transformer converts a low voltage into high voltage. Does it violate the principle of conservation of energy ? Explain.
- (iii) A step-up transformer has 200 and 3000 turns in its primary and secondary coils respectively. The input voltage given to the primary coil is 90 V. Calculate :
- (1) The output voltage across the secondary coil
 - (2) The current in the primary coil if the current in the secondary coil is 2.0 A.



The working principle of transformer is mutual induction.

When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf in it.

Causes of energy losses (Any three)

- (a) Flux leakage
- (b) Resistance of the windings
- (c) Eddy currents
- (d) Hysteresis

1

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

(ii) No

Current changes correspondingly. So, the input power is equal to the output power.

(iii)

(1)

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = \frac{N_s}{N_p} \times V_p = \frac{3000}{200} \times 90$$

$$V_s = 1350 \text{ V}$$

(2)

$$\frac{I_p}{I_s} = \frac{N_s}{N_p}$$

$$I_p = \frac{3000}{200} \times 2 = 30 \text{ A}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

7. The reactance of a capacitor of capacitance C connected to an ac source of frequency ω is 'X'. If the capacitance of the capacitor is doubled and the frequency of the source is tripled, the reactance will become :

(A) $\frac{X}{6}$

(B) $6X$

(C) $\frac{2}{3}X$

(D) $\frac{3}{2}X$

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(A) $\frac{X}{6}$

- 25.** An inductor, a capacitor and a resistor are connected in series with an ac source $v = v_m \sin \omega t$. Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit.

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- 25.** An inductor, a capacitor and a resistor are connected in series with an ac source $v = v_m \sin \omega t$. Derive an expression for the average power dissipated in the circuit. Also obtain the expression for the resonant frequency of the circuit.

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$$v = v_m \sin \omega t$$

$$i = i_m \sin(\omega t + \phi)$$

$$\text{Power, } P = v i = (v_m \sin \omega t) \times [i_m \sin(\omega t + \phi)] \quad \frac{1}{2}$$

$$= \frac{v_m i_m}{2} [\cos \phi - \cos(2\omega t + \phi)] \quad (1) \quad \frac{1}{2}$$

The average power over a cycle is given by the average of the two terms in RHS of eqn (1). It is only the 2nd term which is time dependent. It's average is zero. Therefore, $\frac{1}{2}$

$$P = \frac{v_m i_m}{2} \cos \phi$$

$$P = V I \cos \phi$$

OR

$$P = I^2 Z \cos \phi$$
 $\frac{1}{2}$

At resonance, $X_C = X_L$

$$\frac{1}{\omega C} = \omega L$$
 $\frac{1}{2}$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow v = \frac{1}{2\pi\sqrt{LC}}$$
 $\frac{1}{2}$

14. *Assertion (A) :* The capacitive reactance increases with the increase of the frequency (ω) of applied ac voltage.

Reason (R) : The value of capacitive reactance (X_C) is ωC .

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14. *Assertion (A) :* The capacitive reactance increases with the increase of the frequency (ω) of applied ac voltage.

Reason (R) : The value of capacitive reactance (X_C) is ωC .

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(D) Assertion (A) is false and Reason (R) is also false.)

24. A bulb and an inductor of reactance X_L are connected in series with an ac source. How will the brightness of the bulb change when :

- (a) an iron rod is inserted in the inductor ?
- (b) number of turns in the inductor is decreased ?
- (c) a capacitor of capacitive reactance $X_C (= X_L)$ is connected in series with the inductor ?

3

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3

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- | | |
|--------------------------------------|---|
| (a) Brightness of the bulb decreases | 1 |
| (b) Brightness of the bulb increases | 1 |
| (c) Brightness of the bulb increases | 1 |

- (ii) Calculate the angular frequency at resonance of a series LCR circuit with $R = 12\ \Omega$, $L = 2.0\ \text{H}$ and $C = 32\ \mu\text{F}$, connected to an ac source.

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- (ii) Calculate the angular frequency at resonance of a series LCR circuit with $R = 12\ \Omega$, $L = 2.0\ \text{H}$ and $C = 32\ \mu\text{F}$, connected to an ac source.

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(ii)

$$\begin{aligned}\omega_r &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{2 \times 32 \times 10^{-6}}} \\ \omega_r &= 125\ \text{rad/s}\end{aligned}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

4. In a series LC circuit connected to an ac source, with the increase in the frequency of the source, the net reactance :
- (a) increases linearly
 - (b) decreases linearly
 - (c) first increases to become maximum and then decreases to zero
 - (d) first decreases to become zero and then increases

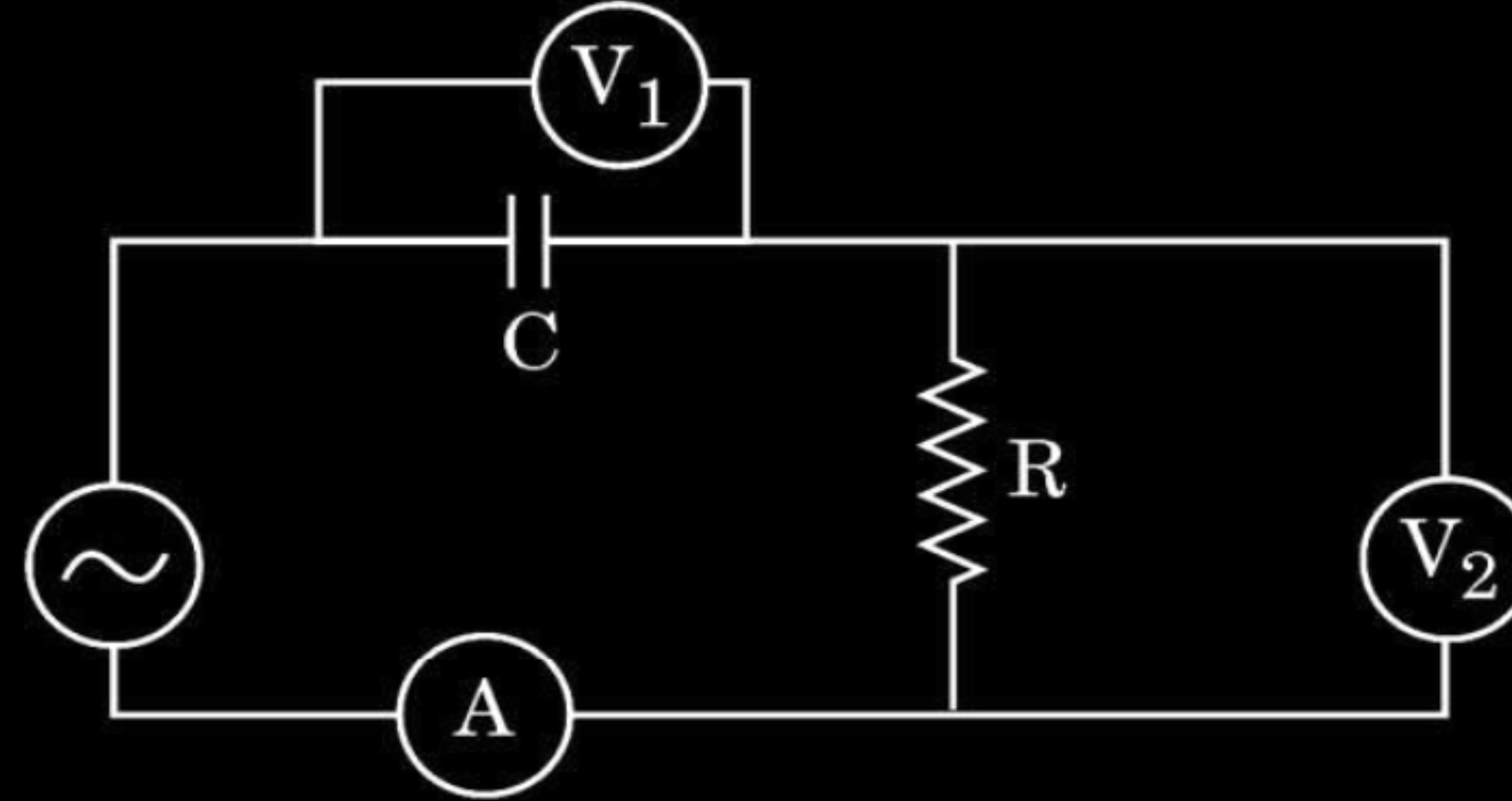
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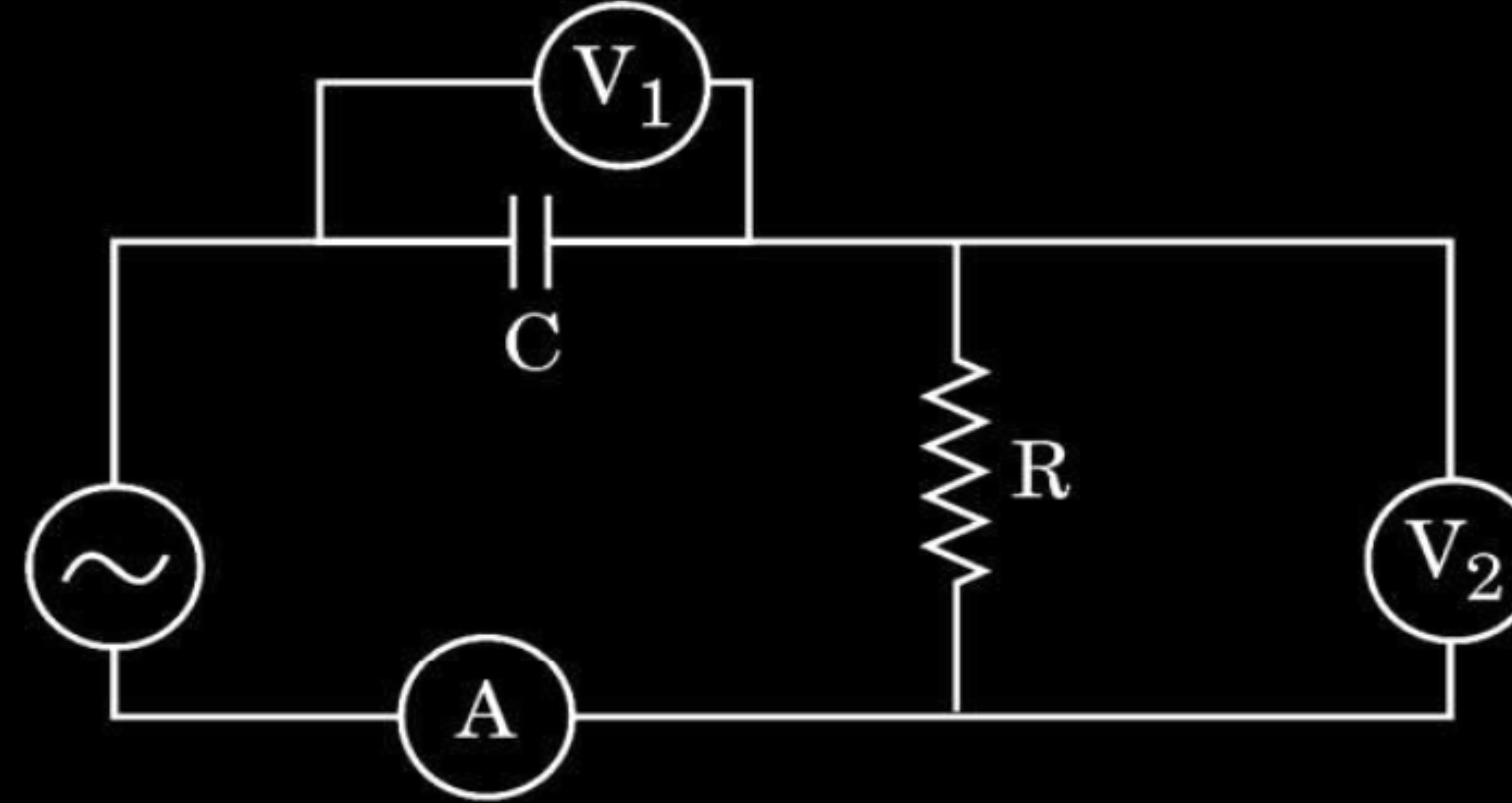
10. The given figure shows a capacitor C and a resistor R connected in series to an ac source. V_1 and V_2 are voltmeters and A is an ammeter.



Which of the following statements is correct ?

- (a) Current in the circuit lags in phase with voltage shown in V_2 .
- (b) The voltage shown in V_1 is ahead in phase with voltage shown in V_2 .
- (c) The current in the circuit and the voltage shown in V_1 are always in phase.
- (d) The voltage shown in V_1 lags behind in phase with the voltage shown in V_2 .

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(d) The voltage shown in V_1 lags behind in phase with the voltage shown in V_2 .

1. In a series LCR circuit, the capacitance is changed from C to $C/4$. For the resonant frequency to remain unchanged, the inductance should be changed from L to nL , where n is :

(a) $\frac{1}{2}$

(b) 2

(c) 4

(d) $\frac{1}{4}$

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1. In a series LCR circuit, the capacitance is changed from C to $C/4$. For the resonant frequency to remain unchanged, the inductance should be changed from L to nL , where n is :

(a) $\frac{1}{2}$

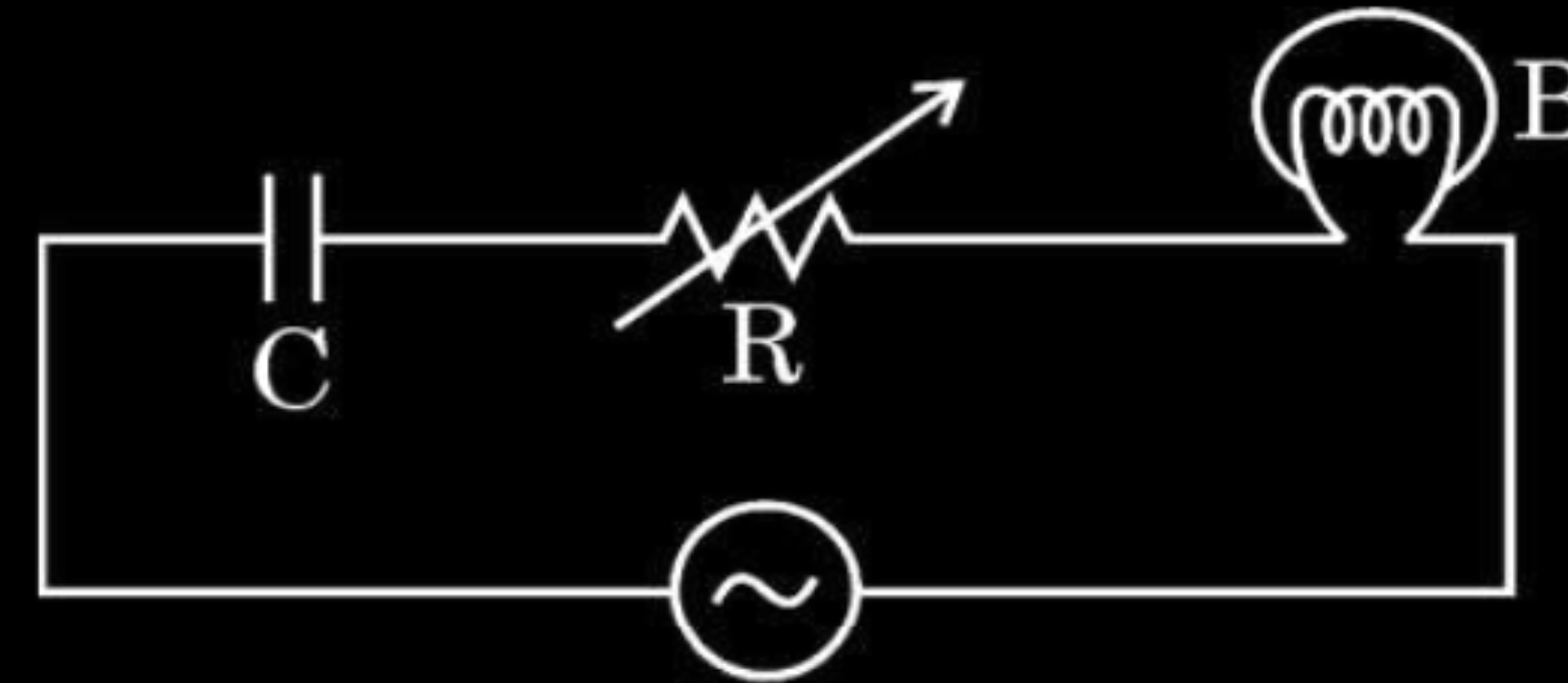
(b) 2

(c) 4

(d) $\frac{1}{4}$

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27. The figure shows a capacitor C , a variable resistor R and a bulb connected in series to the ac source of frequency (ν). The bulb glows with some brightness.



How will the glow of the bulb be affected, if the

- (a) separation between the plates of the capacitor is doubled, keeping resistance R and frequency (ν) the same ?
 - (b) resistance R is decreased keeping the value of capacitance C and frequency (ν) same ?
 - (c) frequency of ac source is decreased for the same value of C and R ?
- Justify your answer in each case.

(a) Glow of bulb will reduce. As capacitance reduces to half the net impedance of the circuit will increase and I decreases.	$\frac{1}{2}$ $\frac{1}{2}$
(b) Glow will enhance as R is decreased, Z decreases and I increases	$\frac{1}{2}$ $\frac{1}{2}$
(c) Glow of bulb will reduce When frequency is decreased, impedance increases which decreases current in the circuit.	$\frac{1}{2}$ $\frac{1}{2}$
(Note: If a student attempt using relevant formulas and conclude correctly give full credit).	

- 27.** (a) (i) Differentiate between the resistance and reactance of a series LCR circuit.
- (ii) Explain how a capacitor blocks a direct current and an inductor opposes an alternating current. CBSE 2023 3

- 27.** (a) (i) Differentiate between the resistance and reactance of a series LCR circuit.
- (ii) Explain how a capacitor blocks a direct current and an inductor opposes an alternating current. CBSE 2023 3

(i)

Resistance

- (a) Power is dissipated in the form of heat.
(b) Voltage and current remain in phase

Any other one point

Reactance

- No power is dissipated
The phase difference between voltage and current is $\frac{\pi}{2}$.

1

(ii) For a capacitor

$$\text{Reactance, } X_c = \frac{1}{\omega C} = \frac{1}{2\pi fC}$$

$\frac{1}{2}$

For direct current frequency $f=0 \quad \therefore X_c = \infty$

$\frac{1}{2}$

Hence direct current is blocked by a capacitor.

For an inductor

$$\text{Reactance, } X_L = \omega L = 2\pi fL$$

$\frac{1}{2}$

With the increase in frequency of alternating current X_L increases, hence alternating current is opposed by an inductor.

$\frac{1}{2}$

- (b) Show mathematically that in an ac circuit containing a pure inductor, the current lags behind the voltage in phase by $\frac{\pi}{2}$.

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Let the voltage across the source be

$$V = V_m \sin \omega t \text{ _____ (i)}$$

Using the Kirchoff's loop rule,

$$V - L \frac{di}{dt} = 0 \text{ _____ (ii)}$$

Where $L \frac{di}{dt}$ is the self induced emf in the inductor and L is self inductance of the inductor. The negative sign follows from Lenz's law.

From(ii)

$$\frac{di}{dt} = \frac{V}{L} = \frac{V_m}{L} \sin \omega t \text{ _____ (iii)}$$

From(iii) it follows that current i as a function of time must be such that $\frac{di}{dt}$ is in phase of the source voltage.

Integrating (iii)

$$i = \frac{V_m}{\omega L} \cos \omega t + \text{constant} \text{ _____ (iv)}$$

since emf oscillates symmetrical about zero, the current it sustains also oscillates symmetrically about zero. Therefore time independent current(constant term in eq. iv) is zero

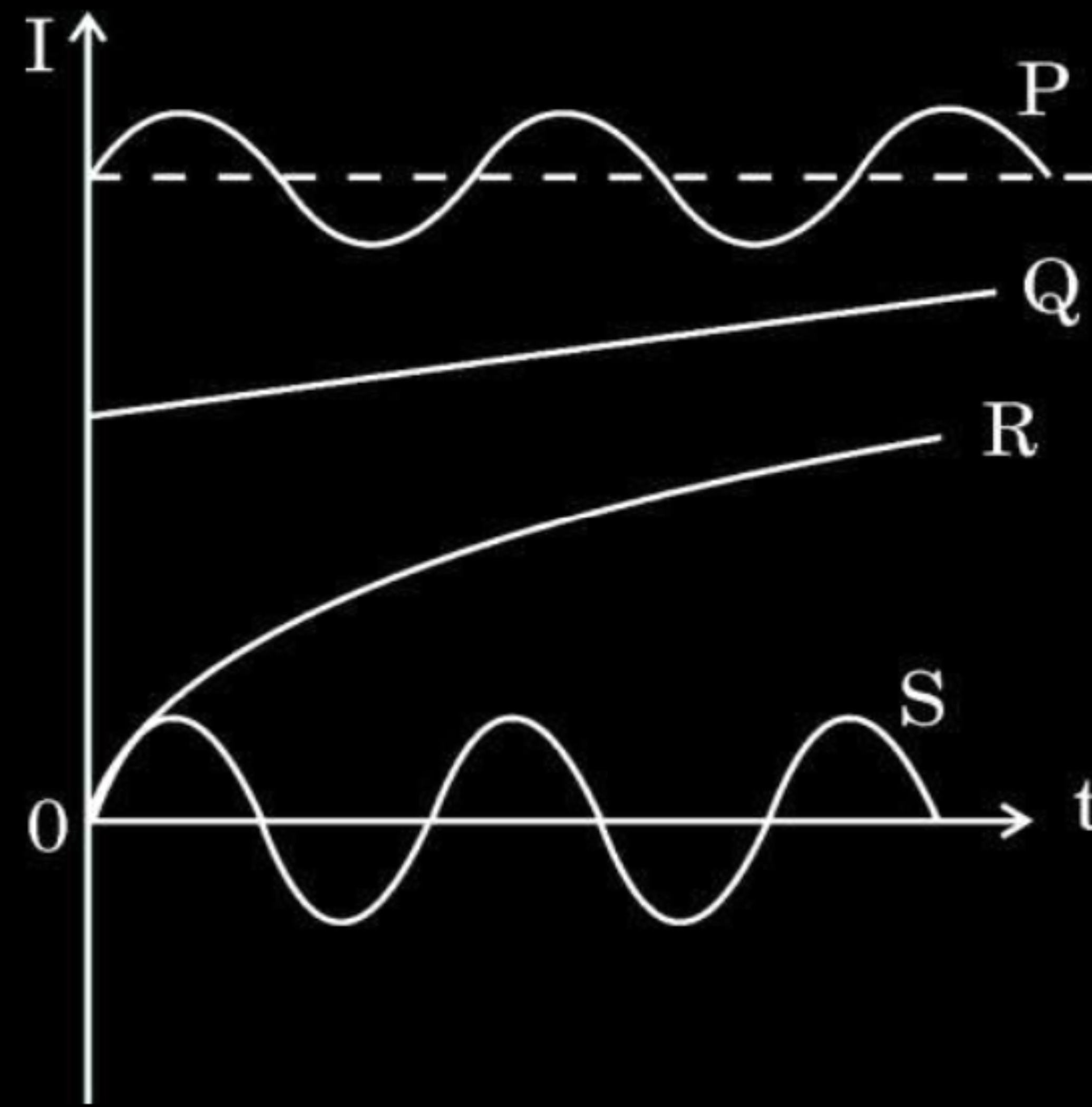
Therefore from eq. (iv) $i = \frac{V_m}{\omega L} \cos \omega t$

$$i = i_m \sin(\omega t - \frac{\pi}{2}) \text{ _____ (v)}$$

$$\text{Current amplitude } i_m = \frac{V_m}{\omega L} = \frac{V_m}{X_L}$$

$$i \text{ lags } V \text{ by } \frac{\pi}{2}$$

7. The figure shows variation of current (I) with time (t) in four devices P, Q, R and S. The device in which an alternating current flows is :



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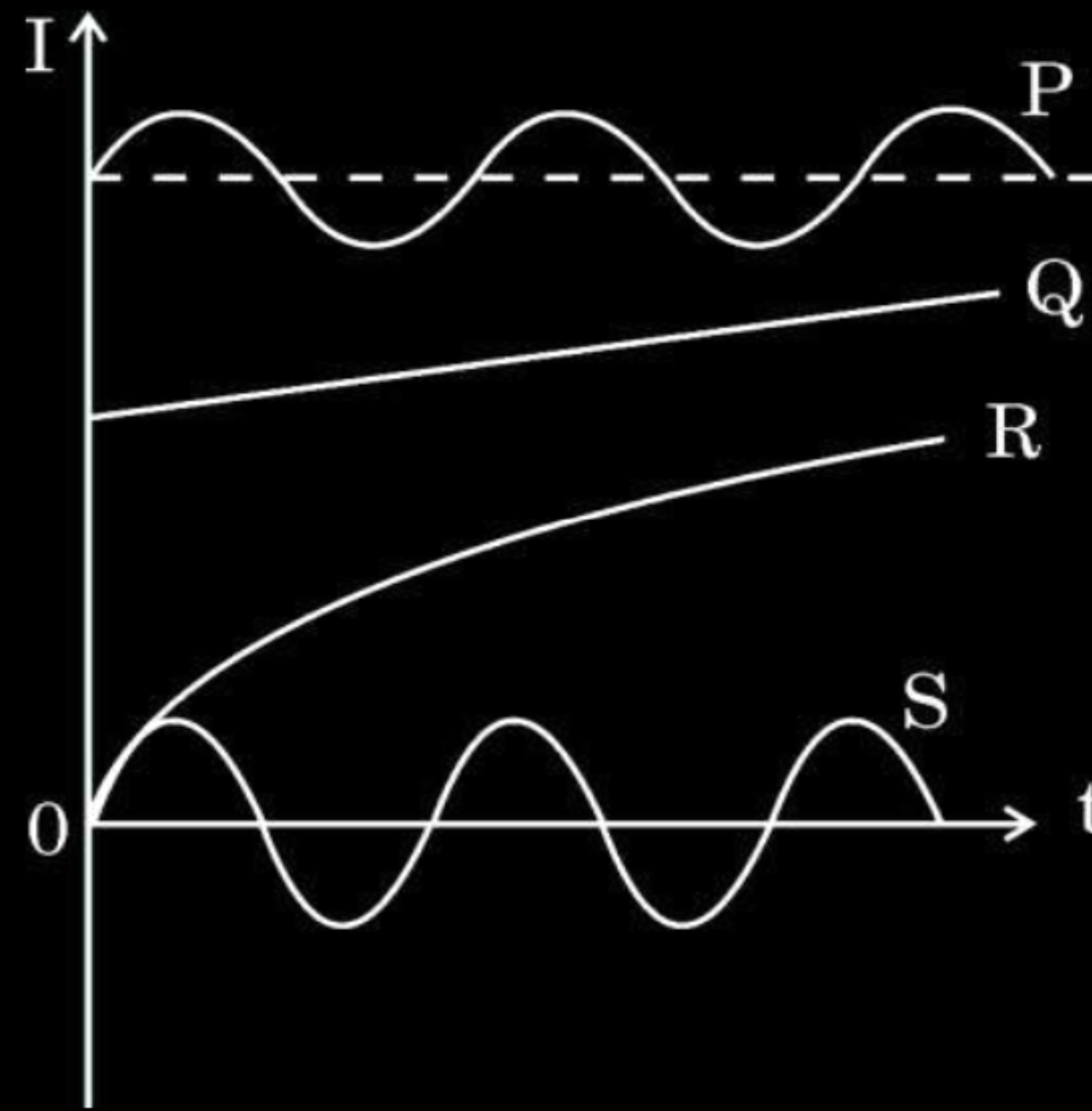
(a) P

(b) Q

(c) R

(d) S

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(a) P

(b) Q

(c) R

(d) S

(d) S

12. An ac voltage $v = v_0 \sin \omega t$ is applied to a series combination of a resistor R and an element X . The instantaneous current in the circuit is $I = I_0 \sin (\omega t + \frac{\pi}{4})$. Then which of the following is correct ?

- (a) X is a capacitor and $X_C = \sqrt{2} R$
- (b) X is an inductor and $X_L = R$
- (c) X is an inductor and $X_L = \sqrt{2} R$
- (d) X is a capacitor and $X_C = R$

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- (c) X is an inductor and $X_L = \sqrt{2} R$
- (d) X is a capacitor and $X_C = R$

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(d) X is capacitor and $X_c = R$

- 29.** (a) A resistor of $30\ \Omega$ and a capacitor of $\frac{250}{\pi}\ \mu\text{F}$ are connected in series to a 200 V, 50 Hz ac source. Calculate (i) the current in the circuit, and (ii) voltage drops across the resistor and the capacitor. (iii) Is the algebraic sum of these voltages more than the source voltage ? If yes, solve the paradox.

3

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$$\therefore X_C = \frac{1}{\omega C}$$

$$\omega = 2\pi\nu = 100\pi$$

$$X_C = \frac{1}{100\pi \times 250/\pi \times 10^{-6}}$$

$$= 40\Omega$$

Impedance of the circuit

$$Z = \sqrt{X_C^2 + R^2}$$

$$= \sqrt{(40)^2 + (30)^2} = 50\Omega$$

(i) Current in the circuit

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{200}{50} = 4A$$

(ii) Voltage drops across the Capacitor,

$$V_C = I_{rms} X_C = 4 \times 40 = 160V$$

Voltage drops across the Resistor,

$$V_R = I_{rms} \times R = 4 \times 30 = 120V$$

(iii) The algebraic sum of the two voltages V_R and V_C is 280V, which is more than the source voltage of 200V. This paradox can be removed by considering impedance triangle because V_R and V_C are out of phase by 90° , therefore

$$V = \sqrt{V_R^2 + V_C^2} = \sqrt{(120)^2 + (160)^2} = \sqrt{14400 + 25600} = 200V$$

This is equal to the source voltage.

1

1/2

1/2

1/2

1/2

- (b) A series LCR circuit with $R = 20 \, \Omega$, $L = 2 \, \text{H}$ and $C = 50 \, \mu\text{F}$ is connected to a 200 volts ac source of variable frequency. What is
- (i) the amplitude of the current, and (ii) the average power transferred to the circuit in one complete cycle, at resonance ?
- (iii) Calculate the potential drop across the capacitor.

3

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- (b) A series LCR circuit with $R = 20 \, \Omega$, $L = 2 \, \text{H}$ and $C = 50 \, \mu\text{F}$ is connected to a 200 volts ac source of variable frequency. What is
- (i) the amplitude of the current, and (ii) the average power transferred to the circuit in one complete cycle, at resonance ?
- (iii) Calculate the potential drop across the capacitor.

3

(i) At resonance, $Z=R$		CBSE 2023
$I_{rms} = \frac{200}{20} = 10\text{A}$	$\frac{1}{2}$	
Amplitude of the current $I_0 = \sqrt{2} \times I_{rms}$ $I_0 = 1.414 \times 10 = 14.14 \, \text{A}$	$\frac{1}{2}$	
(ii) Average power transferred to the circuit in one complete cycle at resonance		
$P = I_{rms}^2 R = (10)^2 \times 20$	$\frac{1}{2}$	
$P = 2000 \, \text{W}$	$\frac{1}{2}$	
(iii) Resonant frequency		
$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 50 \times 10^{-6}}} = 100 \, \text{rad/s}$		
$X_C = \frac{1}{\omega_r C} = \frac{1}{100 \times 50 \times 10^{-6}}$	$\frac{1}{2}$	
$V_c = I_{rms} X_C = 10 \times \frac{1}{100 \times 50 \times 10^{-6}} = 2000 \, \text{V}$	$\frac{1}{2}$	

10. A voltage signal is described by :

$$v = V_0 \quad \text{for } 0 \leq t \leq \frac{T}{2}$$

$$= 0 \quad \text{for } \frac{T}{2} \leq t \leq T$$

for a cycle. Its rms value is :

(a) $\frac{V_0}{\sqrt{2}}$

(b) V_0

(c) $\frac{V_0}{2}$

(d) $\sqrt{2} V_0$

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10. A voltage signal is described by :

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(a) $\frac{V_0}{\sqrt{2}}$

(b) V_0

(c) $\frac{V_0}{2}$

(d) $\sqrt{2} V_0$

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(a) $\frac{V_0}{\sqrt{2}}$

- 15.** An inductor, a capacitor and a resistor are connected in series across an ac source of voltage. If the frequency of the source is decreased gradually, the reactance of :
- (a) both the inductor and the capacitor decreases.
 - (b) inductor decreases and the capacitor increases.
 - (c) both the inductor and the capacitor increases.
 - (d) inductor increases and the capacitor decreases.

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15. An inductor, a capacitor and a resistor are connected in series across an ac source of voltage. If the frequency of the source is decreased gradually, the reactance of :

- (a) both the inductor and the capacitor decreases.
- (b) inductor decreases and the capacitor increases.
- (c) both the inductor and the capacitor increases.
- (d) inductor increases and the capacitor decreases.

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(b) inductor decreases and the capacitor increases.

- 26.** (a) An ac source $v = v_m \sin \omega t$ is connected across an ideal capacitor. Derive the expression for the (i) current flowing in the circuit, and (ii) reactance of the capacitor. Plot a graph of current i versus ωt .

3

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- 26.** (a) An ac source $v = v_m \sin \omega t$ is connected across an ideal capacitor. Derive the expression for the (i) current flowing in the circuit, and (ii) reactance of the capacitor. Plot a graph of current i versus ωt .

3

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$$(i) V_m \sin \omega t = \frac{q}{C}$$

$$I = \frac{dq}{dt} = \frac{d}{dt}(CV_m \sin \omega t)$$

$$I = \omega CV_m \cos \omega t$$

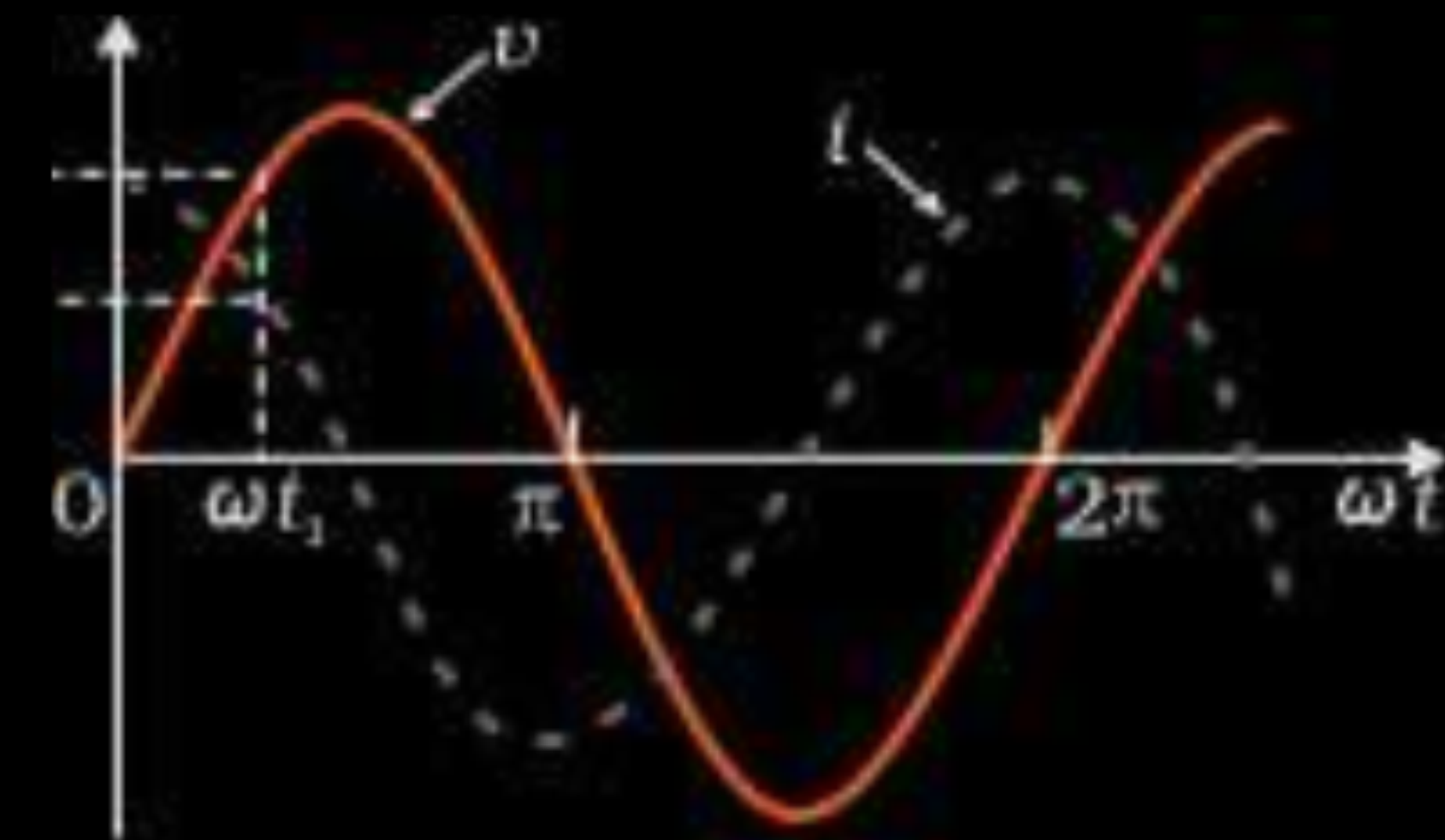
Alternatively:-

$$I = \frac{V_m}{1/\omega C} \cos \omega t = I_m \sin(\omega t + \pi/2)$$

$$(ii) I = \frac{V_m}{1/\omega C} \sin(\omega t + \pi/2) = I_m \sin(\omega t + \pi/2)$$

$$\text{Comparing with } I_m = \frac{V_m}{1/\omega C}$$

$$\text{Reactance of the capacitor; } X_c = \frac{1}{\omega C}$$



$1/2$

$1/2$

1

1

- (b) A series combination of an inductor L , a capacitor C and a resistor R is connected across an ac source of voltage in a circuit. Obtain an expression for the average power consumed by the circuit. Find power factor for (i) purely inductive circuit, and (ii) purely resistive circuit.

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3

- (b) A series combination of an inductor L, a capacitor C and a resistor R is connected across an ac source of voltage in a circuit. Obtain an expression for the average power consumed by the circuit. Find power factor for (i) purely inductive circuit, and (ii) purely resistive circuit.

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3

Instantaneous Power;

$$P = VI = (V_m \sin \omega t) \times i_m \sin(\omega t + \phi)$$

$$P = \frac{V_m i_m}{2} [\cos \phi - \cos(2\omega t + \phi)] \quad \text{----(1)}$$

The average power over a cycle is given by the average of the two terms in the R.H.S of equation (1). It is only the second term which is time dependent .Its average is zero (the positive half of the cosine cancels the negative half).

Therefore,

$$P_{avg} = \frac{V_m I_m}{2} \cos \phi = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi$$

$$P_{avg} = V_{rms} I_{rms} \cos \phi$$

Alternatively:-

If the expression is deduced using integration, then full credit to be given.

(i) Power factor for purely inductive circuit, $\phi = \frac{\pi}{2} \Rightarrow \cos \phi = 0$

(ii) Power factor for purely resistive circuit; $\phi = 0 \Rightarrow \cos \phi = 1$

- 26.** An ac voltage $v = v_m \sin \omega t$ ($v_m = 310 \text{ V}$ and $f = 50 \text{ Hz}$) is connected to a pure capacitor of capacitance $15 \text{ } \mu\text{F}$. Calculate (i) the reactance of the capacitor, and (ii) the amplitude of the current. Write the expression of current through the capacitor as a function of time.

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3

- 26.** An ac voltage $v = v_m \sin \omega t$ ($v_m = 310$ V and $f = 50$ Hz) is connected to a pure capacitor of capacitance $15 \mu\text{F}$. Calculate (i) the reactance of the capacitor, and (ii) the amplitude of the current. Write the expression of current through the capacitor as a function of time.

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3

$V = V_m \sin \omega t$ $V = 310 \sin 100 \pi t$	
$\text{i) } X_c = \frac{1}{\omega C}$	$\frac{1}{2}$
$= \frac{1}{100\pi \times 15 \times 10^{-6}}$	
$= \frac{10^4}{15\pi} = 212 \Omega$	$\frac{1}{2}$
$\text{ii) } i_m = \frac{V_m}{X_c}$	$\frac{1}{2}$
$= \frac{310}{212} = 1.46 \text{ A}$	$\frac{1}{2}$
Equation of current	
$i = i_m \sin(\omega t + \frac{\pi}{2})$	
$= 1.46 \sin(100 \pi t + \frac{\pi}{2})$	1

- 29.** (a) How does the resistance differ from impedance ? With the help of a suitable phasor diagram, obtain an expression for impedance of a series LCR circuit, connected to a source $v = v_m \sin \omega t$.

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3

29. (a) How does the resistance differ from impedance ? With the help of a suitable phasor diagram, obtain an expression for impedance of a series LCR circuit, connected to a source $v = v_m \sin \omega t$.

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3

1. Resistance is opposition offered to both alternating current and direct current while impedance is opposition offered to alternating current only.

2. Resistance is independent of frequency of source while impedance depends on frequency.

3. Resistance is opposition offered by material of the conductor while impedance is combined opposition offered by different electrical components such as resistor, inductor or capacitor.

(Any One)

(Note: Give credit of this part if a student writes any other correct answer.)

$$V_R = i_m R, V_C = i_m X_C, V_L = i_m X_L$$

i_m = Peak value of current in the circuit.

$$\vec{V}_L + \vec{V}_R + \vec{V}_C = \vec{V}_m$$

$$\begin{aligned} (V_m)^2 &= V_R^2 + (V_C - V_L)^2 \\ &= (i_m R)^2 + (i_m X_C - i_m X_L)^2 \\ &= i_m^2 [R^2 + (X_C - X_L)^2] \end{aligned}$$

$$i_m = \frac{V_m}{\sqrt{R^2 + (X_C - X_L)^2}}$$

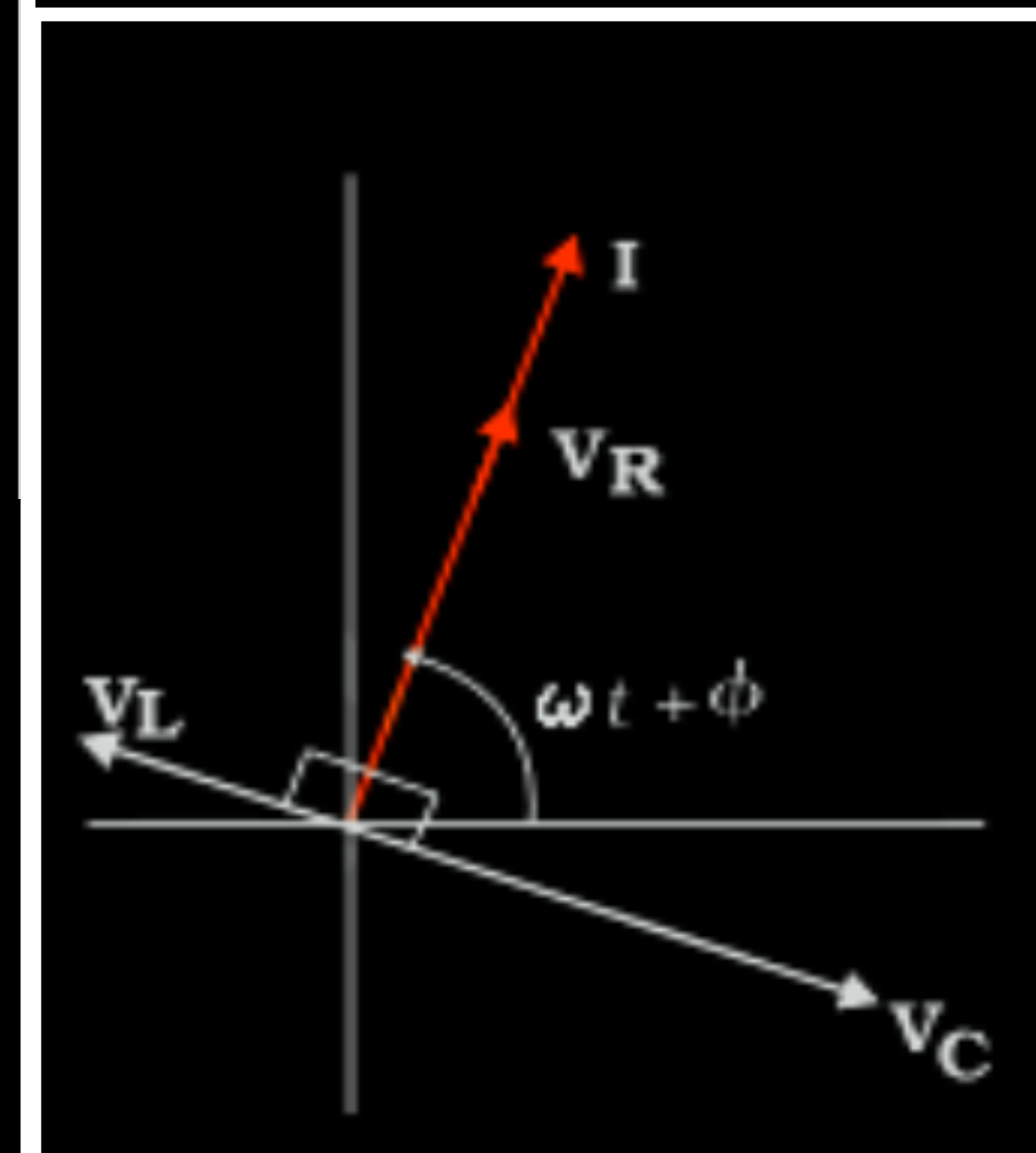
$$i_m = \frac{V_m}{Z} \text{ where } Z = \sqrt{R^2 + (X_C - X_L)^2} = \text{impedance}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



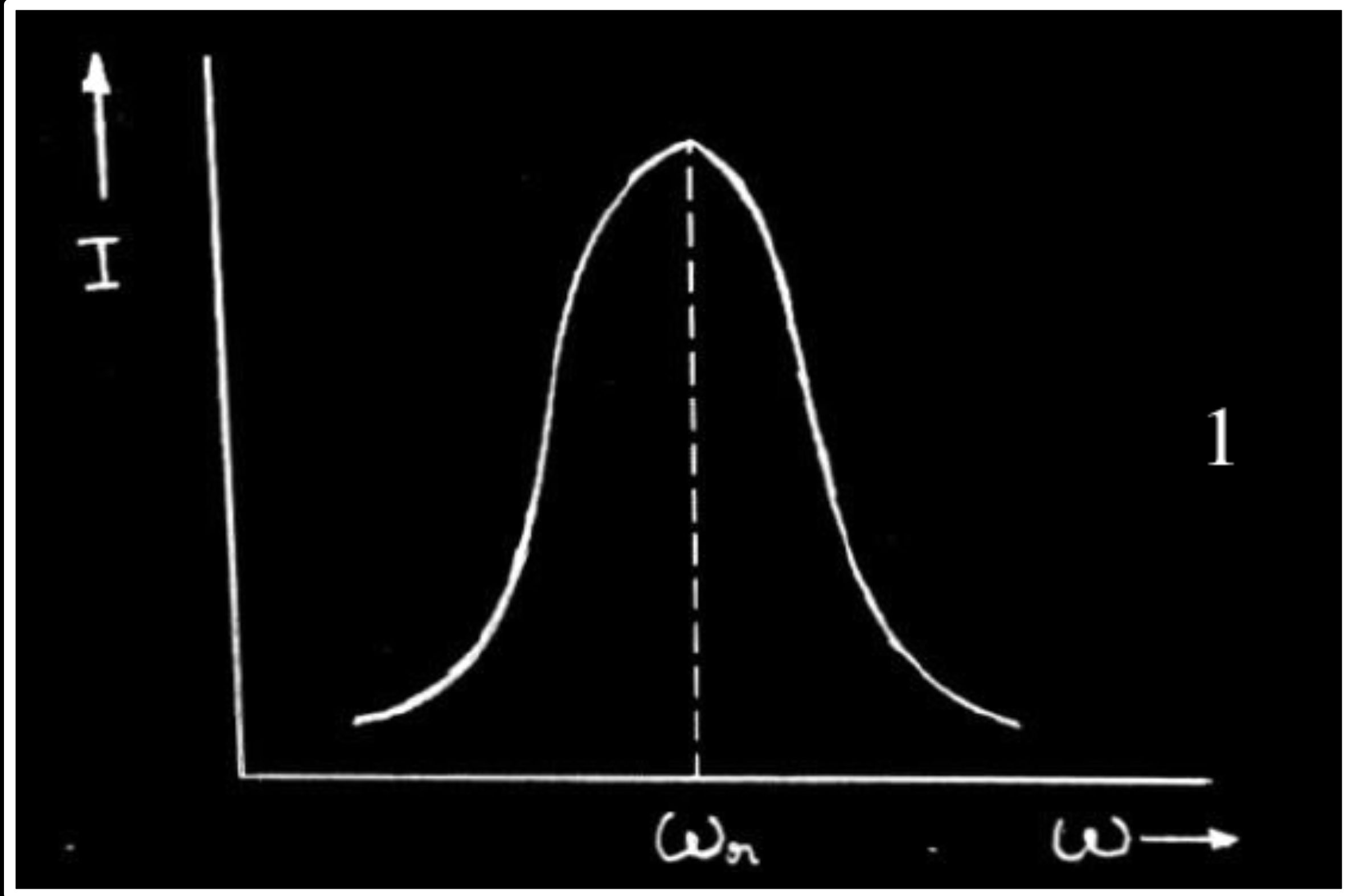
- (b) Find the condition for resonance in a series LCR circuit connected to a source $v = v_m \sin \omega t$, where ω can be varied. Give the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing the variation of electric current with frequency in a series LCR circuit.

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3



$Z = \sqrt{R^2 + (X_L - X_c)^2}$	1/2
For maximum current, Z should be minimum therefore to minimize Z $X_L = X_C$	1/2
Alternatively	
$X_L = X_C$	
$\omega L = \frac{1}{\omega C}$	1/2
$\omega_r = \frac{1}{\sqrt{LC}}$	1/2
Resonant Frequency depends on value of Inductance and Capacitance	1/2 + 1/2

28. The primary and the secondary coils of an ideal step-down transformer consist of 650 and 25 turns respectively. When the primary coil of this transformer is connected to 240 V mains, the current in the primary coil is 1.5 A. Calculate :

3

- (a) the voltage across the secondary coil
- (b) the current in the secondary coil
- (c) the average power delivered to the output circuit

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- (a) the voltage across the secondary coil
- (b) the current in the secondary coil
- (c) the average power delivered to the output circuit

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$$N_p = 650, N_s = 25, V_p = 240V, I_p = 1.5A$$

$$a) \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$V_s = \frac{N_s}{N_p} \times V_p = \frac{25}{650} \times 240$$

$$= \frac{120}{13}$$

$$= 9.23V$$

$$b) \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

$$I_s = \frac{N_p \times I_p}{N_s} = \frac{650 \times 1.5}{25} = 39A$$

$$c) P_{av} = V_s I_s$$

$$= \frac{120}{13} \times 39 = 360W$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

30. A resistor of $50\ \Omega$, a capacitor of $\left(\frac{25}{\pi}\right)\ \mu\text{F}$ and an inductor of $\left(\frac{4}{\pi}\right)\ \text{H}$ are connected in series across an ac source whose voltage (in volt) is given by $V = 70 \sin(100\pi t)$. Calculate :

3

- (a) the net reactance of the circuit,
- (b) the impedance of the circuit
- (c) the effective value of current in the circuit.

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$$R = 50 \, \Omega \quad , \quad C = \frac{25}{\pi} \times 10^{-6} \, F \quad , \quad L = \frac{4}{\pi} \, H$$

$$V_0 = 70 \, V$$

$$\omega = 100 \pi$$

$$\text{a) Net reactance} = |X_L - X_c|$$

$$= \left| \omega L - \frac{1}{\omega C} \right|$$

$$= \left| 100\pi \times \frac{4}{\pi} - \frac{1}{100\pi \times \frac{25}{\pi} \times 10^{-6}} \right|$$

$$= 0$$

$$\text{b) } Z = \sqrt{R^2 + (X_L - X_c)^2}$$

$$= \sqrt{(50)^2 + (0)^2}$$

$$= 50 \, \Omega$$

$$\text{c) } I_{eff} = \frac{V_{eff}}{Z}$$

$$= \frac{70}{\sqrt{2} \times 50}$$

$$\approx 1 \, A$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $\left(\frac{200}{\pi}\right)$ Hz and a DC source is applied across the coil.

AC Source		
S.No.	V (Volts)	I (A)
1	3.0	0.5
2	6.0	1.0
3	9.0	1.5

DC Source		
S.No.	V (Volts)	I (A)
1	4.0	1.0
2	6.0	1.5
3	8.0	2.0

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- (ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $\left(\frac{200}{\pi}\right)$ Hz and a DC source is applied across the coil.

AC Source		
S.No.	V (Volts)	I (A)
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DC Source		
S.No.	V (Volts)	I (A)
1	4.0	1.0
2	6.0	1.5
3	8.0	2.0

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(iii) From the table, $Z = 6 \Omega$, $R = 4 \Omega$

$$Z^2 = R^2 + X_L^2$$

$$X_L^2 = Z^2 - R^2 = 36 - 16 = 20$$

$$X_L = 2\sqrt{5} \approx 4.5 \Omega$$

$$2\pi\nu L = 4.5$$

$$L = \frac{4.5}{2 \times \pi \times \frac{200}{\pi}}$$

$$L = 1.1 \times 10^{-2} H = 11 mH$$

Note : Please do not deduct marks if a student writes answer as

$$0.5\sqrt{5} \times 10^{-2} H$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

26. A series RL circuit with $R = 10 \, \Omega$ and $L = \left(\frac{100}{\pi}\right) \text{ mH}$ is connected to an ac source of voltage $V = 141 \sin (100 \pi t)$, where V is in volts and t is in seconds. Calculate

3

- (a) impedance of the circuit
- (b) phase angle, and
- (c) voltage drop across the inductor

CBSE 2023

26. A series RL circuit with $R = 10 \, \Omega$ and $L = \left(\frac{100}{\pi}\right)$ mH is connected to an ac source of voltage $V = 141 \sin (100 \pi t)$, where V is in volts and t is in seconds. Calculate

3

- (a) impedance of the circuit
(b) phase angle, and
(c) voltage drop across the inductor

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$$(a) \quad Z = \sqrt{R^2 + X_L^2}, \quad X_L = \omega L$$

$$= \sqrt{100 + \left(100\pi \times \frac{100}{\pi} \times 10^{-3}\right)^2}$$

$$= 10\sqrt{2} \, \Omega$$

$$(b) \quad \tan \phi = \frac{X_L}{R}$$

$$= 1$$

$$\therefore \phi = \frac{\pi}{4} \text{ or } 45^\circ$$

$$(c) \quad V_{rms} = I_{rms} X_L$$

$$= \frac{141 \times 10}{\sqrt{2} \times 10\sqrt{2}}$$

$$= \frac{141}{2} \text{ V} = 70.5 \text{ V}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

28. A series CR circuit with $R = 200 \, \Omega$ and $C = (50/\pi) \, \mu\text{F}$ is connected across an ac source of peak voltage $\varepsilon_0 = 100 \, \text{V}$ and frequency $\nu = 50 \, \text{Hz}$. Calculate (a) impedance of the circuit (Z), (b) phase angle (ϕ), and (c) voltage across the resistor.

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3

$$\begin{aligned} \text{(a)} \quad Z &= \sqrt{R^2 + X_c^2} = \sqrt{R^2 + \left(\frac{1}{2\pi\nu C}\right)^2} \\ X_c &= \frac{1}{2\pi\nu C} = \frac{1}{2 \times \pi \times 50 \times \frac{50}{\pi} \times 10^{-6}} = 200\Omega \\ Z &= \sqrt{(200)^2 + (200)^2} = 200\sqrt{2}\Omega \\ &\approx 282 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \tan \phi &= \frac{X_c}{R} = \frac{200}{200} \\ \phi &= 45^\circ \quad \text{or} \quad \frac{\pi}{4} \text{ rad} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad V_{rms} &= I_{rms} R = \frac{V_{rms}}{Z} R \\ &= \frac{100}{\sqrt{2} \times 200\sqrt{2}} \times 200 \\ &= 50\text{V} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

13. What is the ratio of inductive and capacitive reactance in an ac circuit ?

1

(A) $\omega^2 LC$

(B) LC^2

(C) $\frac{LC}{\omega^2}$

(D) $\omega^2 L$

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(B) LC^2

(C) $\frac{LC}{\omega^2}$

(D) $\omega^2 L$

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(A) $\omega^2 LC$

27. An alternating current $I = 14 \sin (100 \pi t)$ A passes through a series combination of a resistor of 30Ω and an inductor of $\left(\frac{2}{5\pi}\right)$ H. Taking $\sqrt{2} = 1.4$, calculate the
- (i) rms value of the voltage drops across the resistor and the inductor, and
 - (ii) power factor of the circuit.

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$$I = 14 \sin(100\pi t) \text{ A}$$

$$I_{rms} = \frac{I_o}{\sqrt{2}} = \frac{14}{1.4} \text{ A}$$

$$I_{rms} = 10 \text{ A}$$

$$R = 30 \Omega$$

$$V_R = 10 \times 30 = 300 \text{ V}$$

$$X_L = \omega L = 100\pi \times \frac{2}{5\pi} = 40 \Omega \quad \left[\because L = \frac{2}{5\pi} \right]$$

$$V_L = 10 \times X_L = 10 \times 40 = 400 \text{ V}$$

$$V_L = 400 \text{ V}$$

$$Z = \sqrt{X_L^2 + R^2}$$

$$Z = \sqrt{(40)^2 + (30)^2}$$

$$= 50 \Omega$$

$$\cos \phi = \frac{30}{50}$$

$$\cos \phi = \frac{3}{5}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

13. An ideal inductor is connected across an AC source of voltage. The current in the circuit

1

- (A) is ahead of the voltage in phase by π .
- (B) lags voltage in phase by π .
- (C) is ahead of voltage in phase by $\pi/2$.
- (D) lags voltage in phase by $\pi/2$.

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- (C) is ahead of voltage in phase by $\pi/2$.
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(D) lags voltage in phase by $\frac{\pi}{2}$

28. The primary coil having N_p turns of an ideal transformer is supplied with an alternating voltage V_p . Obtain an expression for the voltage V_s induced in its secondary coil having N_s turns. Mention two main sources of power loss in real transformers.

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3

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3

When alternating voltage V_p is supplied to the primary coil, an emf is induced in this coil.

$$\epsilon_p = -N_p \frac{d\phi}{dt} = V_p$$

$\frac{1}{2}$

Which causes an induced emf in secondary coil

$$\epsilon_s = -N_s \frac{d\phi}{dt}$$

$\frac{1}{2}$

If the current taken from the secondary coil is small i.e secondary is an open circuit than $\epsilon_s = V_s$

$$V_s = -N_s \frac{d\phi}{dt}$$

$\frac{1}{2}$

$$\Rightarrow V_s = \frac{N_s}{N_p} \times V_p$$

$\frac{1}{2}$

Two main sources of power loss

1. Flux leakage
2. Resistance of the windings
3. Eddy currents
4. Hysteresis

(Any two)

$\frac{1}{2} + \frac{1}{2}$

13. Which of the following statements about a series LCR circuit connected to an ac source is correct ?
- (A) If the frequency of the source is increased, the impedance of the circuit first decreases and then increases.
 - (B) If the net reactance ($X_L - X_C$) of circuit becomes equal to its resistance, then the current leads the voltage by 45° .
 - (C) At resonance, the voltage drop across the inductor is more than that across the capacitor.
 - (D) At resonance, the voltage drop across the capacitor is more than that across the inductor.

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13. Which of the following statements about a series LCR circuit connected to an ac source is correct ?

1

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- (B) If the net reactance ($X_L - X_C$) of circuit becomes equal to its resistance, then the current leads the voltage by 45° .
- (C) At resonance, the voltage drop across the inductor is more than that across the capacitor.
- (D) At resonance, the voltage drop across the capacitor is more than that across the inductor.

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(A) If the frequency of the source is increased, the impedance of the circuit first decreases and then increases.

27. An alternating voltage of 220 V is applied across a device X. A current of 0.22 A flows in the circuit and it lags behind the applied voltage in phase by $\pi/2$ radian. When the same voltage is applied across another device Y, the current in the circuit remains the same and it is in phase with the applied voltage.
- (i) Name the devices X and Y and,
 - (ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y.

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- (i) Name the devices X and Y and,
 (ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y.

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(i)	X is inductor Y is resistor	$\frac{1}{2}$ $\frac{1}{2}$
(ii)	$X_L = \frac{220}{0.22} = 1000\Omega$	$\frac{1}{2}$
	$R = \frac{220}{0.22} = 1000\Omega$	$\frac{1}{2}$
	$I_{rms} = \frac{V_{rms}}{\sqrt{X_L^2 + R^2}}$	$\frac{1}{2}$
	$I_{rms} = \frac{220}{\sqrt{2} \times 1000} = \frac{0.22}{\sqrt{2}} = 0.11 \times \sqrt{2} A$	$\frac{1}{2}$
	$I_{rms} = 0.16 A$	

9. The power factor of a series LCR circuit at resonance will be

(a) 1

(b) 0

(c) $1/2$

(d) $1/\sqrt{2}$

9. The power factor of a series LCR circuit at resonance will be

CBSE 2020

(a) 1

(b) 0

(c) $1/2$

(d) $1/\sqrt{2}$

1

(a) 1

A series combination of an inductor (L), capacitor (C) and a resistor (R) is connected across an ac source of emf of peak value E_0 and angular frequency (ω). Plot a graph to show variation of impedance of the circuit with angular frequency (ω).

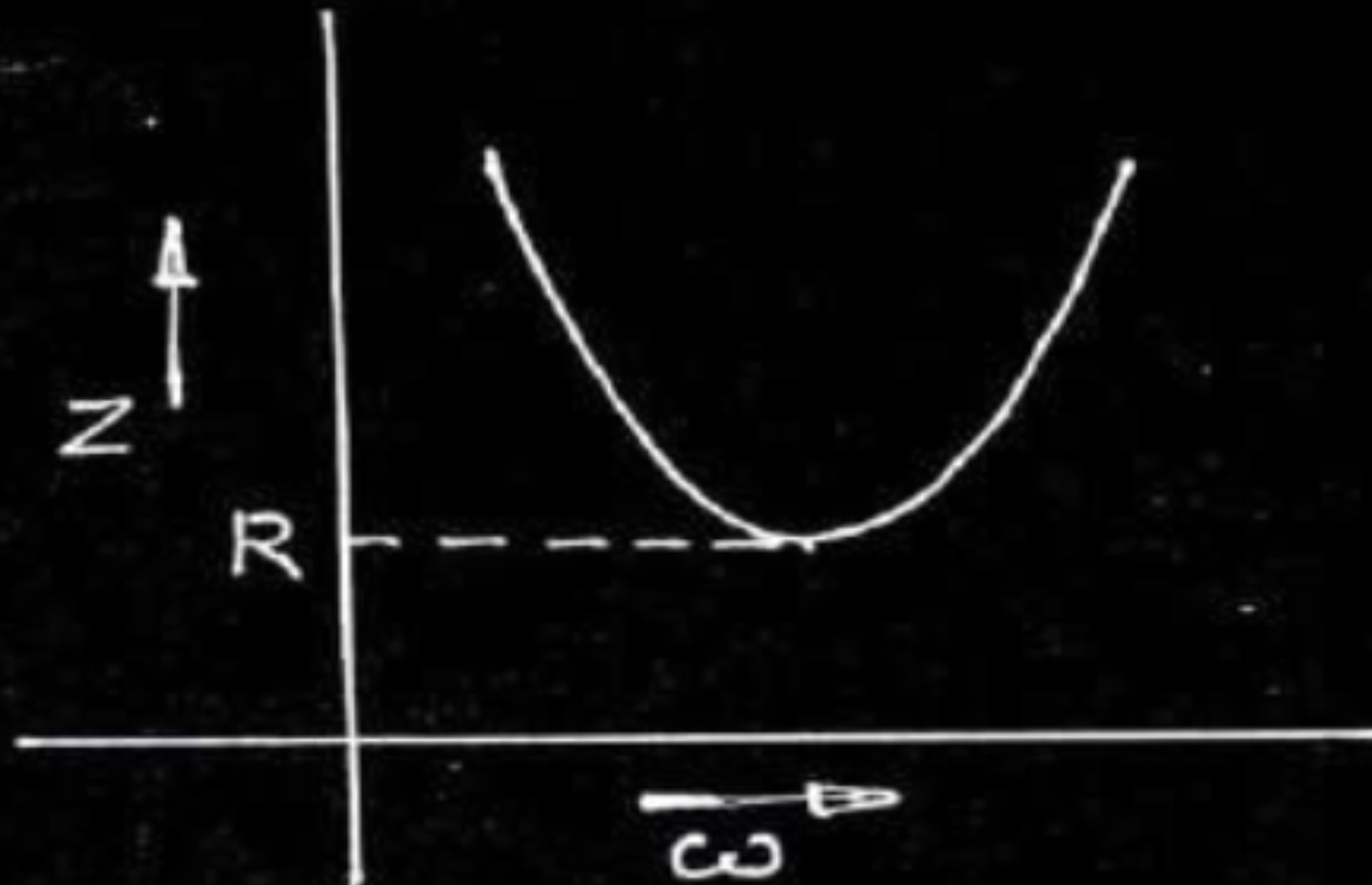
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1

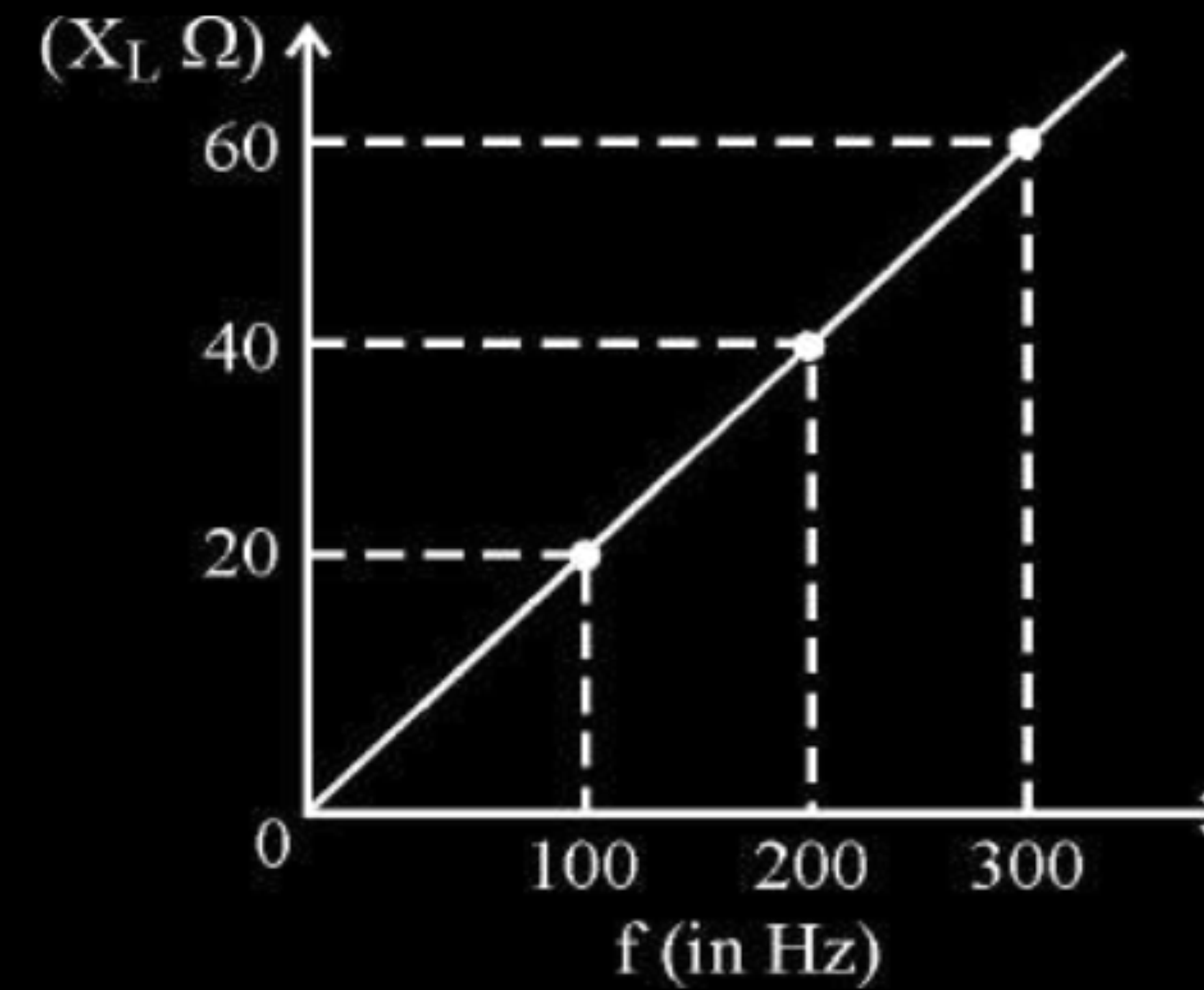
A series combination of an inductor (L), capacitor (C) and a resistor (R) is connected across an ac source of emf of peak value E_0 and angular frequency (ω). Plot a graph to show variation of impedance of the circuit with angular frequency (ω).

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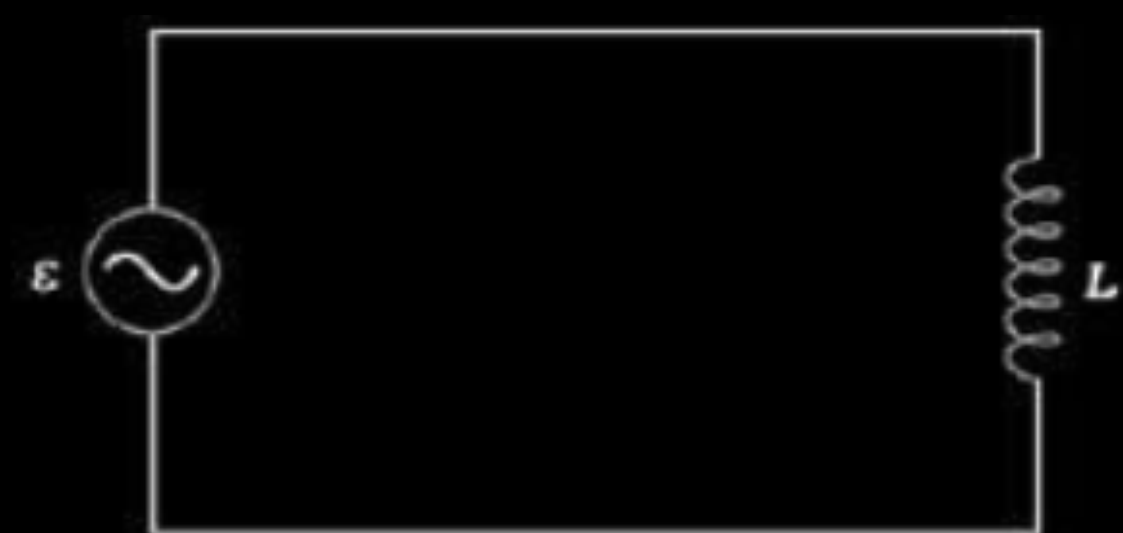
1

	$Z = \sqrt{R^2 + (X_L - X_C)^2}$ [Award one full mark even if the student just draws the graph without writing expression of impedance]	$\frac{1}{2}$ $\frac{1}{2}$
---	--	---

36. (a) Show that an ideal inductor does not dissipate power in an ac circuit.
- (b) The variation of inductive reactance (X_L) of an inductor with the frequency (f) of the ac source of 100 V and variable frequency is shown in the fig.



- (i) Calculate the self-inductance of the inductor.
- (ii) When this inductor is used in series with a capacitor of unknown value and a resistor of $10\ \Omega$ at $300\ \text{s}^{-1}$, maximum power dissipation occurs in the circuit. Calculate the capacitance of the capacitor.



$$V = V_0 \sin \omega t$$

$$I = I_0 \sin(\omega t - \pi / 2)$$

The instantaneous power supplied to the inductor

$$P_L = IV$$

$$= I_0 \sin(\omega t - \pi / 2)(V_0 \sin \omega t)$$

$$= -I_0 V_0 \cos \omega t \sin \omega t$$

$$= -\frac{I_0 V_0}{2} \sin 2\omega t$$

Now average power over a complete cycle,

$$\langle P_L \rangle = \left\langle -\frac{I_0 V_0}{2} \sin 2\omega t \right\rangle$$

$$= -\frac{I_0 V_0}{2} \langle \sin 2\omega t \rangle = 0$$

∴ Average value of $\sin 2\omega t$ over a complete cycle is zero.

Thus average power dissipated over a complete cycle is zero.

1/2

1/2

1/2

1/2

$$(b) (i) X_L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f} = \frac{40}{2\pi \times 200} = 0.1 / \pi \text{ henry} = 0.032H$$

Maximum power dissipation takes place at resonance

$$\nu = \frac{1}{2\pi\sqrt{LC}}$$

$$\therefore C = \frac{1}{L \times 300^2 \times 4\pi^2} F$$

$$C = \frac{\pi}{0.1 \times 9 \times 10^4 \times 4\pi^2} F = 8.8 \mu F$$

1/2

1/2

1/2

1/2

1/2

1/2

23. A resistor R and an inductor L are connected in series to a source of voltage $V = V_0 \sin \omega t$. The voltage is found to lead current in phase by $\pi/4$. If the inductor is replaced by a capacitor C, the voltage lags behind current in phase by $\pi/4$. When L, C and R are connected in series with the same source,

Find the :

- (i) average power dissipated and
- (ii) instantaneous current in the circuit.

CBSE 2020

2

- (i) Average power dissipated

$$P_1 = I_{eff} \times V_{eff} \times \cos 0^\circ$$

$$= I_{eff} \times I_{eff} \times R \times 1 = I_{eff}^2 R = \frac{V_0^2}{R}$$

- (ii) Instantaneous Current

$$I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = I_0 \sin \omega t$$

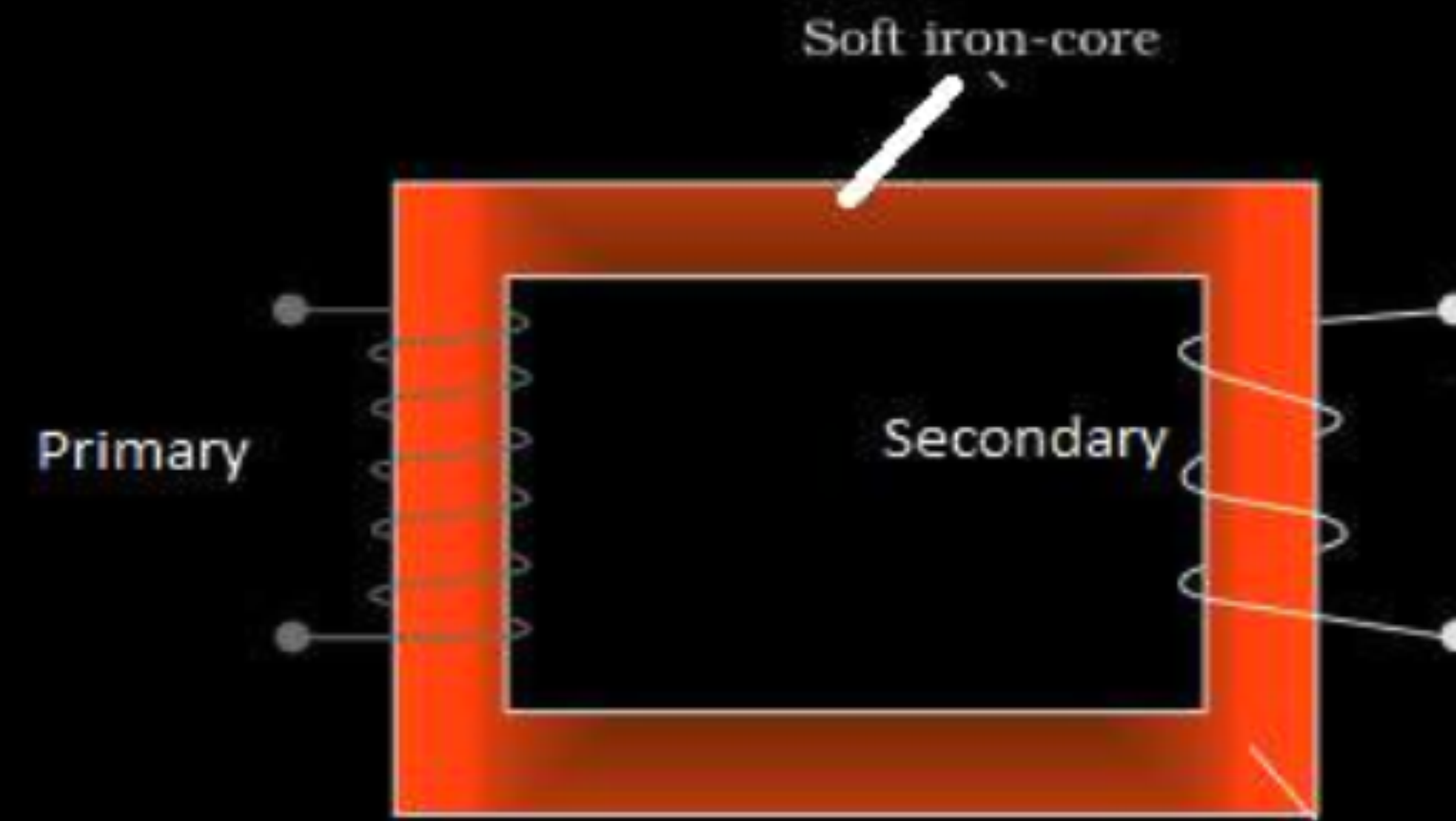
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

31. Explain with the help of a diagram, the working of a step-down transformer. Why is a laminated iron core used in a transformer ? **CBSE 2020** 3

31. Explain with the help of a diagram, the working of a step-down transformer. Why is a laminated iron core used in a transformer ? **CBSE 2020** 3



(b) When an a.c. voltage is applied across the primary coil, the resulting a.c. current in the primary coil changes the magnetic flux linked with the secondary coil, as a result an emf is induced across the secondary coil. As the number of turns in the secondary coil is less than that in the primary coil in the step down transformer, the output voltage is less than the input voltage.

(c) Use of laminated core :-

Use of laminated sheets minimizes the eddy currents, hence the energy loss.

1

1

16. An ac is passed through a series LCR circuit. What is the impedance of the circuit at resonance ?

CBSE 2020

1

16. An ac is passed through a series LCR circuit. What is the impedance of the circuit at resonance ?

CBSE 2020

1

$$Z=R$$

Alternatively, Impedance=Resistance

1

18. In an ac circuit, the applied voltage and flowing current are $E = E_0 \sin \omega t$ and $I = I_0 \sin (\omega t + \frac{\pi}{2})$ respectively. What is the average power consumed in one cycle in this circuit ?

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18. In an ac circuit, the applied voltage and flowing current are $E = E_0 \sin \omega t$ and $I = I_0 \sin (\omega t + \frac{\pi}{2})$ respectively. What is the average power consumed in one cycle in this circuit ?

CBSE 2020

1

Zero

1

29. A series LCR ac circuit has $L = 2.0 \text{ H}$, $C = 32 \mu\text{F}$ and $R = 10 \Omega$.

- (a) At what angular frequency of ac will it resonate ?
- (b) Calculate the Q value of the circuit.

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 (b) Calculate the Q value of the circuit.

3

a)

$$\begin{aligned}\omega_o &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{2 \times 32 \times 10^{-6}}} \\ &= 125 \text{ rad/s}\end{aligned}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

b)

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \text{or} \quad Q = \frac{L\omega}{R}$$

 $\frac{1}{2}$

$$Q = \frac{1}{10} \sqrt{\frac{2}{32 \times 10^{-6}}} = 25 \quad \text{Alternatively } Q = \frac{2 \times 125}{10} = 25$$

1

An ideal inductor of $\frac{5}{\pi}$ H inductance is connected to a 200 V, 50 Hz ac supply.

CBSE 2020

- (a) Calculate the rms and peak value of current in the inductor.
- (b) What is the phase difference between current through the inductor and the applied voltage ? How will it change if a small resistance is connected in series with this inductor in the circuit ?

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CBSE 2020

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3

a)	$X_L = \omega L = 2\pi\nu L$ $\therefore X_L = 2\pi \times 50 \times \frac{5}{\pi} = 500 \Omega$ $I_{rms} = \frac{200}{500} = \frac{2}{5} = 0.4A$ $I_0 = \sqrt{2} I_{rms}$ $= \sqrt{2} \times 0.4$ $= 0.56 A$ [Even if student expresses the answer as $(0.4\sqrt{2})A$ give the last $\frac{1}{2}$ marks]	$\frac{1}{2}$
		$\frac{1}{2}$
		$\frac{1}{2}$
		$\frac{1}{2}$
b)	$\frac{\pi}{2}$ or 90°	$\frac{1}{2}$
	decreases	$\frac{1}{2}$

16. What is the impedance of a capacitor of capacitance C in an ac circuit using source of frequency n Hz ? *1*

OR

What is the value of impedance of a resonant series LCR circuit ? *1*

CBSE 2020

16. What is the impedance of a capacitor of capacitance C in an ac circuit using source of frequency n Hz ? *1*

OR

What is the value of impedance of a resonant series LCR circuit ? *1*

CBSE 2020

$X_c = \frac{1}{2\pi n C}$ OR $Z = R$	<i>1</i>
---------------------------------------	----------

- 23.** (a) Explain the term ‘sharpness of resonance’ in ac circuit.
- (b) In a series LCR circuit, $V_L = V_C \neq V_R$. What is the value of power factor for this circuit ?

23. (a) Explain the term 'sharpness of resonance' in ac circuit.
- (b) In a series LCR circuit, $V_L = V_C \neq V_R$. What is the value of power factor for this circuit ?

2

(a)

Sharpness of resonance is the sharpness of the peak of the resonance curve / a graph between I_m and ω . The sharper or narrower the curve the narrower is the resonance or the resonance lasts over a very small range of frequencies /

Q factor or quality factor is the measure of sharpness of curve.

1

(b)

$$Z = R$$

Hence Power factor

$$\cos \phi = \frac{R}{Z}$$

$$\cos \phi = 1$$

Even if a student just writes power factor is 1, award full 1 mark

1

An ac source of emf $V = V_0 \sin \omega t$ is connected to a capacitor of capacitance C . Deduce the expression for the current (I) flowing in it. Plot the graph of (i) V vs. ωt , and (ii) I vs. ωt .

CBSE 2020

2

An ac source of emf $V = V_0 \sin \omega t$ is connected to a capacitor of capacitance C . Deduce the expression for the current (I) flowing in it. Plot the graph of (i) V vs. ωt , and (ii) I vs. ωt .

CBSE 2020

2



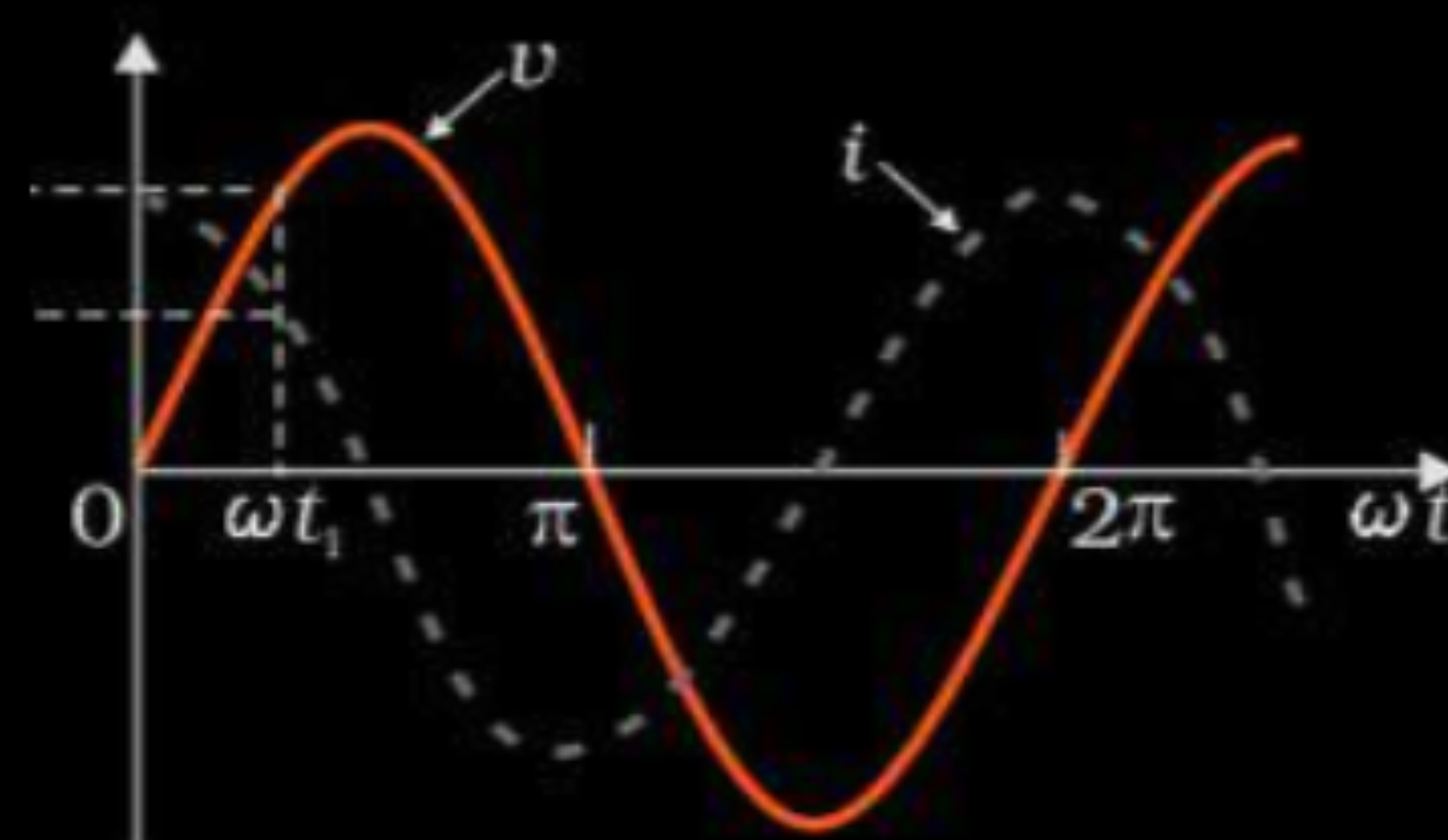
$$I = \frac{dq}{dt} = \frac{d}{dt} C V_0 \sin \omega t = \omega C V_0 \cos \omega t$$

$$= I_0 \cos \omega t$$

$$= I_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

where $I_0 = \frac{V_0}{(1/\omega C)}$

(i)



[Student can draw the two graphs separately also provided the graphs are co-related.]

$\frac{1}{2}$

$\frac{1}{2}$

1

12. Laminated iron sheets are used to minimize _____ currents in the core of a transformer.

CBSE 2020

12. Laminated iron sheets are used to minimize Eddy currents in the core of a transformer.

CBSE 2020

30. A resistance R and a capacitor C are connected in series to a source $V = V_0 \sin \omega t$.

Find :

- (a) The peak value of the voltage across the (i) resistance and (ii) capacitor.
- (b) The phase difference between the applied voltage and current.
Which of them is ahead ?

CBSE 2020

30. A resistance R and a capacitor C are connected in series to a source $V = V_0 \sin \omega t$.

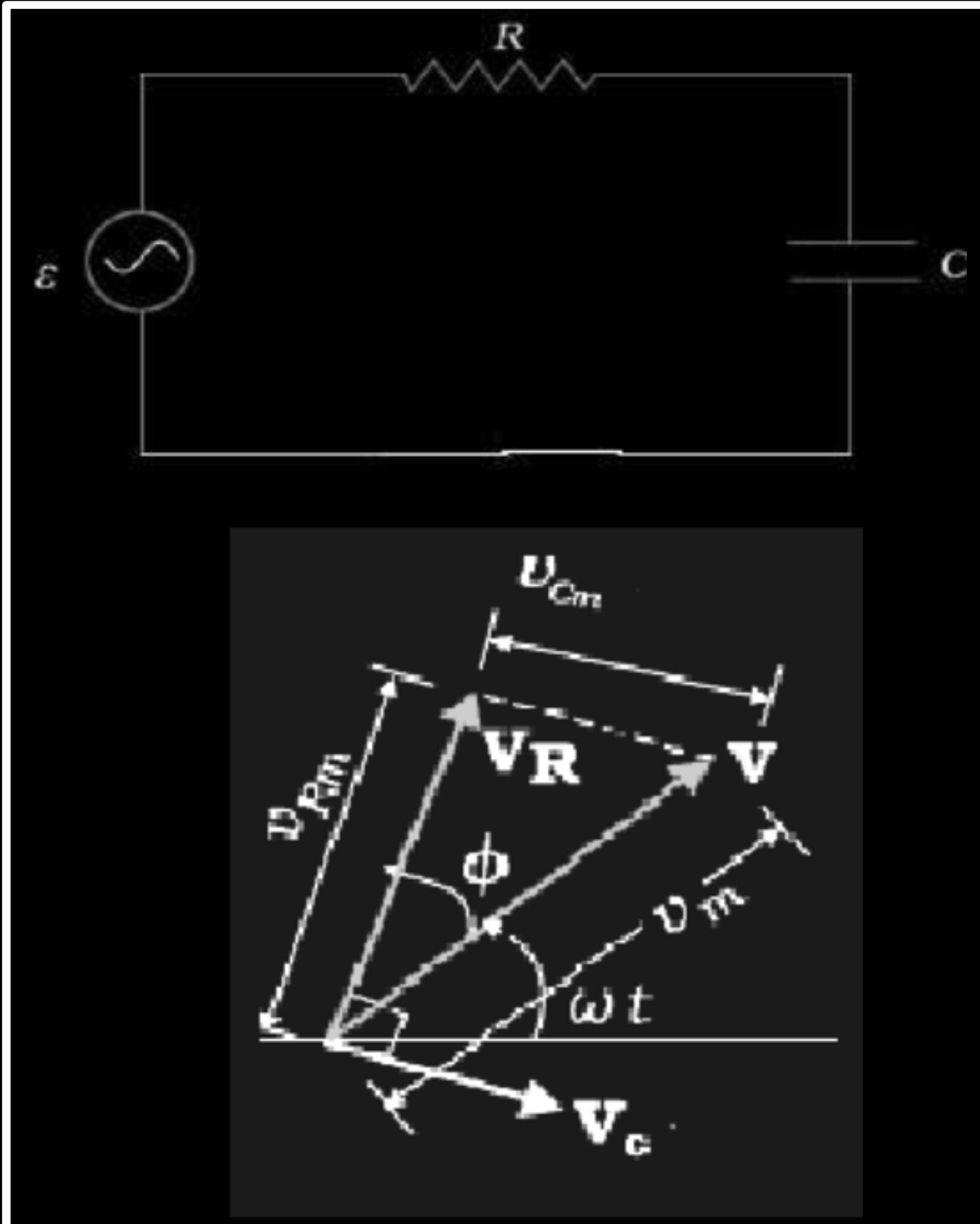
Find :

- (a) The peak value of the voltage across the (i) resistance and (ii) capacitor.
(b) The phase difference between the applied voltage and current.

Which of them is ahead ?

CBSE 2020

3



Peak value of current

$$I_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{X_c^2 + R^2}}$$

$$X_c = \frac{1}{\omega C}$$

$\frac{1}{2}$

$$(i) \quad V_R = I_0 R = \frac{V_0 R}{\sqrt{X_c^2 + R^2}}$$

$\frac{1}{2}$

$$(ii) \quad V_C = I_0 X_c = \left(\frac{V_0}{\sqrt{X_c^2 + R^2}} \right) X_c$$

$\frac{1}{2}$

(b) From phasor

$$\tan \phi = \frac{X_c}{R}$$

$\frac{1}{2}$

Current leads the applied voltage by phase ϕ

$\frac{1}{2}$

- (a) With the help of a labelled diagram, explain the working of a step-up transformer. Give reasons to explain the following :
- (i) The core of the transformer is laminated.
 - (ii) Thick copper wire is used in windings.

CBSE 2020

- (a) With the help of a labelled diagram, explain the working of a step-up transformer. Give reasons to explain the following :
- The core of the transformer is laminated.
 - Thick copper wire is used in windings.

CBSE 2020

[Note: Diagram with different windings can also be drawn]
 When an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf

Induced emf across primary coil

$$e_p = -N_p \frac{d\phi}{dt}$$

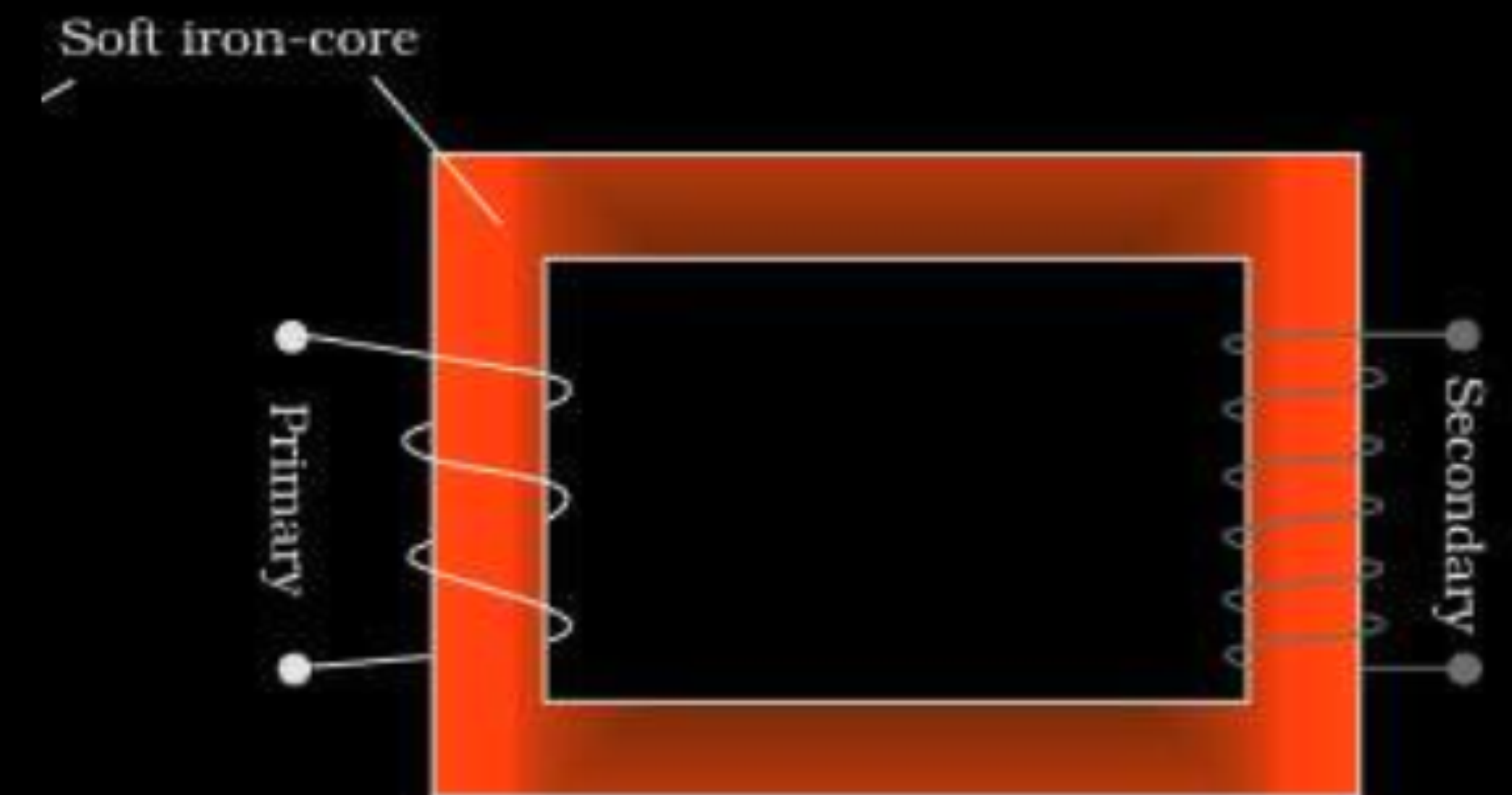
Induced emf across secondary coil

$$e_s = -N_s \frac{d\phi}{dt}$$

$$\frac{e_s}{e_p} = \frac{N_s}{N_p} = r$$

(i) to minimise the eddy currents

(ii) To reduce the heat loss



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

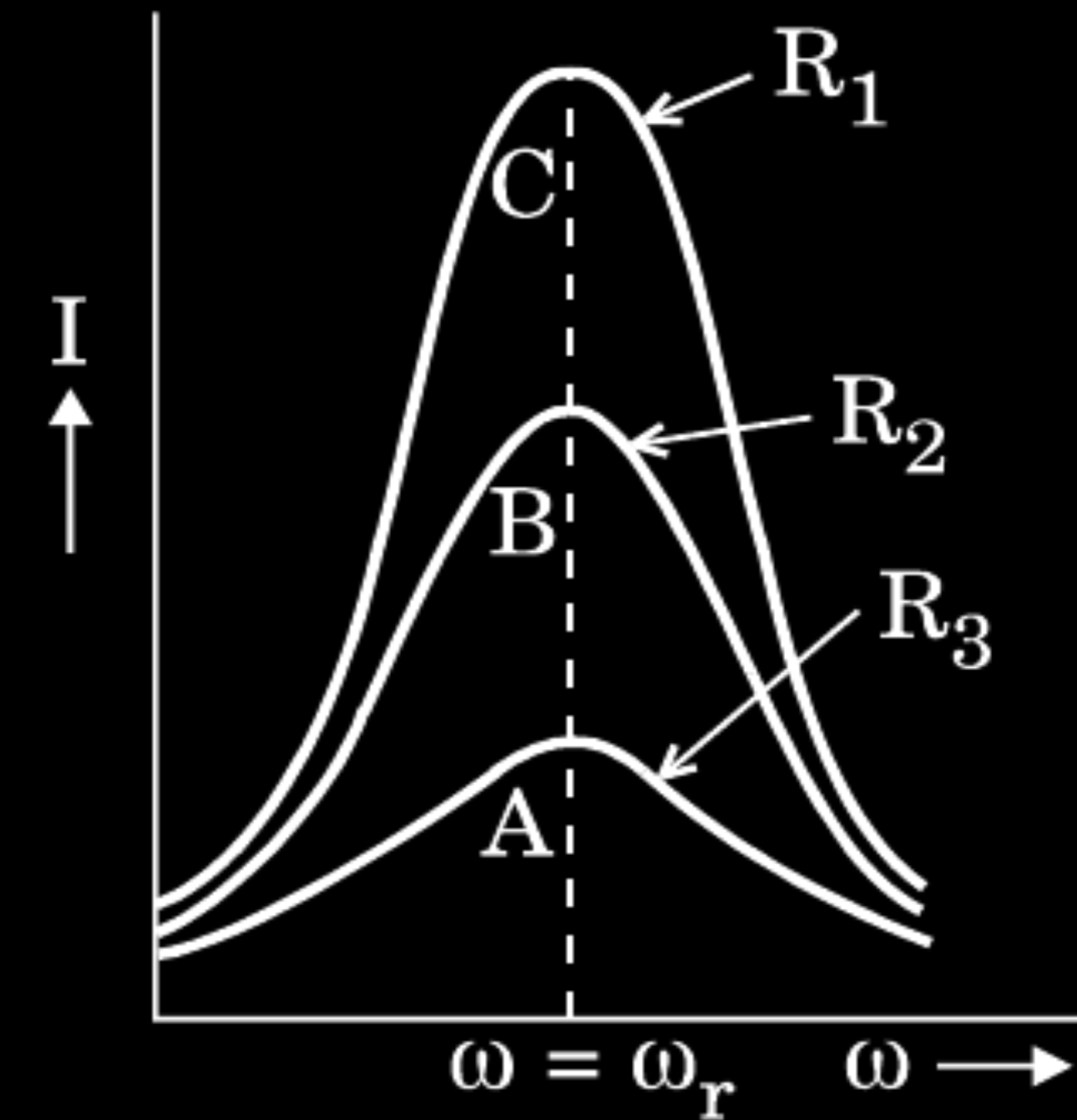
A voltage $v = v_m \sin \omega t$ applied to a series LCR circuit, drives a current in the circuit given $i = i_m \sin (\omega t + \phi)$. Deduce the expression for the instantaneous power supplied by the source. Hence, obtain the expression for the average power.

Define the terms ‘power factor’ and ‘wattless current’, giving the examples where power factor is maximum and the circuit where there is wattless current.

CBSE 2019

$P_i = V_m \sin \omega t I_m \sin(\omega t + \Phi)$ $= V_m I_m \sin \omega t [\sin \omega t \cos \Phi + \cos \omega t \sin \Phi]$ $P = V I \sin^2 \omega t \cos \Phi + \frac{2V_m I_m}{2} \sin \omega t \cos \omega t \sin \Phi$ $V_m I_m \sin \omega t \cos \Phi + \frac{V_m I_m}{2} \sin 2\omega t \sin \Phi$ $P_{av} = \frac{\int_0^T P_i dt}{T}$ $= \frac{1}{T} [V_m I_m \int_0^T (\frac{1 - \cos \Phi}{2}) \cos \Phi dt + \int_0^T V_m I_m \frac{\sin 2\omega t \sin \Phi dt}{2}]$ $P_{av} = \frac{V_m I_m \cos \Phi}{2T} \int_0^T dt = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} \cos \Phi$	1/2
<u>Power factor</u> - $\cos \phi$ factor in P_{av} is power factor.	1/2
[Alternatively: Factor $\cos \phi$ in P_{av} which decides power consumed by an ac circuit] A resistive circuit has maximum power factor.	1/2
[Alternatively : $\cos \phi = \frac{R}{Z}$ $\cos \phi = 1$ $\Rightarrow R = Z$	1/2
<u>Wattless Current</u> - A current which has a phase difference of $\pi/2$ with voltage is	1/2
called wattless current as there is no dissipation of power by this current. Current in purely inductive or purely capacitive circuit is wattless.	1/2

- (a) What do you understand by ‘sharpness of resonance’ for a series LCR resonant circuit ? How is it related with the quality factor ‘Q’ of the circuit ? Using the graphs given in the diagram, explain the factors which affect it. For which graph is the resistance (R) minimum ?



- (b) A $2\ \mu\text{F}$ capacitor, $100\ \Omega$ resistor and $8\ \text{H}$ inductor are connected in series with an ac source. Find the frequency of the ac source for which the current drawn in the circuit is maximum.
- If the peak value of emf of the source is $200\ \text{V}$, calculate the (i) maximum current, and (ii) inductive and capacitive reactance of the circuit at resonance.

CBSE 2019

a) The circuit would be set to have a high Sharpness of Resonance, if the current in the circuit drops rapidly as the frequency of the applied AC source shifts from its resonant value. (Also accept Sharpness of Resonance = $\omega_0/2\Delta\omega$).

1/2

Sharpness of Resonance is measured by the quality factor $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$

1/2

Note : Accept the answer if the student write sharpness of resonance = Q- factor

Sharpness of resonance for given value L and C / value of ω_r depends on R.

1/2

R is minimum for circuit C

1/2

b) $\nu = \frac{1}{2\pi\sqrt{LC}}$

1/2

$$= \frac{1}{2 \times 3.14 \sqrt{8 \times 2 \times 10^{-6}}}$$

$$= \frac{1000}{8 \times 3.14}$$

$$= 39.81 \text{ or } 40 \text{ Hz (Approximately)}$$

1/2

$$V_0 = 200 \text{ V}$$

$$i_0 = \frac{V_0}{Z} = \frac{V_0}{R} \quad (\because Z=R \text{ at resonance})$$

1/2

$$= \frac{200}{100}$$

$$= 2 \text{ Ampere}$$

1/2

At resonance

$$X_L = X_C$$

$$X_L = \omega L = 2\pi\nu L$$

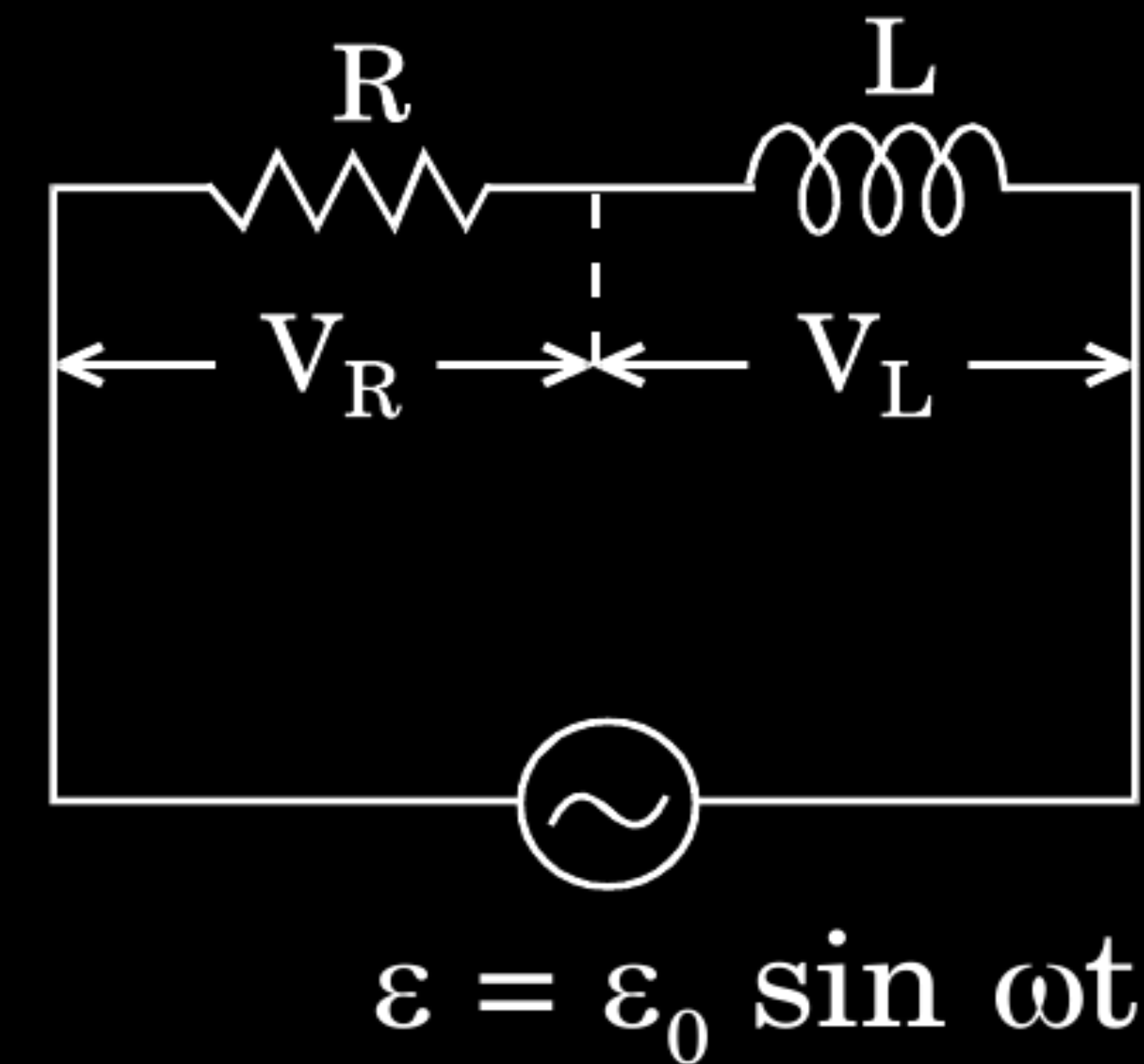
1/2

$$= 2\pi \times 39.81 \times 8$$

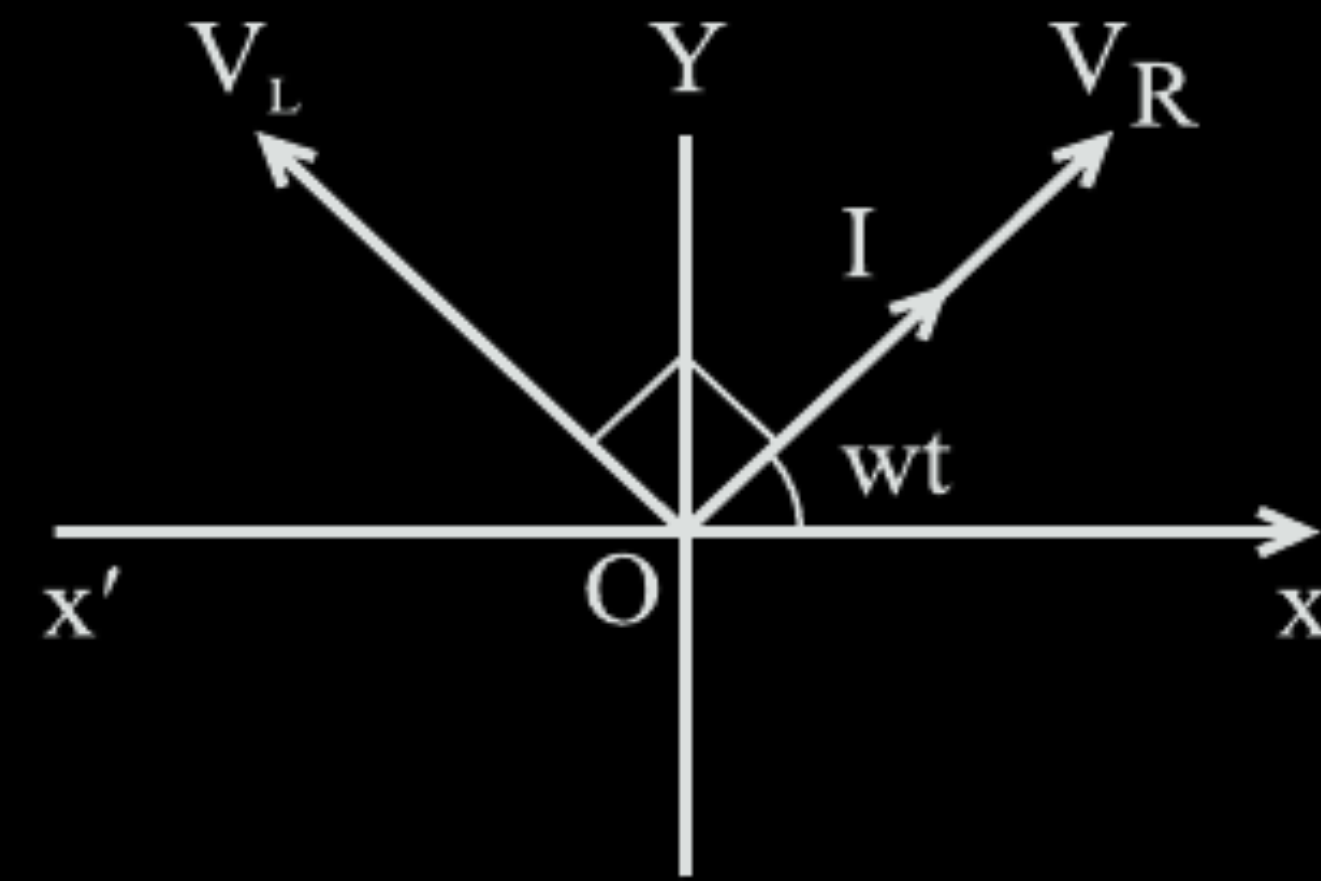
$$= 2000 \Omega$$

1/2

- (a) An ac circuit as shown in the figure has an inductor of inductance L and a resistor of resistance R connected in series. Using the phasor diagram, explain why the voltage in the circuit will lead the current in phase.
- (b) The potential difference across the resistor is 160 V and that across the inductor is 120 V . Find the effective value of the applied voltage. If the effective current in the circuit be 1.0 A , calculate the total impedance of the circuit.
- (c) What will be the potential difference in the circuit when direct current is passed through the circuit ?



CBSE 2019



- a) From phasor diagram it is clear that V_R is in phase with I and V_L is ahead of I in phase by $\pi/2$.
Hence the resultant voltage (= voltage in the circuit) will lead V_R and, therefore, the current in the circuit.

- b) Let V be the effective potential difference across L-R circuit, therefore

$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{(160)^2 + (120)^2} = 200V$$

$$\therefore \text{Impedance of the circuit, } Z = \frac{V}{I} = \frac{200}{1} = 200\Omega$$

- c) For d.c. (constant voltage source) $V_L = 0$, therefore

$$\text{Potential difference in the circuit} = V_R = \text{potential difference across the resistor}$$

(Alternatively, if the student takes the d.c. also as 1A, the potential difference will be $= 160V = V_R$)

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

An ac circuit consists of a series combination of circuit elements X and Y. The current is ahead of the voltage in phase by $\frac{\pi}{4}$. If element X is a pure resistor of $100\ \Omega$,

- (a) name the circuit element Y.
- (b) calculate the rms value of current, if rms value of voltage is 141 V.
- (c) what will happen if the ac source is replaced by a dc source ?

CBSE 2019

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- name the circuit element Y.
- calculate the rms value of current, if rms value of voltage is 141 V.
- what will happen if the ac source is replaced by a dc source ?

CBSE 2019

a) Y is a capacitor	$\frac{1}{2}$
b) Phase angle, $\phi = \pi/4$, Also $\cos\phi = \frac{R}{Z}$	$\frac{1}{2}$
$\Rightarrow Z = \frac{R}{\cos\phi} = \frac{R}{\cos(\pi/4)} = \frac{100}{1/\sqrt{2}} = 100\sqrt{2} = 141.4\Omega$	$\frac{1}{2}$
$I_{\text{r.m.s}} = \frac{V_{\text{r.m.s}}}{Z} = \frac{141\text{V}}{141.4\Omega} \cong 1\text{A}$	$\frac{1}{2}$
c) The current becomes zero.	1

A capacitor (C) and resistor (R) are connected in series with an ac source of voltage of frequency 50 Hz. The potential difference across C and R are respectively 120 V, 90 V, and the current in the circuit is 3 A. Calculate (i) the impedance of the circuit (ii) the value of the inductance, which when connected in series with C and R will make the power factor of the circuit unity.

CBSE 2019

A capacitor (C) and resistor (R) are connected in series with an ac source of voltage of frequency 50 Hz. The potential difference across C and R are respectively 120 V, 90 V, and the current in the circuit is 3 A. Calculate (i) the impedance of the circuit (ii) the value of the inductance, which when connected in series with C and R will make the power factor of the circuit unity.

CBSE 2019

$$Z = \sqrt{R^2 + X_c^2}$$

1/2

$$R = \frac{V_R}{I_R} = 30\Omega$$

1/2

$$X_c = \frac{V_c}{I_c} = \frac{120}{30} = 40\Omega$$

1/2

$$Z = \sqrt{(30)^2 + (40)^2} = 50\Omega$$

1/2

$$X_c = X_L$$

As power factor = 1

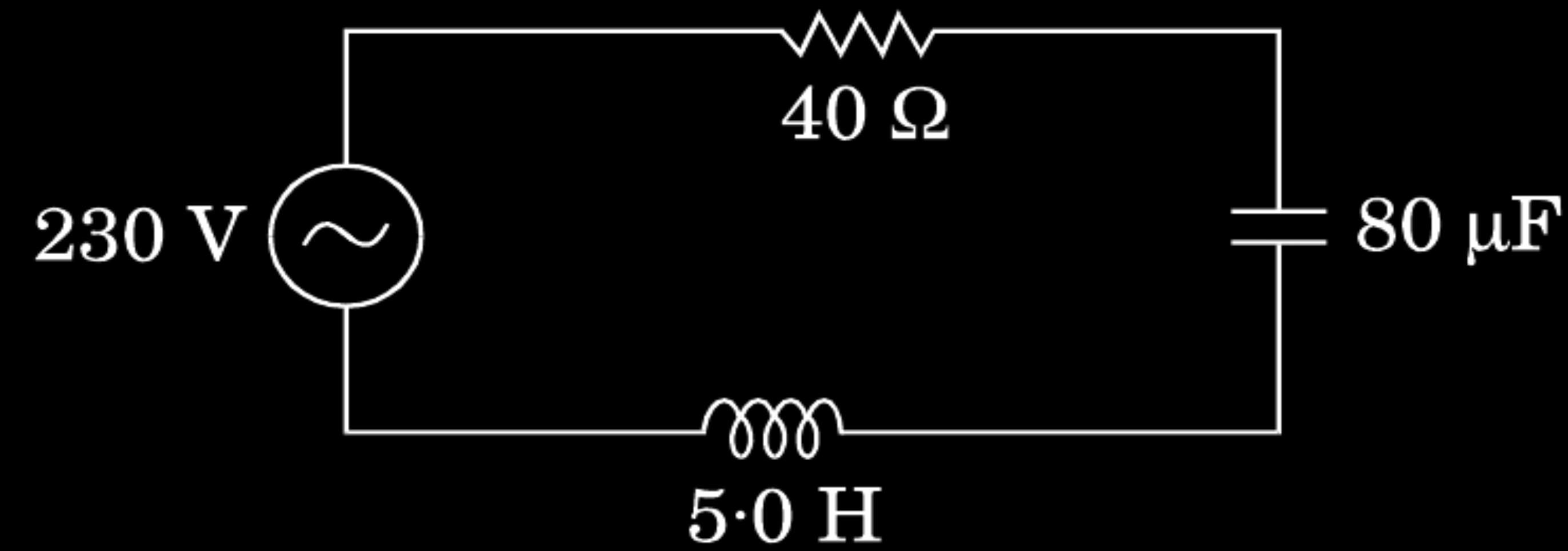
$$100\pi L = 40$$

1/2

$$L = \frac{2}{5\pi} \text{ henry}$$

1/2

The figure shows a series LCR circuit connected to a variable frequency 230 V source.



CBSE 2019

- (a) Determine the source frequency which drives the circuit in resonance.
- (b) Calculate the impedance of the circuit and amplitude of current at resonance.
- (c) Show that potential drop across LC combination is zero at resonating frequency.

$$(a) \omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = \frac{1}{\sqrt{400 \times 10^{-6}}}$$

$$\omega = \frac{1000}{20} = 50 \text{ Hz}$$

$$(b) Z = R = 40 \Omega$$

$$I_m^{\max} = \frac{230\sqrt{2}}{R} = \frac{230\sqrt{2}}{40} = 8.1 \text{ A}$$

$$V_c = I_m^{\max} X_c = \frac{230\sqrt{2}}{40} \times \frac{1}{\omega C} = 2033 \text{ volt}$$

$$V_L = I_m^{\max} X_L = \frac{230\sqrt{2}}{40} \times 2\pi\nu L = 2033 \text{ volt}$$

$$(c) V_c - V_L = 0$$

1

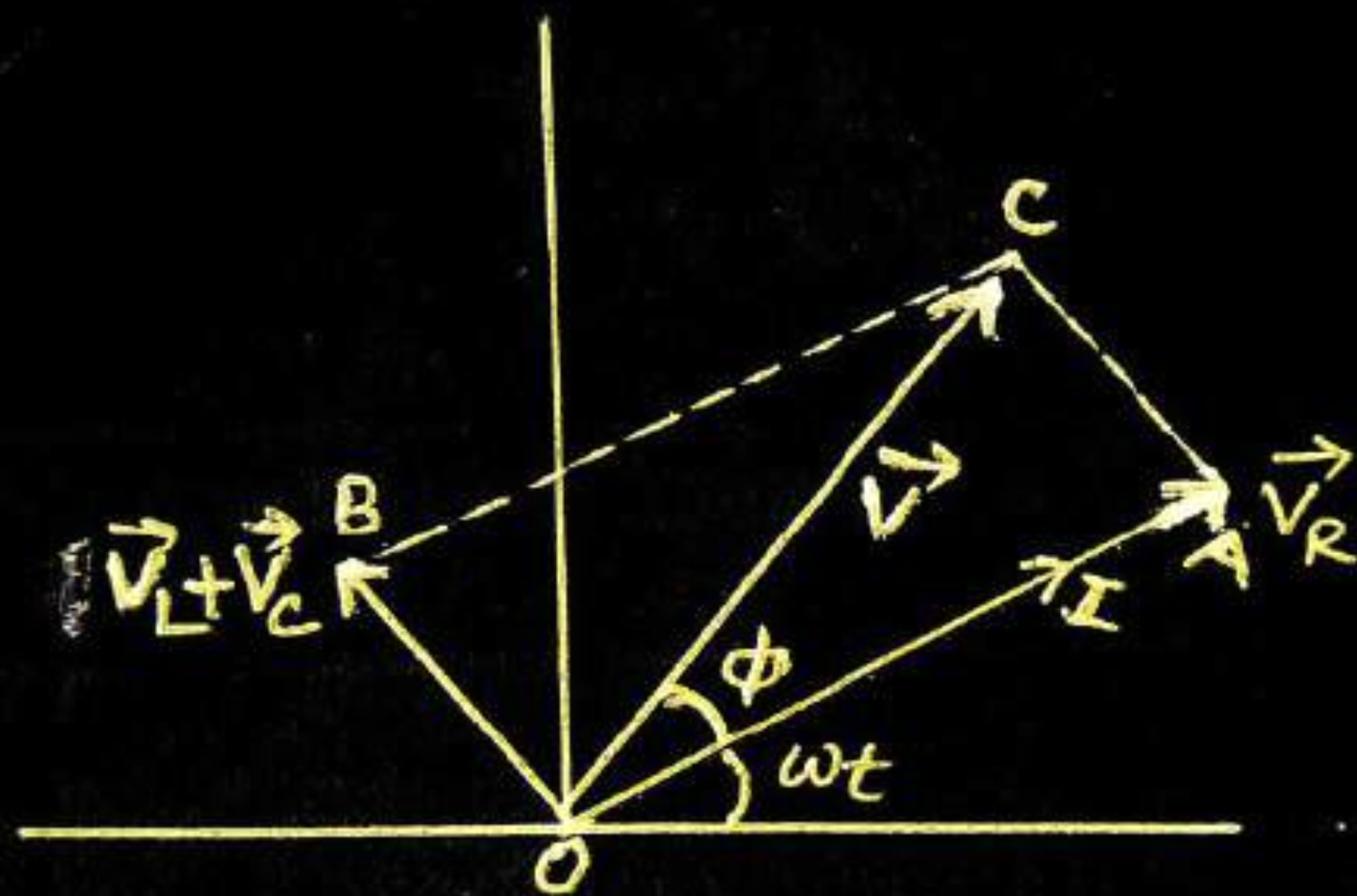
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) In a series LCR circuit connected across an ac source of variable frequency, obtain the expression for its impedance and draw a plot showing its variation with frequency of the ac source.
- (b) What is the phase difference between the voltages across inductor and the capacitor at resonance in the LCR circuit ?
- (c) When an inductor is connected to a 200 V dc voltage, a current of 1A flows through it. When the same inductor is connected to a 200 V, 50 Hz ac source, only 0.5 A current flows. Explain, why ? Also, calculate the self inductance of the inductor.



$$|\vec{V}| = V_m$$

$$|V_R| = V_{Rm}$$

$$|V_L| = V_{Lm}$$

From the figure, the pythagorean theorem gives

$$V_m^2 = V_{Rm}^2 + (V_{Lm} - V_{cm})^2$$

$$V_{Rm} = i_m R, V_{Lm} = i_m X_L, V_{cm} = i_m X_C,$$

$$V_m = i_m Z$$

$$= (i_m Z)^2 = (I_m R)^2 + (i_m X_L - i_m X_C)^2$$

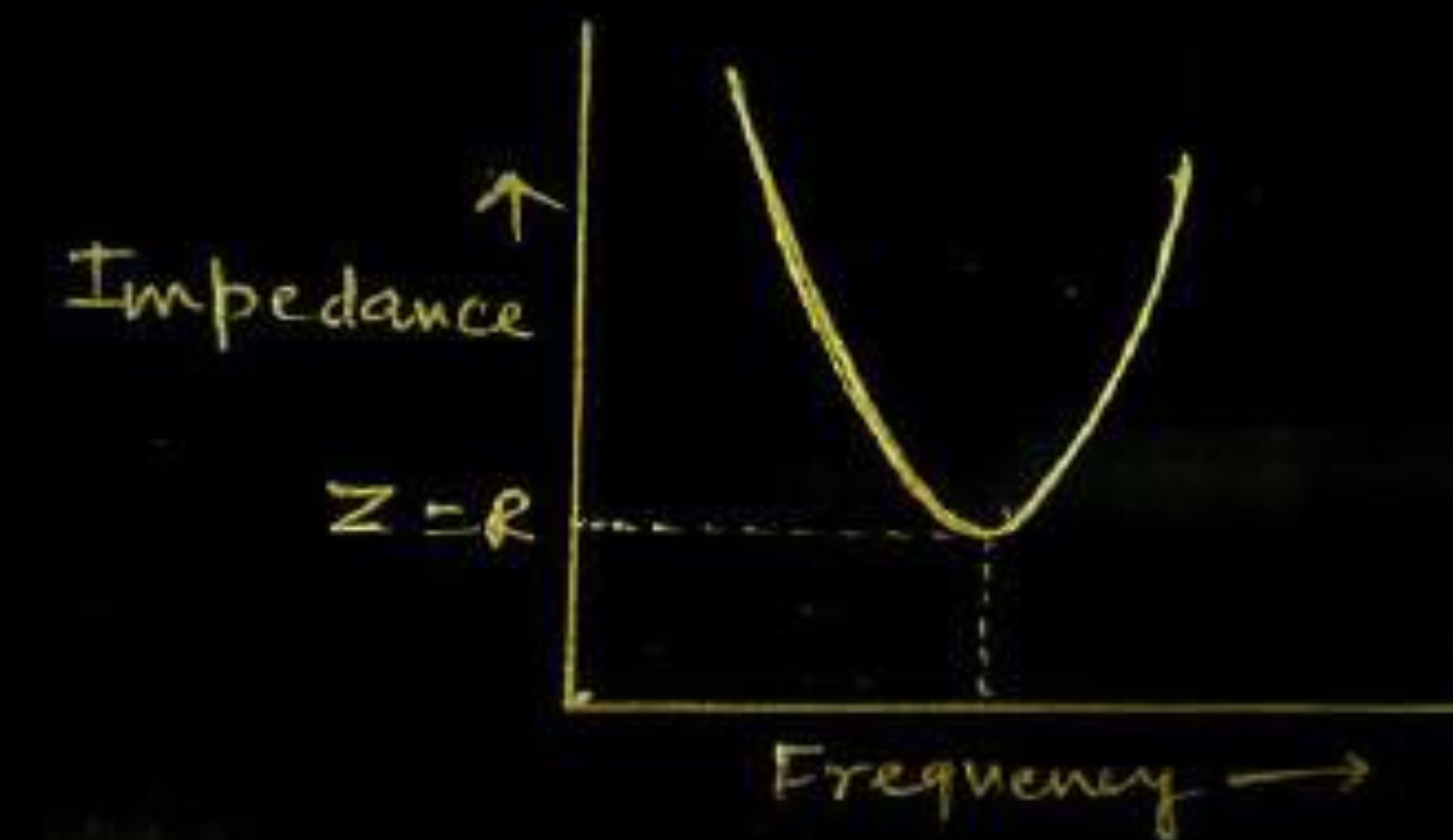
$$Z^2 = R^2 + (X_L - X_C)^2$$

$$\therefore Z = \sqrt{R^2 + (X_L - X_C)^2}$$

1

[note: award these two marks, If a student does it correctly for the other case i.e

$(V_c > V_L)$]



(b) Phase difference between voltage across inductor and the capacitor at resonance is 180°

(c) Inductor will offer an additional impedance to ac due to its self inductance.

$$R = \frac{V_{rm}}{I_{rms}} = \frac{200}{1} = 200 \Omega$$

Impedance of the inductor

$$Z = \frac{V_{rms}}{I_{rms}} = \frac{200}{0.5} = 400 \Omega$$

$$\text{Since } Z = \sqrt{R^2 + (X_L)^2}$$

$$\therefore (400)^2 - (200)^2 = (X_L)^2$$

$$X_L = \sqrt{600 \times 200} = 346.4 \Omega$$

$$\text{Inductance (L)} = \frac{X_L}{\omega} = \frac{364.4}{2 \times 3.14 \times 50} = 1.1 \text{H}$$

½

½

½

½

½

½

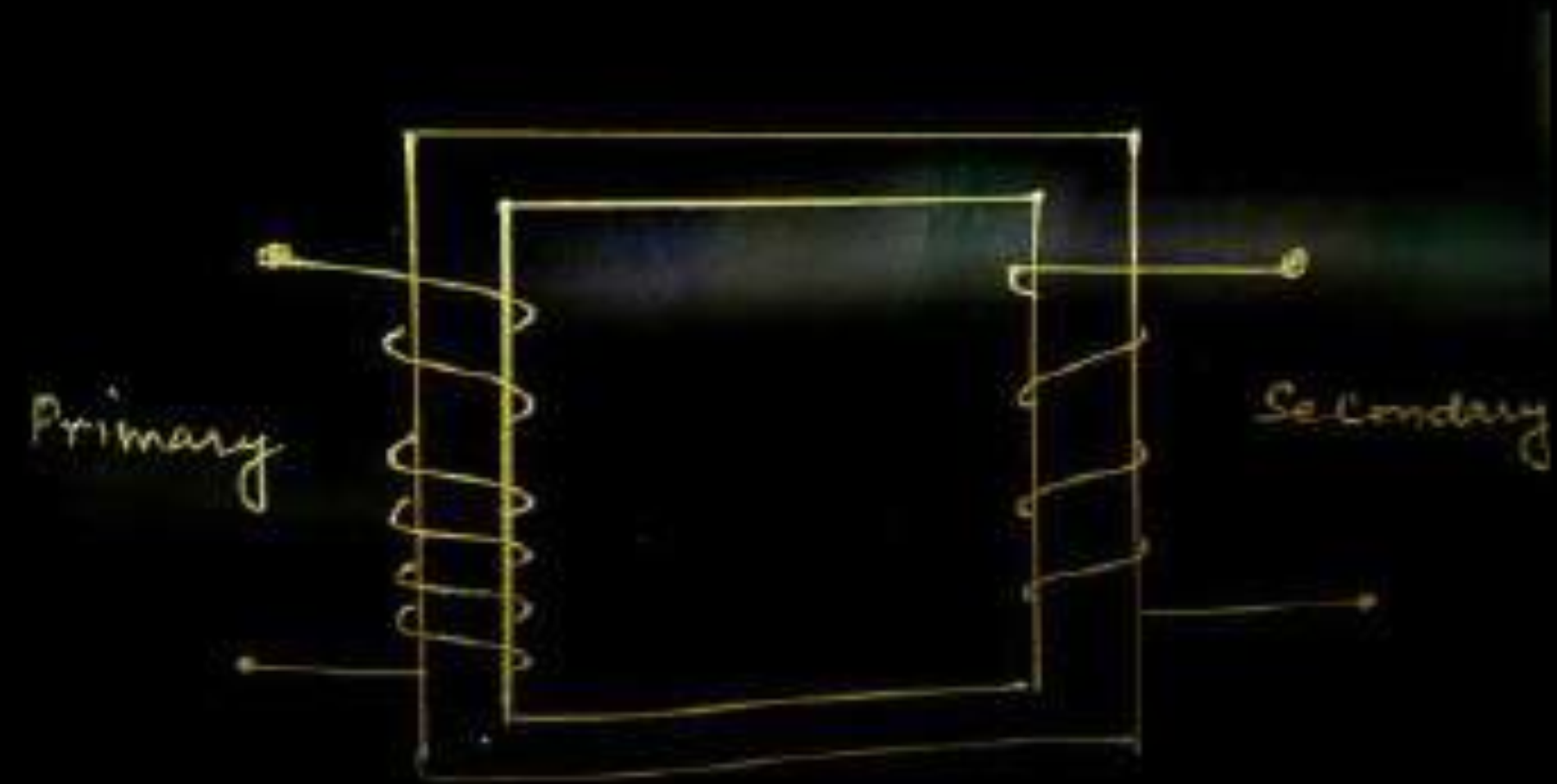
½

½

- (a) Draw the diagram of a device which is used to decrease high ac voltage into a low ac voltage and state its working principle. Write four sources of energy loss in this device.
- (b) A small town with a demand of 1200 kW of electric power at 220 V is situated 20 km away from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is $0.5\ \Omega$ per km. The town gets the power from the line through a 4000-220 V step-down transformer at a sub-station in the town. Estimate the line power loss in the form of heat.

CBSE 2019

(a)



Working Principle : When the alternating voltage is applied to the primary , the resulting current produces an alternating magnetic flux in secondary and induces an emf in it./It works on the mutual induction.

Four sources of energy loss

- (i) Flux leakage between primary and secondary windings
- (ii) Resistance of the windings
- (iii) Production of eddy currents in the iron core.
- (iv) Magnetization of the core.

(b) Total resistance of the line = length X resistance per unit length
= 40 km x 0.5 Ω /km
= 20 Ω

1

1/2

1/2

1/2

1/2

1/2

1/2

Current flowing in the line $I = P/V$

$$I = \frac{1200 \times 10^3}{4000}$$

$$= 300A$$

$\therefore$ Line power loss in the form of heat

$$P = I^2 R$$
$$= (300)^2 \times 20$$
$$= 1800 \text{ kW}$$

1/2

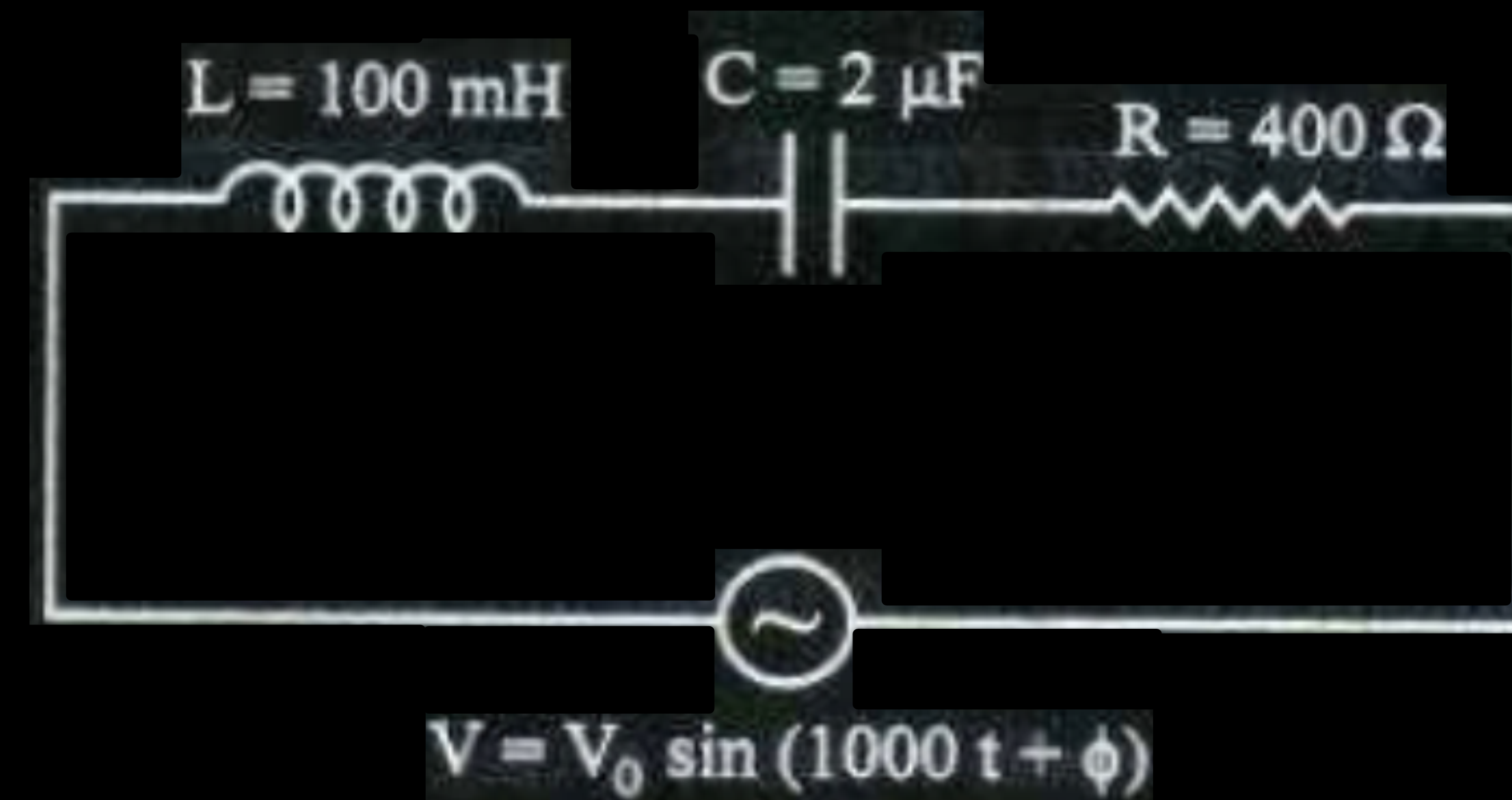
1/2

- (a) Draw graphs showing the variations of inductive reactance and capacitive reactance with frequency of the applied ac source.
- (b) Draw the phasor diagram for a series RC circuit connected to an ac source.
- (c) An alternating voltage of 220 V is applied across a device X, a current of 0.25 A flows, which lag behind the applied voltage in phase by $\frac{\pi}{2}$ radian. If the same voltage is applied across another device Y, the same current flows but now it is in phase with the applied voltage.
- (i) Name the devices X and Y.
- (ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y.

- (a) State the principle of working of a transformer.
- (b) Define efficiency of a transformer.
- (c) State any two factors that reduce the efficiency of a transformer.
- (d) Calculate the current drawn by the primary of a 90% efficient transformer which steps down 220 V to 22 V, if the output resistance is $440\ \Omega$.

CBSE 2018

- (i) Find the value of the phase difference between the current and the voltage in the series LCR circuit shown below. Which one leads in phase : current or voltage ?
- (ii) Without making any other change, find the value of the additional capacitor C_1 , to be connected in parallel with the capacitor C , in order to make the power factor of the circuit unity.

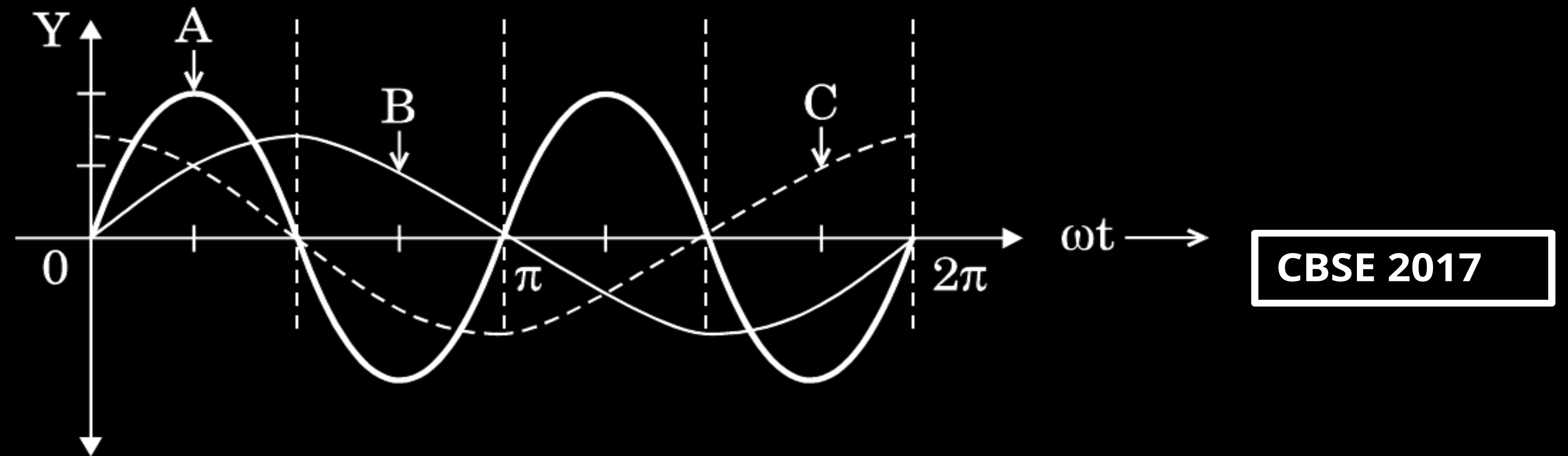


CBSE 2017

- (a) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary to primary voltage in terms of number of turns and currents in the two coils
- (b) A power transmission line feeds input power at 2200 V to a step-down transformer with its primary windings having 3000 turns. Find the number of turns in the secondary to get the power output at 220 V.

CBSE 2017

A device 'X' is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph :



- Identify the device 'X'.
- Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit ? Justify your answer.
- How does its impedance vary with frequency of the ac source ? Show graphically.
- Obtain an expression for the current in the circuit and its phase relation with ac voltage.

a) The device X is a capacitor

1/2

b) Curve B → voltage

1/2

Curve C → current

1/2

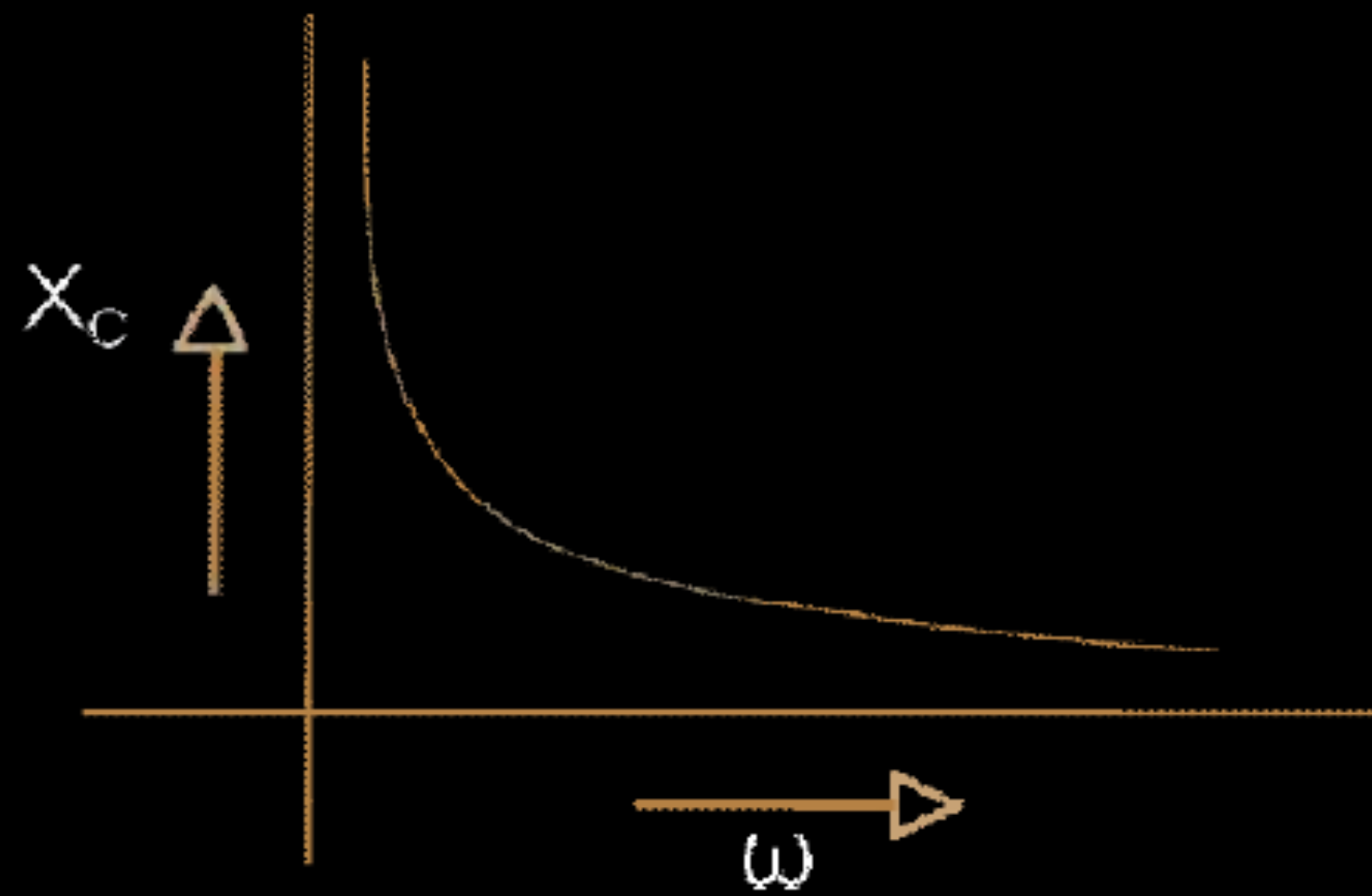
Curve A → power

Reason: The current leads the voltage in phase, by $\pi/2$, for a capacitor.

1/2

c) $X_c = \frac{1}{\omega C}$ ($X_c \propto \frac{1}{\omega}$)

1/2



1/2

d) $V = V_o \sin \omega t$

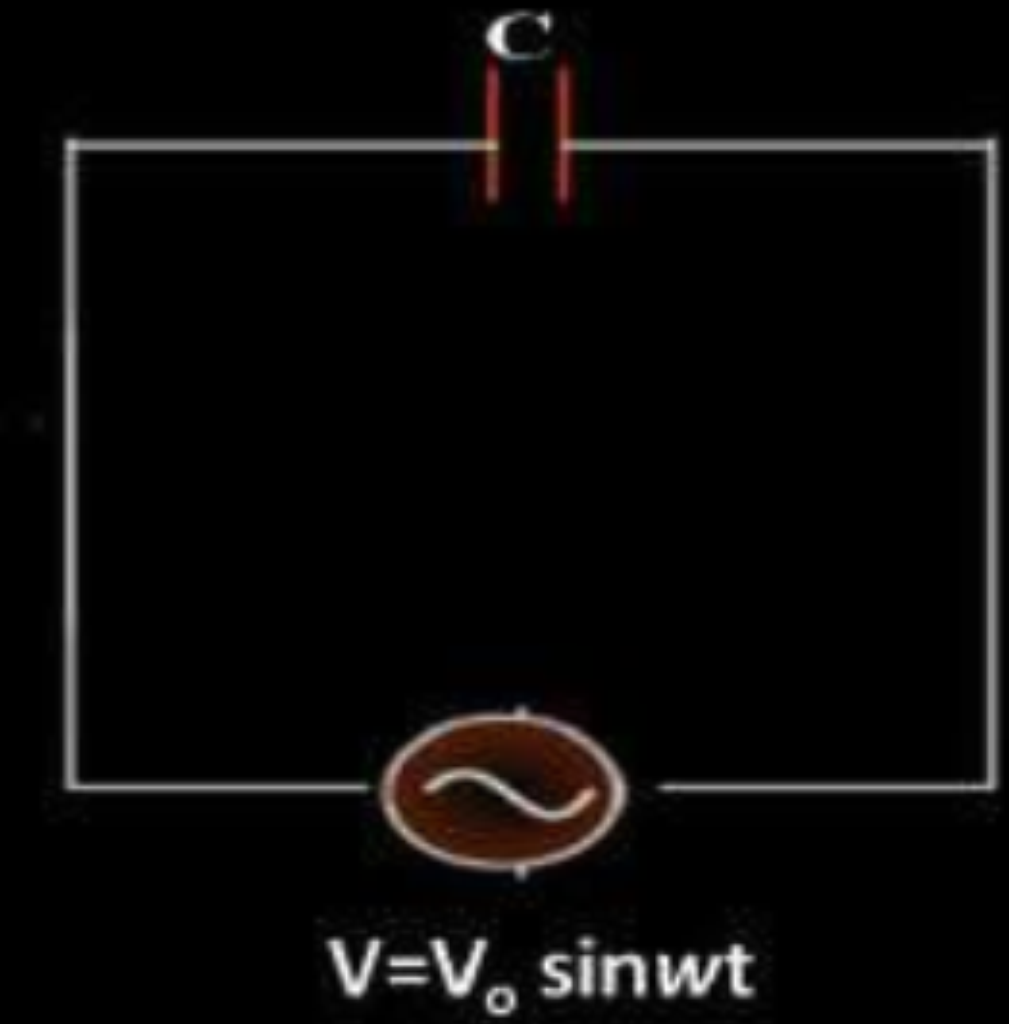
$$Q = CV = CV_o \sin \omega t$$

$$I = \frac{dq}{dt} = \omega C V_o \cos \omega t$$

$$= I_o \sin(\omega t + \pi/2)$$

Current leads the voltage, in phase, by $\pi/2$

(Note : If the student identifies the device X as an Inductor but writes correct answers to parts (c) and (d) (in terms of an inductor), the student be given full marks for (only) these two parts)



1/2

1/2

1/2

1/2

- (i) An a.c. source of voltage $V = V_0 \sin \omega t$ is connected to a series combination of L, C and R. Use the phasor diagram to obtain expressions for impedance of the circuit and phase angle between voltage and current. Find the condition when current will be in phase with the voltage. What is the circuit in this condition called ?
- (ii) In a series LR circuit $X_L = R$ and power factor of the circuit is P_1 . When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes P_2 . Calculate P_1/P_2 .

CBSE 2016

From Figure

$$\vec{V} = \vec{V}_L + \vec{V}_R + \vec{V}_C$$

where $|\vec{V}_R| = i_m R$

$$|\vec{V}_L + \vec{V}_C| = V_{Cm} - V_{Lm}$$

$$= i_m(X_C - X_L)$$

$$\Rightarrow V_m^2 = V_{Rm}^2 + (V_{Cm} - V_{Lm})^2$$

$$I_m^2 Z^2 = I_m^2 R^2 + I_m^2 (X_C - X_L)^2$$

$$\Rightarrow Z = \sqrt{R^2 + (X_C - X_L)^2}$$

From Figure

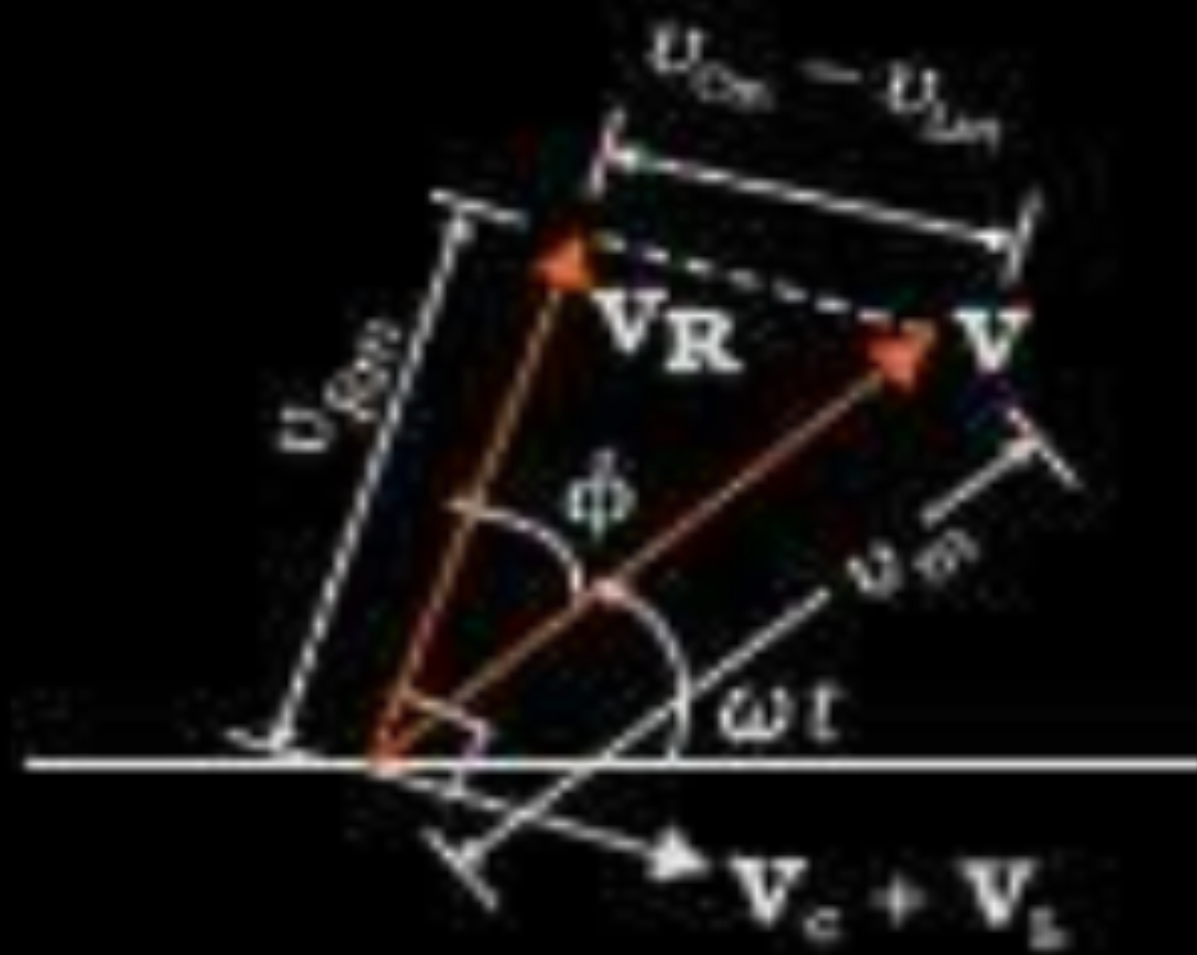
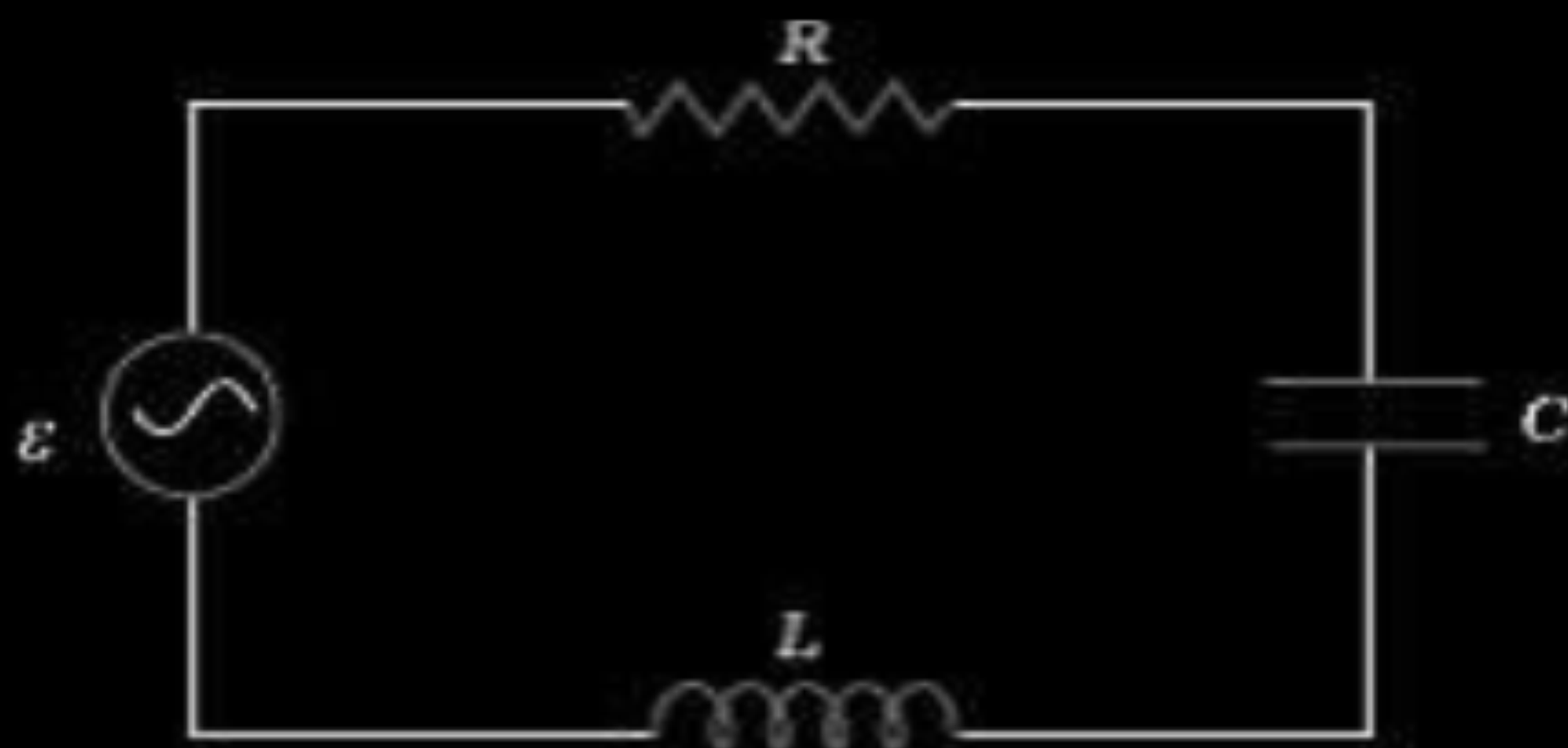
$$\tan \phi = \frac{V_{Cm} - V_{Lm}}{V_{Rm}} = \frac{i_m(X_C - X_L)}{i_m R}$$

$$\phi = \tan^{-1}\left(\frac{X_C - X_L}{R}\right)$$

Condition for current and voltage are in phase :

$$V_L = V_C \text{ or } X_L = X_C$$

Circuit is called Resonant circuit.



ii) Power factor $P_1 = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + R^2}} = \frac{1}{\sqrt{2}}$

(as $X_L = R$)

Power factor when capacitor C of Reactance $X_C = X_L$ is put in series in the circuit

$$P_2 = \frac{R}{Z} = \frac{R}{R} = 1$$

as $Z=R$ at resonance

$$\therefore \frac{P_1}{P_2} = \frac{\frac{1}{\sqrt{2}}}{1} = \frac{1}{\sqrt{2}}$$

- (i) Write the function of a transformer. State its principle of working with the help of a diagram. Mention various energy losses in this device.
- (ii) The primary coil of an ideal step up transformer has 100 turns and transformation ratio is also 100. The input voltage and power are respectively 220 V and 1100 W. Calculate
- (a) number of turns in secondary
 - (b) current in primary
 - (c) voltage across secondary
 - (d) current in secondary
 - (e) power in secondary

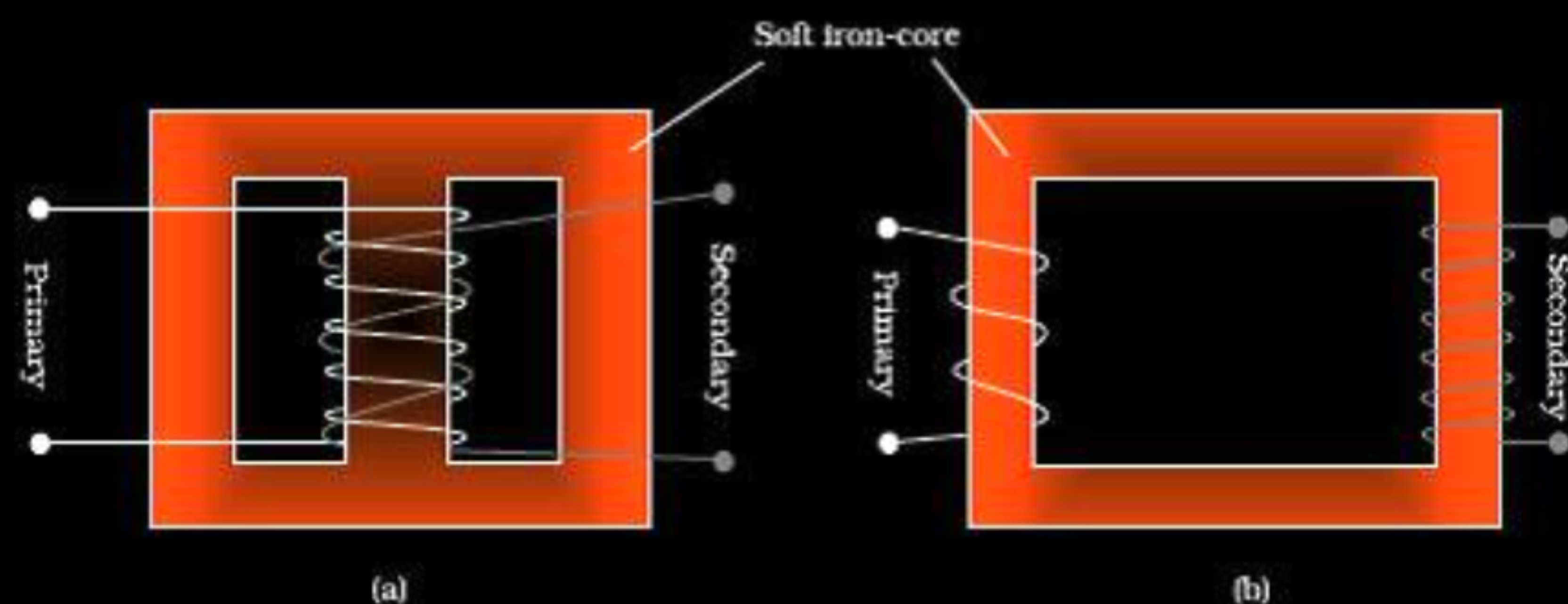
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(i) Conversion of ac of low voltage into ac of high voltage & vice versa

1/2

Mutual induction: When alternating voltage is applied to primary windings, emf is induced in the secondary windings.

1/2



1/2

(Any one of the above diagram)

Energy losses:

- Leakage of magnetic flux
- Eddy currents
- Hysteresis loss
- Copper loss

(Any two)

1/2 + 1/2

ii)

$$N_p = 100$$

Transformation ratio = 100

a) Number of turns in secondary coil

$$N_s = 100 \times 100 = 10000$$

1/2

b) Input Power = Input voltage x current in primary

$$1100 = 220 \times I_p$$

$$\Rightarrow I_p = 5A$$

1/2

$$c) \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\frac{V_s}{220} = 100$$

$$\Rightarrow V_s = 2.2 \times 10^4 \text{ volts}$$

1/2

$$d) \frac{I_p}{I_s} = \frac{N_s}{N_p}$$

$$\frac{5}{I_s} = 100$$

$$\Rightarrow I_s = \frac{5}{100} = 0.05 A$$

1/2

e) Power in secondary = Power in Primary

$$= 1100 W$$

1/2

The power factor of an a.c. circuit is 0.5. What is the phase difference between voltage and current in the circuit ?

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The power factor of an a.c. circuit is 0.5. What is the phase difference between voltage and current in the circuit ?

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Phase angle = 60°

[Note : If the student only writes, $[\cos \varphi = 0.5]$, give $\frac{1}{2}$ mark]

1

A capacitor of unknown capacitance, a resistor of $100\ \Omega$ and an inductor of self inductance $L = (4/\pi^2)$ henry are connected in series to an ac source of 200V and 50 Hz. Calculate the value of the capacitance and impedance of the circuit when the current is in phase with the voltage. Calculate the power dissipated in the circuit.

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$$\begin{aligned}\text{Capacitance} = C &= \frac{1}{L\omega^2} \\ &= \frac{1}{\frac{4}{\pi^2}(2\pi \times 50)^2} \text{F} \\ &= 2.5 \times 10^{-5} \text{ F}\end{aligned}$$

Impedance = resistance(since V and I are in phase)

$\therefore$ Impedance = $100\ \Omega$

$$\begin{aligned}\text{Power dissipated} &= \frac{E_{rms}^2}{R} \\ &= \frac{(200)^2}{100} \text{ W} = 400 \text{ watt}\end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

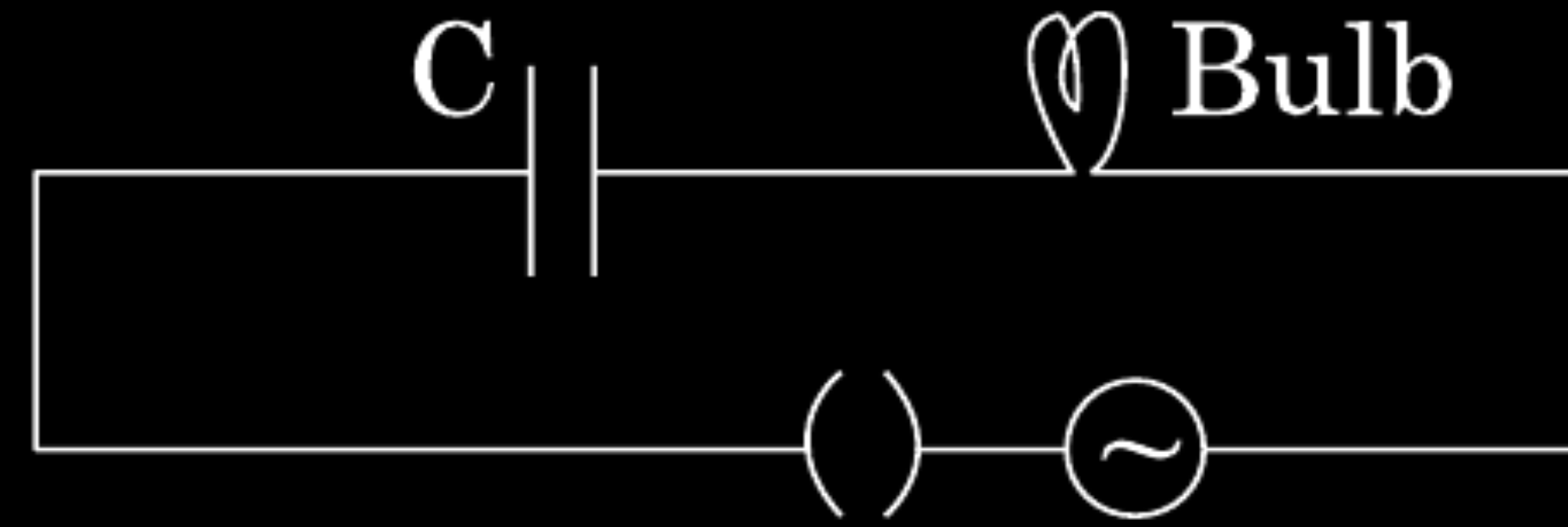
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (i) When an AC source is connected to an ideal capacitor, show that the average power supplied by the source over a complete cycle is zero.
- (ii) A bulb is connected in series with a variable capacitor and an A.C. source as shown. What happens to the brightness of the bulb when the key is plugged in and capacitance of the capacitor is gradually reduced ?



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(i) Let the applied voltage be

$$V = V_0 \sin \omega t$$

The current through an ideal capacitor, would then be

$$I = I_0 \sin \left(\omega t + \frac{\pi}{2} \right) = I_0 \cos \omega t$$

1/2

$$\therefore P_{inst} = VI$$

$$\therefore P_{AV} = \frac{1}{T} \int_0^T VI dt$$

1/2

$$\begin{aligned} \therefore P_{AV} &= \frac{V_0 I_0}{2} \langle \sin 2\omega t \rangle \\ &= 0 \end{aligned}$$

1/2

1/2

(Alternatively,

For an ideal capacitor, the current leads voltage in phase by $\pi/2$.

$$\begin{aligned} \therefore P &= \frac{E_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi = \frac{E_0 I_0}{2} \cos \frac{\pi}{2} \\ &= 0 \end{aligned}$$

(ii) The brightness of the bulb would also reduce gradually.

1

(Alternatively:

$$X_c = \frac{1}{\omega C}$$

$\therefore X_c$ increases as C decreases. Hence, with decreasing C , the brightness of the bulb would decrease.)

In a series LCR circuit connected to an a.c. source of voltage $v = v_m \sin \omega t$, use phasor diagram to derive an expression for the current in the circuit. Hence obtain the expression for the power dissipated in the circuit. Show that power dissipated at resonance is maximum.

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Using the phasor diagram, we get

$$v_m^2 = v_{Rm}^2 + (v_{Cm} - v_{Lm})^2$$

$$\text{Or } v_m^2 = i_m^2 [R^2 + (X_C - X_L)^2]$$

$$\therefore i_m = \frac{v_m}{\sqrt{R^2 + (X_C - X_L)^2}}$$

$$\text{Also, } \tan \phi = \frac{v_C - v_L}{v_R} = \frac{X_C - X_L}{R}$$

$\therefore$ the expression, for current, is

$$i = i_m \sin(\omega t + \phi)$$

(Note: Award these two marks even if the student draws the phasor diagram / does the derivation of $i = i_m \sin(\omega t - \phi)$ for $X_C < X_L$)

Power dissipated:

The instantaneous power, p, supplied by the source, is

$$p = v i$$

$$= (v_m \sin \omega t)(i_m \sin(\omega t + \phi))$$

$$= \frac{v_m i_m}{2} [\cos \phi - \cos(2\omega t + \phi)]$$

The average power, over a cycle, is, therefore

$$P = \langle p \rangle = \frac{V_m i_m}{2} (\cos \phi)$$

$$= VI \cos \phi$$

At resonance, we have

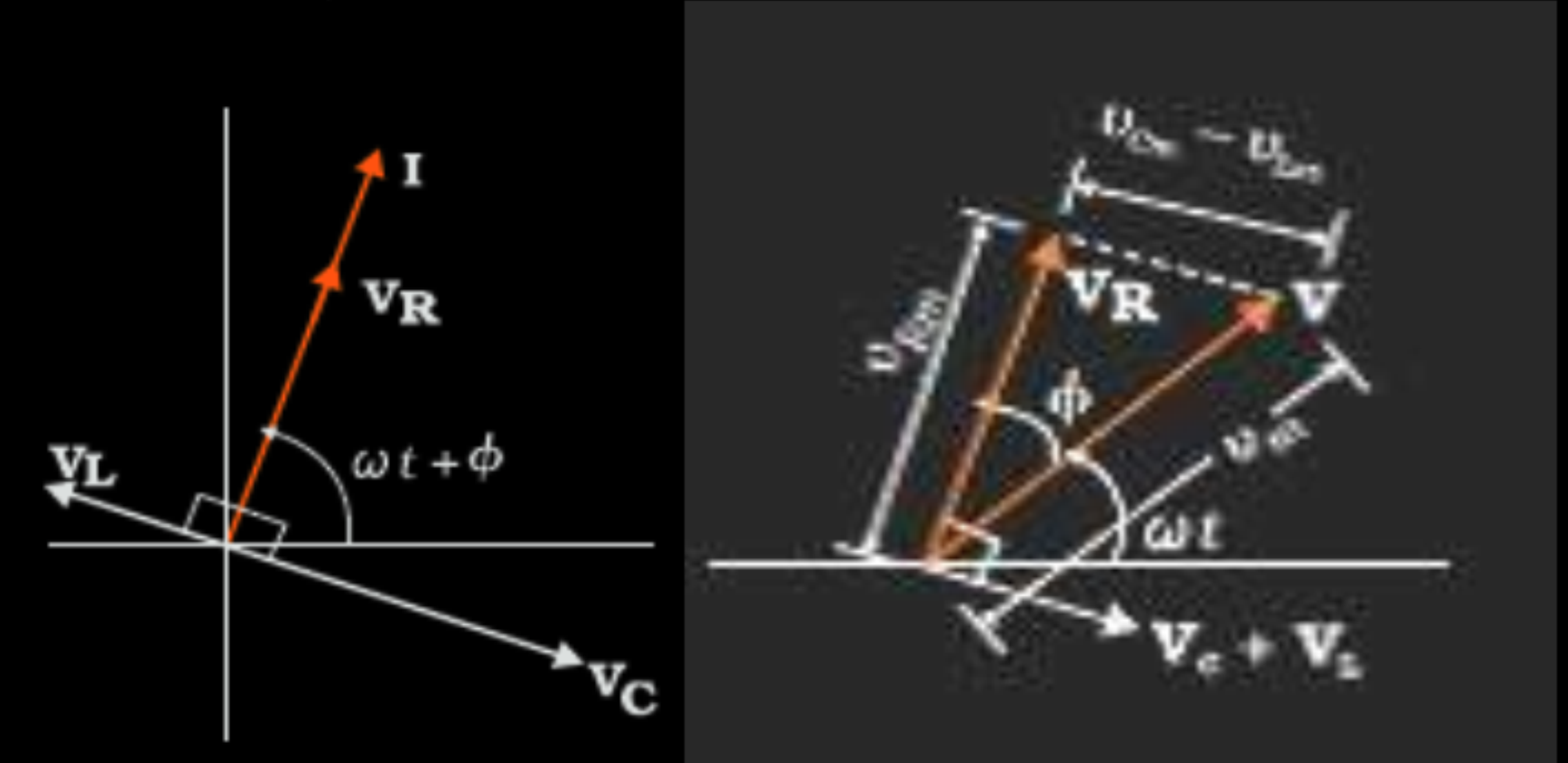
$$X_C = X_L$$

$$\tan \phi = 0 \Rightarrow \phi = 0^\circ$$

$$\therefore \cos \phi = 1, \text{ its maximum value.}$$

Hence $P (= VI \cos \phi)$ has its maximum value at resonance.

Phasor diagram



1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

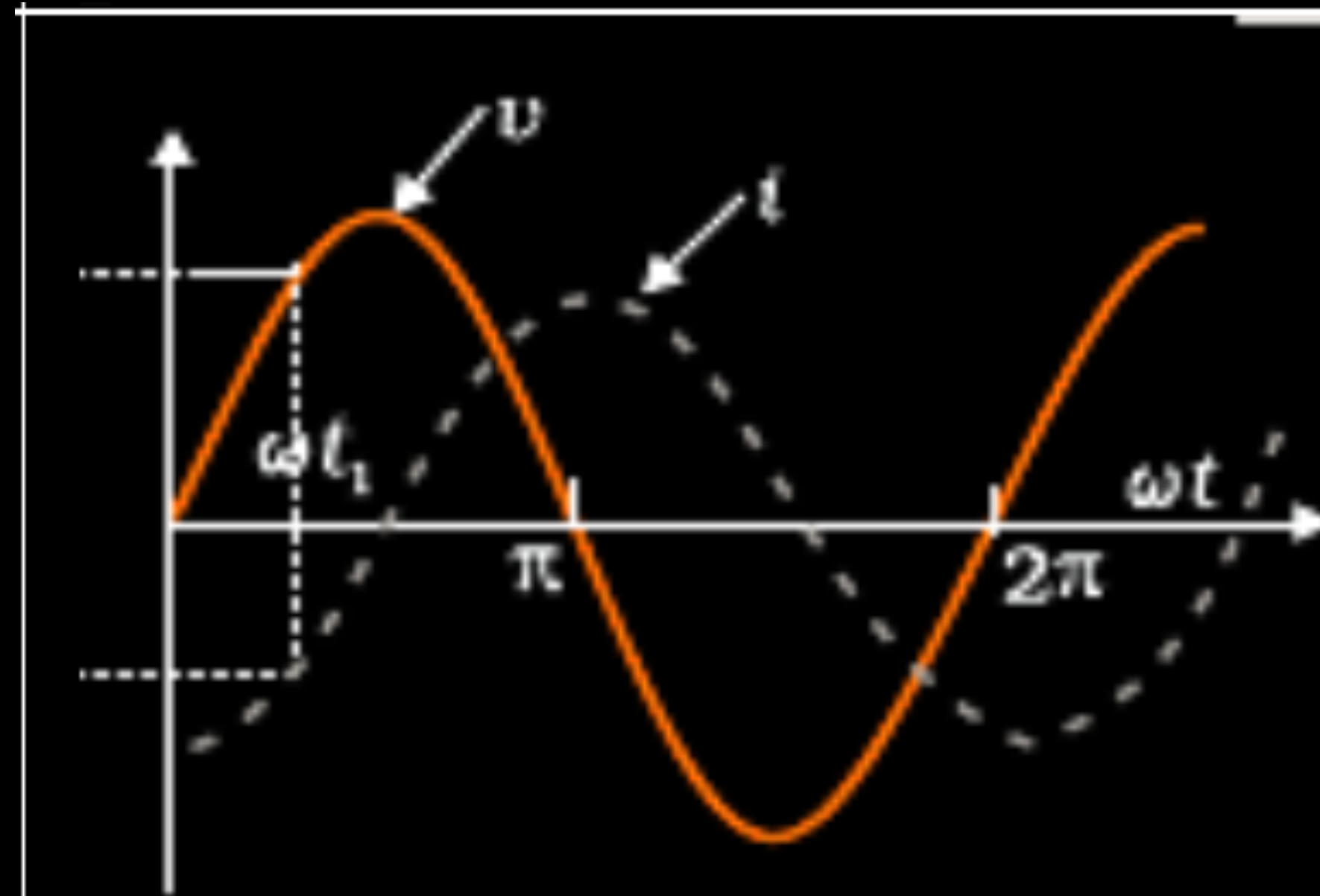
1/2

An a.c. source of voltage $V = V_0 \sin \omega t$ is connected to an ideal inductor. Draw graphs of voltage V and current i versus ωt .

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An a.c. source of voltage $V = V_0 \sin \omega t$ is connected to an ideal inductor. Draw graphs of voltage V and current i versus ωt .

CBSE 2016



Graph of V
Graph of I

$\frac{1}{2}$
 $\frac{1}{2}$

Explain why current flows through an ideal capacitor when it is connected to an a.c. source but not when it is connected to a d.c. source in a steady state.

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Explain why current flows through an ideal capacitor when it is connected to an a.c. source but not when it is connected to a d.c. source in a steady state.

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For a.c. source, circuit is complete due to the presence of displacement current in the capacitor. For steady dc, there is no displacement current, therefore, circuit is not complete.

$\frac{1}{2} + \frac{1}{2}$

[Alternatively, Capacitive reactance $X_c = \frac{1}{2\pi fC} = \frac{1}{\omega C}$

So, capacitor allows easy path for a.c. source.

For d.c, $f = 0$, so $X_c = \text{infinity}$,

So capacitor blocks d.c]

$\frac{1}{2} + \frac{1}{2}$

Shiv had a high tension tower erected on his farm land. He kept complaining to the authorities to remove it since it occupied a large portion of his land. His uncle, who was a teacher, explained to him the need for erecting these towers for efficient transmission of power. As Shiv got convinced and realized its significance, he stopped complaining.

Based on the above paragraph, answer the following questions :

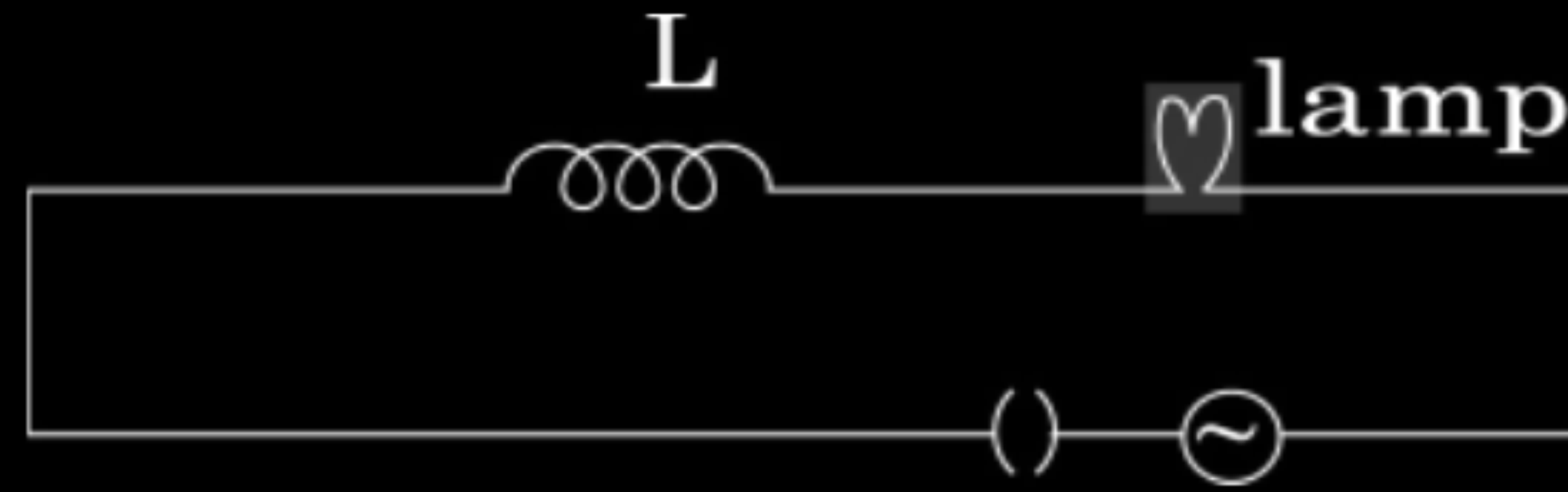
- (a) Why is it necessary to transport power at high voltages ?
- (b) 'A low power factor implies large power loss'. Explain.
- (c) Write the two values displayed by Shiv and his Uncle.

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(a) To reduce power losses in the transmission line.	1
(b) Since power loss is inversely proportional to power factor $(P = VI \cos \varphi \text{ where } \cos \varphi \text{ is power factor}). \text{ To supply a given power at a given voltage, if } \cos \varphi \text{ is small, we have to increase current accordingly. This will lead to large power loss } (I^2 R) \text{ in transmission /}$ $(Effective\ Power = \frac{True\ Power}{\cos \varphi})$	1
(c) Values displayed by	
(i) Shiv – understanding nature/ respecting elders/ helping nature/ caring/ etc.	$\frac{1}{2} + \frac{1}{2}$
(ii) Uncle– knowledgeable/ helping nature/ caring/ etc.(Any two each)	$\frac{1}{2} + \frac{1}{2}$

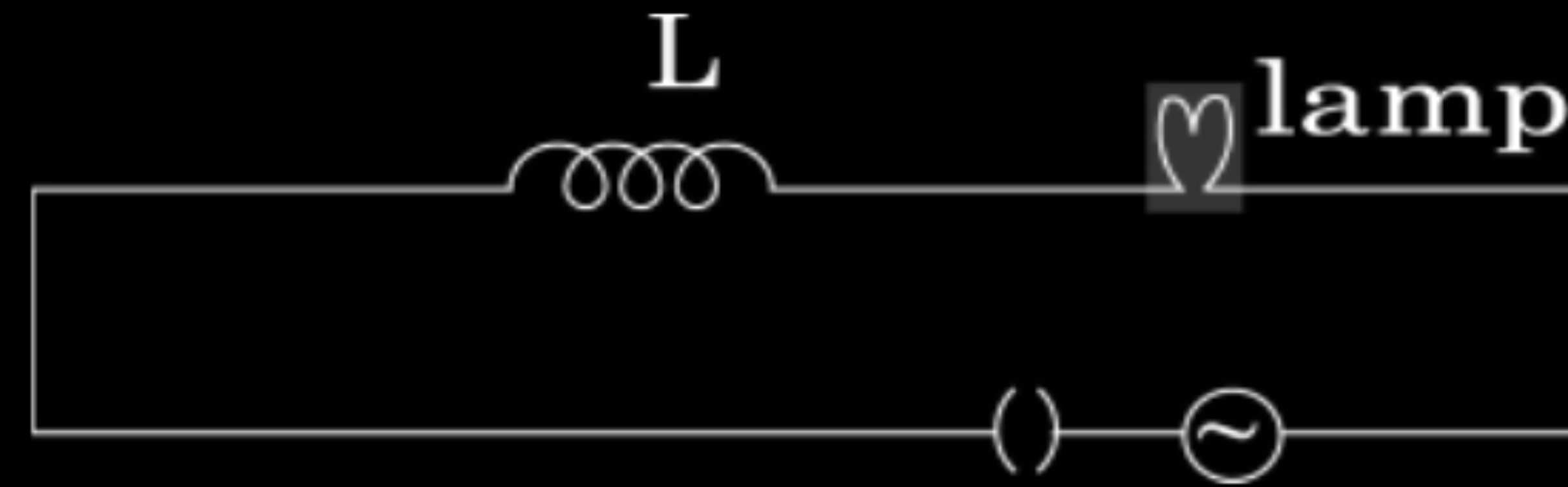
- (i) When an AC source is connected to an ideal inductor show that the average power supplied by the source over a complete cycle is zero.
- (ii) A lamp is connected in series with an inductor and an AC source. What happens to the brightness of the lamp when the key is plugged in and an iron rod is inserted inside the inductor ? Explain.

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i) $P_{av} = I_{av} \times e_{av} \cos \phi$

For an ideal inductor, $\phi = \frac{\pi}{2}$

$$\therefore P_{av} = I_{av} \times e_{av} \cos \frac{\pi}{2}$$

$$P_{av} = 0$$

ii) Brightness decreases

Because as iron rod is inserted inductance increases.
Thus, current decreases and brightness decreases.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (i) Draw a labelled diagram of a step-down transformer. State the principle of its working.
- (ii) Express the turn ratio in terms of voltages.
- (iii) Find the ratio of primary and secondary currents in terms of turn ratio in an ideal transformer.
- (iv) How much current is drawn by the primary of a transformer connected to 220 V supply when it delivers power to a 110 V – 550 W refrigerator ?

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Principle-

When the current flowing through the primary coil changes, an emf is induced in the secondary coil due to the change in magnetic flux linked with it.

[Note- Give $\frac{1}{2}$ mark to the student who writes only mutual induction only.]

ii. $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

iii. For an ideal transformer,

$$i_p V_p = i_s V_s \quad \therefore \frac{i_p}{i_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

iv. We have

$$i_p V_p = i_s V_s = 550 \text{ W}$$

$$V_p = 220 \text{ V}$$

$$i_p = \frac{550}{220} = \frac{5}{2} = 2.5 \text{ A}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

